Tax Policy and Capital Maintenance

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The tax elasticity of investment is commonly treated as a sufficient statistic for the aggregate effects of tax reform. This breaks down when firms maintain existing capital: cheaper investment leads firms to substitute away from maintenance, raising depreciation, so investment promotes capital churn rather than growth. Combining asset-level railroad data and industry-level corporate tax returns, I show maintenance is large and elastic, implying new investment translates into 57% as much capital as typically assumed. Applied to the 2017 Tax Cuts and Jobs Act, corporate capital grows by $248 billion less than standard models predict, with corresponding output and wage haircuts.

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1 Introduction

Policymakers repeatedly cut corporate taxes and subsidize investment to raise growth and wages (Romer and Romer 2010), and the case for these policies rests in part on the tax elasticity of investment. In workhorse models, the tax elasticity of investment is equal to the tax elasticity of capital. Since tax cuts do increase investment (Hassett and Hubbard 2002; Zwick and Mahon 2017; Kennedy et al. 2023), the observed investment can be mapped one-for-one into capital and, under further general equilibrium assumptions, into output and wages (Hall and Jorgenson 1967; Chodorow-Reich et al. 2025). However, that inference rests on investment being the only margin through which firms adjust their capital stock. It is not. Firms also *maintain* existing capital, and this overlooked margin interrupts both the transmission of tax cuts to capital accumulation and the mapping of investment into capital. That is because maintenance has always been fully tax-deductible whereas investment typically is not. This paper's contribution is to show theoretically, empirically, and quantitatively this difference in tax treatment is critically important for the evaluation of tax reform.

Theoretical Mechanisms. In models with fixed depreciation, the cost of capital is simply the cost of purchasing new assets—the familiar Hall and Jorgenson (1967) user cost of capital. However, the true cost of capital services includes both the initial purchase and the ongoing cost of maintaining assets over their lifetime. Accounting for that is important because the tax treatment of maintenance differs from investment. Whereas historically investment and its return are taxed at the corporate tax rate, maintenance has always been fully tax-deductible. As a result, the cost of capital services is a weighted average of the cost of new capital (which responds to tax incentives) and the cost of maintenance (which does not). Consequently, when policymakers introduce investment subsidies, only one component of this weighted average falls. I call this a *capital preservation effect*. The scale response is therefore smaller than standard models predict, with the magnitude of attenuation equal to the maintenance share of user cost. Standard models implicitly assume this share is zero.

Beyond the capital preservation effect, tax incentives create a relative price distortion in the choice to maintain existing capital or buy new capital. By reducing the cost of new capital while leaving maintenance costs unchanged, they make replacement relatively cheaper than preservation. If maintenance demand is elastic, firms substitute away from maintenance and toward investment, and economic depreciation rises. This creates an *investment-capital gap*: gross investment overstates net capital accumulation because more of the observed investment offsets higher depreciation rather than expanding the capital stock. Both mechanisms survive in general equilibrium. Equilibrium forces plausibly reinforce, rather than undo, the attenuation.

Maintenance Demand in the Data. The theoretical mechanisms depend on two parameters. First, the capital preservation effect requires knowing the *level* of maintenance relative to the user cost of capital. McGrattan and Schmitz Jr. (1999) points out that this share is plausibly large in Canada, but it remains underexplored in the United States. Second, the investment-capital gap depends on the *elasticity* of maintenance demand. Goolsbee (1998b) shows that the decision to *replace* capital is price-sensitive, but we lack a direct estimate of the demand elasticity.[1] I estimate this demand function twice, in two settings that exploit firms' differential exposure to a common shock and recover comparably large elasticities: a hand-collected, firm-asset-level panel of freight railroads, and economy-wide industry data from corporate tax returns. Because the two rely on entirely different variation (input costs in one, tax policy in the other), their agreement is difficult to attribute to confounding in either setting alone.

My first source of data comes from the regulatory filings of large freight railroads. Class I freight railroads—those with revenue over roughly $1 billion—must submit detailed accounts of their financial and real activities to the Surface Transportation Board. I hand-collect and digitize these going back several decades. Although these filings are an unparalleled window into firms' maintenance and investment behavior at the asset level, they have been largely unused in economics since Bitros (1976). I observe the quantity and value of each firm's locomotives and freight cars, what they spend maintaining them, and how much of that maintenance is done in-house rather than contracted out. To study maintenance demand, I focus on the period 1999-2023.

Three facts emerge. First, maintenance is large: the maintenance share of user cost averages 60%, implying standard user cost formulas overstate tax responsiveness by a factor of 2.5. Second, maintenance spending typically exceeds investment in new assets. Third, about two-thirds of maintenance is performed in-house, with labor costs accounting for 40% of total expenditures. I exploit cross-firm and cross-asset variation in this labor share for identification.

To identify the maintenance demand elasticity, I construct a Bartik-style instrument that leverages variation in the geographic distribution and labor composition of maintenance costs by firm. By interacting pre-determined, firm-specific maintenance labor cost shares with plausibly exogenous state-level shocks to maintenance wages, this strategy isolates cost-driven shifts in the relative price of maintenance from unobserved firm-level demand shocks. The analysis yields a maintenance demand elasticity of approximately 4.8. Crucially, this response is driven entirely by adjustments to in-house maintenance, while outsourced maintenance services do not

1. Goolsbee (1998b) provides important precedent for the mechanism I study. He shows that airlines effectively increase their depreciation rates by retiring planes more quickly in response to investment subsidies. Though his setting involves scrappage and resale rather than maintenance decisions, the economic insight is the same: investment incentives can endogenously raise depreciation, breaking the standard one-to-one mapping from investment to capital. In a putty-clay model with vintages, his replacement effect is formally equivalent to the endogenous depreciation I model through maintenance.

respond. This divergence supports a causal interpretation: the instrument isolates a supply-side cost shock rather than a confounding demand shock, which would move both margins in tandem.

Because freight rail may not be representative of capital more broadly and the firm-level analysis cannot capture general equilibrium forces, I turn to industry-level corporate tax returns from the Internal Revenue Service's Statistics of Income (SOI). These cover the entire corporate sector and net out within-industry reallocation the railroad data cannot. There, the maintenance share of user cost is less than half as large as in the railroad data, and only about half as large as new investment in levels; the demand elasticity, however, is again large and positive. Following the empirical *investment* literature (Zwick and Mahon 2017; Curtis et al. 2021), I use a difference-in-differences across industries' exposure to major investment incentives—bonus depreciation and the 2017 Tax Cuts and Jobs Act—to identify the *maintenance* demand elasticity. Because industries differ in their capital composition, these national policies create differential, plausibly exogenous shocks to the after-tax wedge between new investment and maintenance. This yields an estimated elasticity of approximately 2.6—on the same order of magnitude as the railroad estimate—and the elasticity is positive only for taxable firms, exactly as theory predicts.

Quantitative Implications. The empirical findings—positive and elastic maintenance demand—alter both how we interpret existing evidence on tax reform and how we conduct counterfactual analysis of it.

First, accounting for maintenance changes how we should read existing estimates. The standard approach in public finance estimates investment elasticities and treats them as equivalent to capital elasticities, implicitly assuming a one-to-one correspondence between investment and capital accumulation (Hartley, Hassett, and Rauh 2025; Chodorow-Reich et al. 2025). But accounting for maintenance breaks this correspondence. When investment incentives alter relative prices, firms substitute away from maintenance, raising depreciation endogenously. The same observed investment response therefore corresponds to a smaller capital response: a portion of gross investment now simply replaces capital that depreciates faster rather than expanding the productive stock. I derive a closed-form correction showing the tax elasticity of capital is roughly half of the observed investment elasticity at empirically estimated maintenance parameters, and a second, distinct correction shows that maintenance also biases the measured investment elasticity itself toward zero. Because both are ratios, they apply to whatever investment elasticity one adopts.[2] This substantial wedge, driven by the maintenance margin, matters for both policy analysis and for models calibrated to match observed investment moments.

2. The capital preservation effect also implies that standard investment regressions suffer from omitted variable bias: because maintenance costs are tax-shielded, the measured change in user cost overstates the true change, causing the estimated coefficient to be biased downward. I provide corrections for both channels.

Second, the two mechanisms imply smaller macroeconomic effects than standard models predict. The capital preservation effect operates through the level of maintenance: when maintenance accounts for 25-60% of user cost (as I find empirically), tax cuts reduce total capital costs by only 40-75% as much as standard models predict. The steady-state capital stock therefore rises proportionally less, with correspondingly attenuated effects on output and wages. The investment-capital gap reinforces this attenuation along the transition path. Because elastic maintenance demand breaks the one-to-one mapping between gross investment and net capital accumulation, matching observed investment responses requires higher adjustment costs in models with maintenance than in standard models. This slows convergence to the new steady state, so growth effects within policy-relevant horizons—such as the ten-year budget window used for tax scoring—are even smaller than the steady-state haircut alone would suggest.

To quantify these forces, I embed the estimated maintenance function into a general equilibrium model calibrated to the U.S. economy, following the framework of Chodorow-Reich et al. (2025). I calibrate the maintenance demand function (both its level and its elasticity) to my main, economy-wide SOI specification. Simulating the TCJA, I compare outcomes to an otherwise identical model without maintenance, isolating the haircut maintenance imposes. The results confirm the first-order importance of the maintenance margin. Across a variety of model closures, including those that account for general equilibrium effects through wages, interest rates, and maintenance prices, the ten-year responses of macroeconomic aggregates are roughly 70-75% as large as in the standard framework, and smaller still (closer to one-half) once interest-rate crowding out is included. The implied overestimate of corporate capital (roughly $248 billion over the decade, about a quarter of the gain the standard model predicts) is the gap I identify at the outset. And because this muted capital response generates correspondingly less growth, tax reform proves meaningfully less self-financing than the standard framework implies.

Related Literature. This paper relates to four main strands of literature.

First, my paper builds on the literature on multiple margins of capital adjustment. The closest paper is McGrattan and Schmitz Jr. (1999). In a model with endogenous depreciation and the asymmetric tax treatment of maintenance and investment, they show that a tax cut raises steady-state capital intensity by less than a standard model implies, and their Canadian survey evidence makes the case that maintenance is too large to ignore. Their result is a sign, not a magnitude. They establish the attenuation as a comparative static and never measure it, in part because no comparable U.S. data exist. I provide the magnitude they leave open, and I add a second distortion their steady-state comparison misses. Two mechanisms result. The capital preservation effect works through the *level* of maintenance. Because maintenance is already expensed, only a fraction $(1-s_m)$ of the user cost responds to investment incentives, so the response is attenuated even when maintenance demand is inelastic. This is the magnitude of their attenuation, equal to the maintenance share of user cost. The investment-capital gap, the new distortion, works through its *elasticity*. Because tax cuts endogenously raise depreciation, observed investment responses overstate capital accumulation, which breaks the one-to-one mapping from investment to capital and slows convergence. To discipline both, I provide the first estimates of the maintenance demand function for the United States, then trace them through general equilibrium to scoring the TCJA. Other papers, including Kabir, Tan, and Vardishvili (2024), Boucekkine, Fabbri, and Gozzi (2010), and Albonico, Kalyvitis, and Pappa (2014), study maintenance and capital utilization in non-tax settings. Feldstein and Rothschild (1974) and Cooley, Greenwood, and Yorukoglu (1997) study how tax policy shapes capital replacement in vintage-capital settings, with broadly similar logic: tax cuts can lower the value of existing capital relative to newer vintages and so accelerate replacement. In addition, Feldstein and Rothschild (1974) and Cooley, Greenwood, and Yorukoglu (1997) study how tax policy influences capital replacement decisions in vintage capital settings. Although their models are substantially different, their points are broadly similar: tax cuts may reduce the value of existing capital relative to newer vintages and therefore lead to faster replacement.

Second, maintenance's empirical relevance is well documented for housing but not for firms. A large literature documents the determinants and effects of residential housing maintenance, showing it is economically large (Knight and Sirmans 1996; Harding, Rosenthal, and Sirmans 2007; Hernandez and Trupkin 2021).[3] On the firm side, tax cuts appear to induce firms to replace old, higher-maintenance capital with younger, lower-maintenance capital. For example, Goolsbee (1998b) shows that airlines retire their airplanes more quickly when tax policy makes it favorable to buy new ones. Similarly, Goolsbee (2004) shows that firms buy capital with lower maintenance requirements following tax cuts. Some studies—which abstract from tax distortions—rely on maintenance data from India (Kabir, Tan, and Vardishvili 2024; Kabir and Tan 2024) or Canada (Albonico, Kalyvitis, and Pappa 2014; Angelopoulou and Kalyvitis 2012), but none, to my knowledge, estimate a maintenance demand function or use U.S. data. This lack of direct evidence has forced prior models to rely on assumptions. My work provides the first empirical estimate of the U.S. maintenance demand function.

Third, the user cost of capital is the primary transmission channel for business tax policy in virtually all modern frameworks, whatever complications they add—explicit demographics (PWBM 2019), heterogeneous capital (Barro and Furman 2018), heterogeneous firms (Sedlacek

3. The investment-maintenance distortion goes the other way in the housing tax code. Whereas improvements are deductible from the capital gains tax basis, maintenance is not, which creates a distortion in favor of the former. There is no direct evidence of the importance of that margin, but Cunningham and Engelhardt (2008) and Shan (2011) show that the 1997 Taxpayer Relief Act, which lowered the capital gains tax, increased housing mobility, which is akin to increasing the renewal rate of housing.

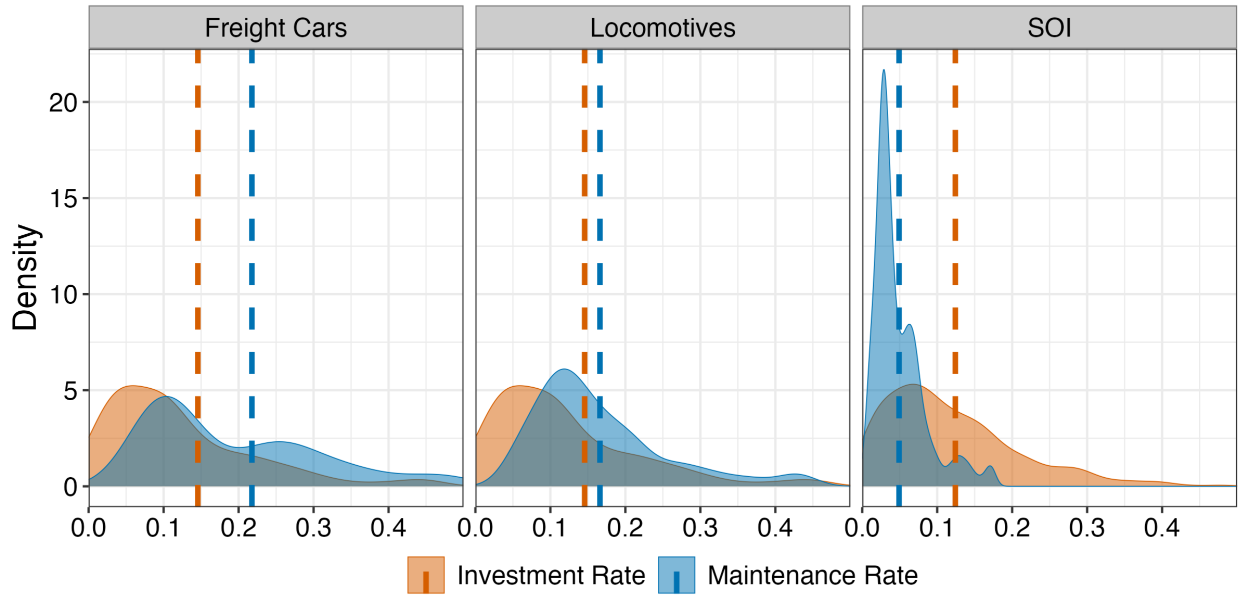
and Sterk 2019; Zeida 2022), lumpy adjustment (Winberry 2021), financial frictions (Occhino 2023), or global tax considerations (Chodorow-Reich et al. 2025). Because my key mechanisms alter this user cost directly, the haircut I identify in a neoclassical model should carry over to these richer settings: it furnishes a general adjustment factor for the quantitative tax reform literature, dampening the central channel of capital taxation regardless of other model features.

Finally, the elasticity of *capital* need not equal the elasticity of *investment*, yet the investment-regression literature routinely reads one as the other. Since Hall and Jorgenson (1967) and Summers (1981), that literature has produced a consensus short-run semi-elasticity of the investment rate to the user cost of roughly -0.5 to -1 (Hassett and Hubbard 2002; Chodorow-Reich, Zidar, and Zwick 2024), with modern quasi-experimental studies finding robust, sometimes larger, responses (Zwick and Mahon 2017; Kennedy et al. 2023; Chodorow-Reich et al. 2025; Hartley, Hassett, and Rauh 2025); Chodorow-Reich (2025) stresses that these are conceptually distinct objects. I identify a tax-driven source of the divergence: when investment becomes cheaper, firms substitute away from maintenance, depreciation rises, and observed investment overstates capital accumulation. A second, separate distortion runs through the regressions themselves: because maintenance is tax-shielded, the change in the user cost these studies measure overstates the true change, biasing the estimated semi-elasticities toward zero, so the underlying responses are larger than the published coefficients suggest. Because both corrections are ratios, they apply to whatever investment elasticity one adopts, leaving my conclusions largely orthogonal to ongoing debates over its magnitude.

2 Maintenance vs. Investment

Firms engage in a wide range of activities to change the capital stock. Those activities are distinguished by maintenance, which preserves or restores capital services without improving quality, and investment, which creates or upgrades capital. For example, a ground shipping company may make its vehicle fleet live longer through diligent attention to routine maintenance like oil changes or proactively changing the tires early to avoid worse damage through a highway tire blowout. By contrast, investment would be purchasing an entirely new fleet of vehicles or replacing the engine in an existing vehicle. Thus, the key distinction between investment and maintenance is that the former adds new capital to the stock, while the latter simply keeps old capital around for longer in its existing quality. This distinction is not merely semantic. It maps directly to firm expenditures and how the tax code treats them.

Figure 1: Density plots for maintenance and investment rates



Note: Each density plot is constructed with beginning of period book capital in the denominator. The dashed lines are mean maintenance and investment rates. The rolling stock data span from 1999-2023, with 172 observations per asset type. There are 1116 observations in the SOI data, covering 49 major industries from 1999-2021.

Before turning to tax rules, it is useful to establish magnitudes. How large is maintenance, relative to investment? Figure 1 plots the distribution of maintenance and investment rates—measured as shares of beginning-of-period book capital—across two major railroad asset types and across all major industries from the Statistics of Income (SOI). I describe the underlying data in detail in Section 4, but include this preview here to illustrate the economic importance of maintenance expenditures. In each case, maintenance rates are substantial, suggesting that firms allocate considerable resources toward preserving existing capital, on the same order of magnitude as investment.

The tax code reinforces the distinction between maintenance and investment through differential treatment under Sections 162 and 263 of the Internal Revenue Code. In general, maintenance is immediately deductible as an operating expense, while investment must be capitalized and depreciated over time. That matters for a firm deciding whether to invest an extra dollar in new capital or spend the dollar on maintaining existing capital. Letting τ^c denote the firm's tax rate, the after-tax cost of maintenance is $1 - \tau^c$ because it is deductible. On the other hand, a new capital expenditure must be depreciated over time.[4] Let z denote the net present value

4. Firms cannot easily game this distinction because routine maintenance expenditures typically fall below a *de minimis* safe-harbor threshold, while capital improvements do not, and established case law enforces consistent application of the rules. For example, in the 2003 court case *FedEx Corp. v. United States*, FedEx disputed whether

of the depreciation deductions, and c denote any additional investment incentives like investment tax credits. The after-tax cost of investment is $1 - \tau^c z - c$. This implies a wedge in the maintenance-investment decision given by

$$\text{Wedge} = \frac{\text{After-tax Cost of \$1 of Maintenance}}{\text{After-tax Cost of \$1 of Investment}} = \frac{1-\tau^c}{1-\tau^c z - c}. \tag{1}$$

By shifting the wedge, tax reforms can shift the incentive to maintain existing capital, or invest in new capital. I make that statement precise in the following section.[5]

3 A Model of Capital Maintenance

This section develops a model where firms choose both investment and maintenance to manage their capital stock, building on Hall and Jorgenson (1967) and McGrattan and Schmitz Jr. (1999). The starting point is the standard law of motion for capital:

$$K_{t+1} = I_t + (1-\delta)K_t \tag{2}$$

In steady state, investment replaces depreciated capital, so $I = \delta K$. Taking elasticities with respect to a tax reform:

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} \tag{3}$$

Standard models assume depreciation is fixed, so $\varepsilon_{\delta,\tau} = 0$ and $\varepsilon_{I,\tau} = \varepsilon_{K,\tau}$. This assumption justifies interpreting observed investment responses as capital responses. But when firms can maintain capital, depreciation becomes a choice variable, and this equivalence breaks down.

The model delivers two mechanisms, each with a direct empirical consequence. The first is a *capital preservation effect*: because maintenance is already fully expensed, only fraction $(1-s_m)$ of user cost responds to investment incentives, where s_m is the maintenance share. The same logic implies that standard investment regressions, which omit maintenance from user cost, understate the investment elasticity—a measurement consequence I use to correct existing estimates. The second is an *investment-capital gap*: because tax cuts endogenously raise depreciation, observed investment responses overstate capital accumulation. Its dynamic counterpart is slower convergence—matching a given short-run investment response requires larger adjustment costs when maintenance is present, so the economy approaches its (smaller) steady state more gradually. Both mechanisms survive in general equilibrium under plausible conditions. I conclude by

$66M worth of expenditures on maintaining existing aircraft engines in 1993 should be considered maintenance or investment. The question decided ten percent of their tax bill for that year.

5. Appendix Figure A.1 shows how the wedge varies by asset and over time.

deriving an empirical specification for estimating the maintenance demand function.

3.1 A Model with Endogenous Depreciation

Consider a simplified version of the standard neoclassical model in which output is produced with capital alone $Y_t = F(K_t)$, where $F(K_t)$ has standard properties. In that model, capital evolves according to

$$K_{t+1} = I_t + (1 - \delta)K_t.$$

However, in practice, firms can influence how quickly their capital wears out by spending on maintenance, repair, and upkeep. Let M_t denote spending on maintenance and $m_t \equiv M_t/K_t$ denote the maintenance rate—maintenance expenditure per unit of capital. Depreciation depends on maintenance effort through a technology $\delta(m_t)$. In that case, the capital accumulation equation becomes:

$$K_{t+1} = I_t + (1 - \delta(m_t))K_t \tag{4}$$

I make two assumptions on the depreciation technology $\delta(m)$.

Assumption 1 (Depreciation Technology)**.** *The depreciation function* $\delta : [0, \bar{m}) \to (0, 1)$ *satisfies:*

(i) $\delta'(m) < 0$ *for all* $m > 0$*: maintenance reduces depreciation;*

(ii) $\delta''(m) > 0$ *for all* $m > 0$*: diminishing returns to maintenance.*

.

Hence, there are essentially two ways to produce capital: by investing in new capital or by maintaining existing capital. However, their tax treatment differs to the degree defined by the wedge in the previous section. This asymmetry is the key feature of the model. Investment is capitalized and benefits from accelerated depreciation (z) and tax credits (c). Maintenance is simply expensed at the statutory rate. When investment incentives are generous, investment becomes tax-advantaged relative to maintenance. The firm's after-tax dividends are:

$$d_t = (1 - \tau^c) \left[F(K_t) - p_t^M M_t\right] - (1 - \tau^c z - c)p_t^I I_t \tag{5}$$

where p_t^M is the price of maintenance services. The firm chooses sequences $\{I_t, M_t, K_{t+1}\}_{t=0}^{\infty}$ to maximize:

$$V_0 = \max \sum_{t=0}^{\infty} \left(\prod_{s=0}^{t-1} \frac{1}{1 + r_s} \right) d_t \quad \text{s.t.} \quad (4) \text{ and } (5) \tag{6}$$

The firm's first-order conditions are:

$$
\begin{aligned}
(1-\tau_t^c z_t - c_t)p_t^I = \frac{1}{1+r_t}\Big\{&(1-\tau_{t+1}^c)F'(K_{t+1}) \\
&+ (1-\tau_{t+1}^c z_{t+1} - c_{t+1})p_{t+1}^I\big[1-\delta(m_{t+1})+\delta'(m_{t+1})m_{t+1}\big]\Big\}
\end{aligned} \tag{7}
$$

$$
(1-\tau_t^c)p_t^M = (1-\tau_t^c z_t - c_t)p_t^I\big[-\delta'(m_t)\big] \tag{8}
$$

The capital condition (7) incorporates the maintenance FOC (8): the raw derivative of dividends with respect to K_{t+1} includes a term $-(1-\tau_{t+1}^c)p_{t+1}^M m_{t+1}$ from the maintenance cost $M_{t+1} = m_{t+1}K_{t+1}$, which the maintenance FOC transforms into the $\delta'(m_{t+1})m_{t+1}$ term inside the brackets. I evaluate and discuss the maintenance and capital conditions in the following sections, which then deliver the main theoretical results.

3.2 The Capital Preservation Effect

The first mechanism operates through the level of maintenance. In steady state, only a fraction of user cost is sensitive to tax incentives; the rest—the maintenance component—is already fully expensed and receives no additional benefit from policies like bonus depreciation or investment tax credits.

Evaluating the capital FOC (7) in steady state and using the optimal maintenance condition (8) to substitute out the $\delta'(m)m$ term yields:

$$
F'(K) = \frac{p^I(r+\delta(m^*))}{1-\tau} + p^M m^* \equiv \Psi \tag{9}
$$

where $(1-\tau) \equiv (1-\tau^c)/(1-\tau^c z - c)$ collects the investment incentives into a single tax-adjusted price of investment, falling as bonus depreciation z or the credit c rises. The user cost Ψ has two components: an investment component capturing financing and depreciation replacement, and a maintenance component capturing upkeep expenditure.

Definition 1 (User Cost Components). The investment component and maintenance component of user cost are

$$
\Psi^I \equiv \frac{p^I(r+\delta(m^*))}{1-\tau}, \qquad \Psi^M \equiv p^M m^*
$$

so that $\Psi = \Psi^I + \Psi^M$.

Definition 2 (Maintenance Share). The maintenance share of user cost is $s_m \equiv \Psi^M/\Psi$. The investment share is $1 - s_m = \Psi^I/\Psi$.

This share governs the attenuation of user cost responses. At a common steady state where the standard model has depreciation $\bar{\delta} = \delta(m^*)$, the standard Hall-Jorgenson user cost equals only the investment component: $\Psi^I = (1-s_m)\Psi$. The standard model ignores maintenance entirely—it captures only a fraction $(1 - s_m)$ of the true user cost. How does this affect the response to tax policy?

Proposition 1 (Capital Preservation Effect)**.** *At a common steady state where $\bar{\delta} = \delta(m^*)$:*

$$\varepsilon_{\Psi,\tau} = (1 - s_m)\frac{\tau}{1 - \tau} \tag{10}$$

Similarly for elasticities with respect to r and p^I. The elasticity with respect to p^M is $\varepsilon_{\Psi,p^M} = s_m$.
Proof: Appendix B.1.

The proof applies the envelope theorem: at the optimum, the firm has already adjusted maintenance to minimize costs, so a small change in tax policy affects user cost only through the direct channel. The indirect effect through induced changes in m^* vanishes. Since the direct effect operates only on Ψ^I, which is fraction $(1 - s_m)$ of total user cost, the elasticity is attenuated by $(1 - s_m)$.

The intuition is straightforward. Tax incentives operate through the tax wedge τ (via z, c, or τ^c), but only the investment component Ψ^I responds—the maintenance component Ψ^M is already fully expensed and receives no additional benefit. In the standard model, the user cost elasticity is $\frac{\tau}{1-\tau}$. With maintenance, the elasticity is $(1 - s_m)\frac{\tau}{1-\tau}$. If maintenance accounts for half of user cost, the response is halved. This is particularly important in the context of long-lived assets which require heavy maintenance. If maintenance dominates user cost, the user-cost elasticity would collapse to zero, and capital would not respond to taxes at all. Thus long-lived assets that also require heavy upkeep can be less tax-elastic than short-lived ones, reversing the standard "long-lived assets are more price-elastic" result in House (2014) and Koby and Wolf (2020).

Remark (General Equilibrium). The $(1 - s_m)$ attenuation survives general equilibrium. When maintenance prices are more sensitive to capital accumulation than investment prices and interest rates combined—plausible since maintenance is labor-intensive and locally supplied while investment goods are globally traded—general equilibrium amplifies the attenuation. Appendix B.2 provides the formal treatment.

Bias in Investment Regressions. The same maintenance share that attenuates the structural response also distorts its measurement. The canonical investment regression is derived from Hall-Jorgenson user cost theory, which guides both the specification and the interpretation of estimates. When that theory omits maintenance, the regression inherits the omission—and the resulting coefficients are biased.

The canonical regression in the investment literature estimates the short-run response of investment to user cost:

$$f(I_{i,t}, K_{i,t-1}) = \beta \times \log(\Psi_{i,t}) + X_{i,t} + u_{i,t}, \tag{11}$$

where $f(\cdot)$ is either $\log I_{i,t}$ or the investment rate $I_{i,t}/K_{i,t-1}$, and $X_{i,t}$ includes controls and fixed effects.[6] This specification is not atheoretical: researchers regress on $\log \Psi$ because Hall-Jorgenson theory says investment responds to user cost, and they construct user cost as $\frac{p^I(r+\delta)}{1-\tau}$ because theory says that is the relevant price. When theory omits maintenance, the regressor becomes Ψ^I rather than $\Psi = \Psi^I + \Psi^M$—and the measured price change overstates the true price change.

The intuition is straightforward. When policymakers cut taxes on investment, the measured user cost Ψ^I falls by more than the true user cost Ψ because maintenance expenditures are already fully deductible and do not benefit from the tax cut. The econometrician observes a large measured reduction in user cost but only a modest investment response, and concludes that investment demand is relatively inelastic. In reality, investment demand may be quite elastic—the problem is that the measured price change overstates the true price change.

This parallels Goolsbee (1998a), who shows that regressing investment on tax terms alone implicitly assumes perfectly elastic supply of investment goods. When supply is upward-sloping, tax cuts raise investment prices, and the observed quantity response understates demand elasticity. The maintenance bias operates through similar logic but on a different margin: by omitting maintenance from user cost, standard regressions overstate the price reduction from tax cuts, biasing estimated demand elasticities downward.

Proposition 2 (Bias in Investment Regressions)**.** *Suppose the econometrician regresses investment on $\log \Psi^I$ when the true user cost is $\Psi = \Psi^I + \Psi^M$. The estimated short-run coefficient is biased toward zero, with the true short-run coefficient β related to the estimated coefficient $\hat{\beta}$ by*

$$\beta \approx \frac{\hat{\beta}}{1 - s_m} \tag{12}$$

This holds because the maintenance component Ψ^M does not respond to investment tax provisions, so the true user cost change is $(1 - s_m)$ times the measured change. The correction factor is independent of the choice of $f(\cdot)$ on the left-hand side.
Proof: Appendix B.4.

The correction can be substantial. If maintenance accounts for 25% of user cost, the true short-run elasticity is one-third larger than the estimated elasticity.

6. See Cummins, Hassett, and Hubbard (1996), Desai and Goolsbee (2004), Edgerton (2010), and Hartley, Hassett, and Rauh (2025) for examples.

3.3 The Investment-Capital Gap

The second mechanism operates through the elasticity of maintenance demand. In the standard model, depreciation is fixed at $\bar{\delta}$, so steady-state investment is $I = \bar{\delta}K$ and $\varepsilon_{I,\tau} = \varepsilon_{K,\tau}$. With endogenous maintenance, this identity no longer holds. To see why, consider the firm's optimal maintenance choice. Because maintenance is chosen each period without adjustment costs—i.e., there is no "stock of repair" that accumulates over time—the decision is static. The firm equates the marginal benefit of maintenance (reduced depreciation) with its marginal cost:

$$-\delta'(m^*) = \frac{p^M}{p^I}(1-\tau)$$

There are three comparative statics which follow immediately from the first-order condition.

1. $\partial m^*/\partial p^M < 0$: higher maintenance prices reduce maintenance intensity
2. $\partial m^*/\partial p^I > 0$: higher investment prices increase maintenance intensity
3. $\partial m^*/\partial(1-\tau) < 0$: tax cuts reduce maintenance intensity

Part (iii) is central. When investment becomes tax-advantaged—higher z or c, which raises $(1-\tau)$—the firm substitutes away from maintenance toward investment. This raises the depreciation rate $\delta(m^*)$ endogenously.

The consequence for the investment-capital relationship is immediate. In steady state, investment replaces depreciated capital: $I = \delta(m^*)K$. Unlike the standard model, $\delta(m^*)$ now depends on tax policy through m^*. Taking logs and differentiating:

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau}$$

The investment response has two components: the capital accumulation effect ε_K, which reflects changes in the desired capital stock, and the depreciation effect ε_δ, which reflects changes in the rate at which capital must be replaced.

Proposition 3 (Investment-Capital Gap)**.** *Tax cuts raise depreciation ($\varepsilon_{\delta,\tau} > 0$) and raise capital ($\varepsilon_{K,\tau} > 0$), where elasticities are with respect to the net-of-tax rate $(1-\tau)$. Therefore:*

$$\varepsilon_{I,\tau} > \varepsilon_{K,\tau}$$

Proof: Appendix B.1.

In the standard model, $\varepsilon_{\delta,\tau} = 0$ and the proposition collapses to $\varepsilon_{I,\tau} = \varepsilon_{K,\tau}$. With maintenance, the depreciation effect drives a wedge. When policymakers introduce tax cuts, firms respond on two margins. First, the lower user cost raises desired capital, generating new investment. Second, the tax advantage of investment over maintenance induces substitution: firms reduce upkeep, depreciation rises, and more investment is required simply to maintain the higher capital stock. Observed investment responses capture both effects, but only the first represents net capital accumulation.

The gap implies that investment subsidies promote capital churn, not necessarily capital growth. In steady state, $I/K = \delta(m)$. A tax cut that makes investment cheaper also makes maintenance relatively expensive, raising $\delta(m)$ as firms let capital wear out faster. The investment rate increases permanently, but this reflects faster replacement of a younger capital stock, not a larger capital stock.[7] Standard models with fixed depreciation miss this distinction entirely.

The magnitude of the gap depends on the elasticity of maintenance demand. If maintenance demand is inelastic, the depreciation effect is small and investment responses are good proxies for capital responses. If maintenance demand is elastic, a substantial fraction of observed investment replaces endogenously higher depreciation rather than building new capital. Like the preservation effect, the gap is reinforced rather than undone in general equilibrium: the same wage sensitivity developed in Appendix B.2—labor-intensive maintenance growing more expensive as capital accumulates—further discourages upkeep and raises depreciation.

Transition Dynamics. The gap shapes not only the magnitude of the long-run response but the speed of reaching it. In practice, researchers often estimate short-run investment responses to tax changes and use these to infer long-run effects on capital. This requires a model of transition dynamics. With adjustment costs, capital converges gradually to its new steady state. Suppose the transition path for investment following a permanent tax change takes the form:

$$\varepsilon_{I,\tau}(t) = \varepsilon_{I,\tau}^{LR}\left[1 - (1-\theta)e^{-\kappa t}\right] \tag{13}$$

Definition 3 (Transition Parameters). The *impact effect* $\theta \in (0,1)$ is the fraction of the long-run response achieved on impact: $\theta = \varepsilon_{I,\tau}(0)/\varepsilon_{I,\tau}^{LR}$. The *convergence rate* $\kappa > 0$ governs the speed of subsequent adjustment.

Different adjustment cost specifications generate different feasible (θ, κ) pairs, but standard specifications share a common property: higher θ is associated with higher κ.

7. Firms also have a scrappage margin, which Goolsbee (1998b) shows is empirically important for tax reform. Endogenous depreciation captures this channel in reduced form: tax cuts induce firms to shed old capital more quickly. In this sense, the maintenance demand elasticity can be interpreted as a reduced-form tax elasticity for the age distribution of capital.

Proposition 4 (Slower Convergence)**.** *Suppose the set of feasible* (θ, κ) *pairs is co-monotonic. If both the standard model and the maintenance model are calibrated to match the same short-run investment elasticity at horizon* T*, then the maintenance model has a lower convergence rate.*
Proof: Appendix B.5.

The intuition follows from the investment-capital gap. In steady state, investment replaces depreciated capital: $I = \delta(m)K$. The long-run investment elasticity with maintenance therefore exceeds that without:

$$\varepsilon_{I,\tau}^{LR} = \varepsilon_{K,\tau} + \varepsilon_{\delta,\tau} > \varepsilon_{K,\tau}$$

Both models are calibrated to match the same ε_I^{SR} at horizon T. From (13), this requires $(1 - \theta)e^{-\kappa T} = 1 - \varepsilon_I^{SR}/\varepsilon_I^{LR}$. Since the maintenance model has a higher long-run target, the right-hand side is larger, requiring a larger $(1 - \theta)e^{-\kappa T}$. Co-monotonicity then implies lower κ. Appendix B.5 illustrates the magnitude under the Summers (1981) specification, which remains commonly used (Koby and Wolf 2020; Chodorow-Reich et al. 2025).

The combination—slower convergence to a smaller steady state—has direct implications for dynamic scoring of tax reforms. Budget scores are typically computed over ten-year windows, well before the economy reaches its new steady state. The maintenance channel reduces both where the economy is heading and how quickly it gets there, compressing growth effects within policy-relevant horizons.

3.4 Taking the Model to Data

Omitting maintenance from user cost theory distorts empirical work in three places—the regressor, the interpretation, and the calibration—all traceable to the two mechanisms above and governed by the same small set of primitives. The capital preservation effect attenuates user cost elasticities by $(1 - s_m)$ (Proposition 1) and, in regressions that omit maintenance, biases the estimated investment elasticity by the same factor (Proposition 2). The investment-capital gap drives a wedge $\varepsilon_{I,\tau} - \varepsilon_{K,\tau} = \varepsilon_{\delta,\tau} > 0$ between investment and capital responses (Proposition 3) and, along the transition, slows convergence to the new steady state (Proposition 4).

Two parameters govern the quantitative importance of these mechanisms:

1. **The maintenance share** s_m: This determines the capital preservation effect and the OVB correction directly. Higher s_m means a larger fraction of user cost is tax-insensitive, so the attenuation $(1 - s_m)$ is more severe and the bias in standard regressions is larger.

2. **The elasticity of maintenance demand** ω: This determines how much firms substitute between maintenance and investment when relative prices change. Appendix B.3 shows that the

depreciation elasticity is

$$\varepsilon_{\delta,\tau} = \frac{\gamma\omega}{\delta_0 + \frac{\gamma\omega}{1-\omega}} \tag{14}$$

which governs both the investment-capital gap ($\varepsilon_{I,\tau} - \varepsilon_{K,\tau} = \varepsilon_{\delta,\tau}$) and slower convergence.

To estimate these parameters, I adopt a constant-elasticity specification for the depreciation technology:

$$\delta(m) = \delta_0 - \frac{\gamma^{1/\omega}}{1 - 1/\omega} m^{1-1/\omega} \tag{15}$$

where $\delta_0 > 0$ is baseline depreciation without maintenance, $\gamma > 0$ shifts the level of maintenance, and $\omega > 1$ governs the curvature. This specification satisfies Assumption 1: it is decreasing and convex in m.

Proposition 5 (Maintenance Demand)**.** *Under the constant-elasticity depreciation technology, optimal maintenance is:*

$$m^* = \gamma \left(\frac{p^M}{p^I} (1 - \tau) \right)^{-\omega} \tag{16}$$

The parameter ω is the price elasticity of maintenance demand.

The proof follows directly from the maintenance first-order condition. The level parameter γ determines the maintenance share: at prevailing prices, optimal maintenance is $m^* = \gamma(\bar{p}^M/\bar{p}^I(1-\bar{\tau}))^{-\omega}$, and the maintenance share is $s_m = p^M m^*/\Psi$. Given observed (m^*, s_m) and prices, one can recover γ by inverting the demand function. Thus γ and s_m are interchangeable for quantifying the capital preservation effect and the OVB correction.

Taking logs of the demand function yields the estimating equation:

$$\log m_{it} = \alpha_i + \lambda_t - \omega \log \left(\frac{p^M_{it}}{p^I_{it}} (1 - \tau_{it}) \right) + u_{it} \tag{17}$$

where α_i and λ_t are firm and time fixed effects. The coefficient ω is identified from within-firm variation in the after-tax relative price of maintenance. The fixed effects absorb $\log \gamma$ along with other time-invariant and common factors.

The empirical task is therefore twofold: measure s_m from the level of maintenance expenditures, and estimate ω from variation in relative prices.

4 Data

This section introduces two novel sources of maintenance data for the United States. First, I digitize regulatory filings from Class I freight railroads spanning 1999-2023, providing firm-asset level detail on maintenance expenditures, capital stocks, and input costs. To my knowledge, this is the first use of R-1 data in modern economics since Bitros (1976). Second, I construct industry-level measures from IRS Statistics of Income (SOI) corporate tax returns covering 1999-2021. While the SOI has been used to study investment (e.g., Zwick and Mahon (2017)), this is the first application to maintenance. Given the theory, each dataset must provide two objects: the maintenance share of user cost s_m, which requires data on maintenance expenditures, capital stocks, depreciation rates, and interest rates; and variation in relative prices sufficient to estimate the elasticity of maintenance demand ω. These datasets provide complementary evidence: the railroad data offer precision and detailed variation in firm behavior, while the SOI provides economy-wide coverage.

4.1 R-1 Data from the Surface Transportation Board

Class I freight railroads—defined as having revenue exceeding roughly $1 billion—account for about 40% of U.S. freight transportation. Following industry consolidation in the 1980s and early 1990s, the sector has consisted of approximately seven large firms in stable competitive equilibrium since the late 1990s.[8] By regulation, these firms must file annual R-1 reports with the Surface Transportation Board (STB). These reports detail the size and composition of firms' capital in value and quantities, trackage by state, taxes paid, capital expenditures, and maintenance expenditures broken down by capital type and input source (materials, labor, or purchased services). Reports are independently audited and provide an unparalleled window into firm capital structure and maintenance behavior. The data span 1999-2023.

For this paper, I maintain a narrow focus on freight cars and locomotives. These assets offer two advantages for studying maintenance demand. First, maintenance activities and associated prices can be straightforwardly identified in the data, whereas this is not true for other capital types. Second, track maintenance is strictly regulated by the Federal Railroad Administration, while locomotive and freight car maintenance faces considerably less regulatory constraint. This provides meaningful variation in firm maintenance choices, which is essential for identification.

The main measure of the maintenance rate is the ratio of maintenance expenditures to lagged book capital. I also construct two alternative metrics: a physical maintenance rate (maintenance expenditures per horsepower for locomotives or per freight ton capacity for freight cars), and separate internal and external maintenance rates. The distribution of maintenance rates for both asset types appears in Figure 1.

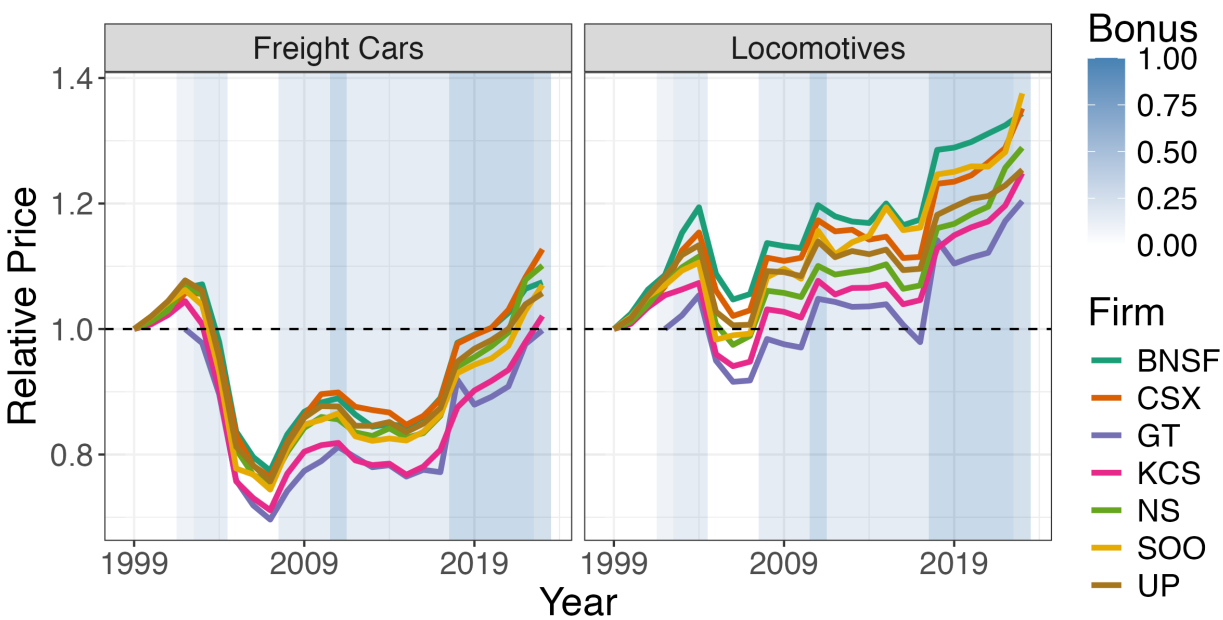
8. The seven firms are CSX, Burlington Northern & Santa Fe, Union Pacific, Norfolk Southern, Kansas City Southern, Soo Line, and Grand Trunk (operated by Canadian National Railway). I end the sample in 2023 due to subsequent industry consolidation.

Next, I construct firm- and asset-specific relative prices as

$$P_{i,j,t} = \frac{p^M_{i,j,t}}{p^I_{j,t}} \frac{1-\tau_{i,t}}{1-\tau_{i,t} z_t}, \tag{18}$$

where $p^M_{i,j,t}$ is the pre-tax maintenance price of capital good j for firm i at time t, $p^I_{j,t}$ is the investment price of asset j, and $\tau_{i,t}$ is the firm-specific tax rate. Due to data availability, only the pre-tax price of maintenance varies by firm and capital type, whereas tax rates vary by firm and investment prices by capital type. Tax rates do not vary between assets because the IRS places locomotives and freight cars in the same depreciation category. Further details on data construction and summary statistics are in Appendix C.1.

Figure 2: After-tax relative price of maintenance to investment



Notes: I construct the after-tax relative price of maintenance to investment as described in the main text and in Appendix C.1. Background shading is for bonus depreciation.

Figure 2 plots the after-tax relative price of maintenance to investment for all Class I railroads since 1999. The price of maintaining locomotives has persistently increased since 2000, while the pattern is more U-shaped for freight cars. Background shading indicates periods of bonus depreciation, which reduced the relative price of investment and thus increased the maintenance-investment wedge. However, since all equipment here is in the same tax category, there is no variation between assets or across firms. The variation I later exploit for the railroad data is therefore not the tax wedge but the cost of maintenance itself: the pre-tax price p^M, which moves with maintenance wages and so differs across firms with their geographic footprint. Section 6.2

draws on this cost-side variation to estimate the elasticity.

To construct the maintenance share of user cost, I also require depreciation rates and a cost of capital. I use economic depreciation rates for railroad equipment (4% for locomotives, 3% for freight cars) and the industry-average discount rate for NAICS 48 from Gormsen and Huber (2022). Appendix C.1 provides details.

4.2 Industry Data from the Statistics of Income

The IRS Statistics of Income (SOI) aggregates corporate tax returns into industry-level samples at approximately the three-digit NAICS level. This is the only economy-wide collection of maintenance data at annual frequency in the United States. Corporations report maintenance expenditures and book capital as line items on their tax forms, which the SOI aggregates across firms within industries. I use SOI data from 1999-2021, aggregating to 49 industries to correspond with Bureau of Economic Analysis (BEA) classifications.[9] For some analyses, I separate the sample into taxable firms (those with positive net income) and non-taxable firms.

The SOI maintenance rate—the ratio of maintenance expenditures to lagged book capital—is noisier than the R-1 measure for two reasons. First, because labor expenditures appear as a separate line item on tax forms, reported maintenance largely reflects spending on materials and external services, understating total maintenance costs.[10] Second, the capital stock denominator uses tax depreciation schedules to construct book capital, which may not accurately reflect economic capital stocks. Despite these measurement issues, the resulting maintenance rates are similar to those observed in aggregate Canadian data.[11] Figure 1 plots the distribution of SOI maintenance rates across industries and years. Because there is no easily identifiable measure of relative prices at the industry level, my identification strategy for the SOI data relies on the policy wedge defined in equation 1. I construct asset-specific wedges for every BEA asset using the mapping from assets to the tax code in House and Shapiro (2008). I then aggregate the asset-specific wedges into a capital-weighted industry-specific wedge, where the weights reflect each industry's capital composition. Industries with more equipment-intensive capital structures face larger policy-driven changes in the maintenance-investment wedge when tax incentives vary.[12]

Figure 3 plots the distribution of industry-specific wedges over time. Tax reforms throughout the 2000s—particularly expansions of bonus depreciation—tended to reduce the wedge proportionally more for equipment-intensive industries, while structures-intensive industries saw little

9. I exclude filings made with Forms 1120S, 1120-REIT, and 1120-RIC, and remove the financial sector. See Appendix C.2 for details on the SOI-BEA industry mapping and why I use the BEA definition.

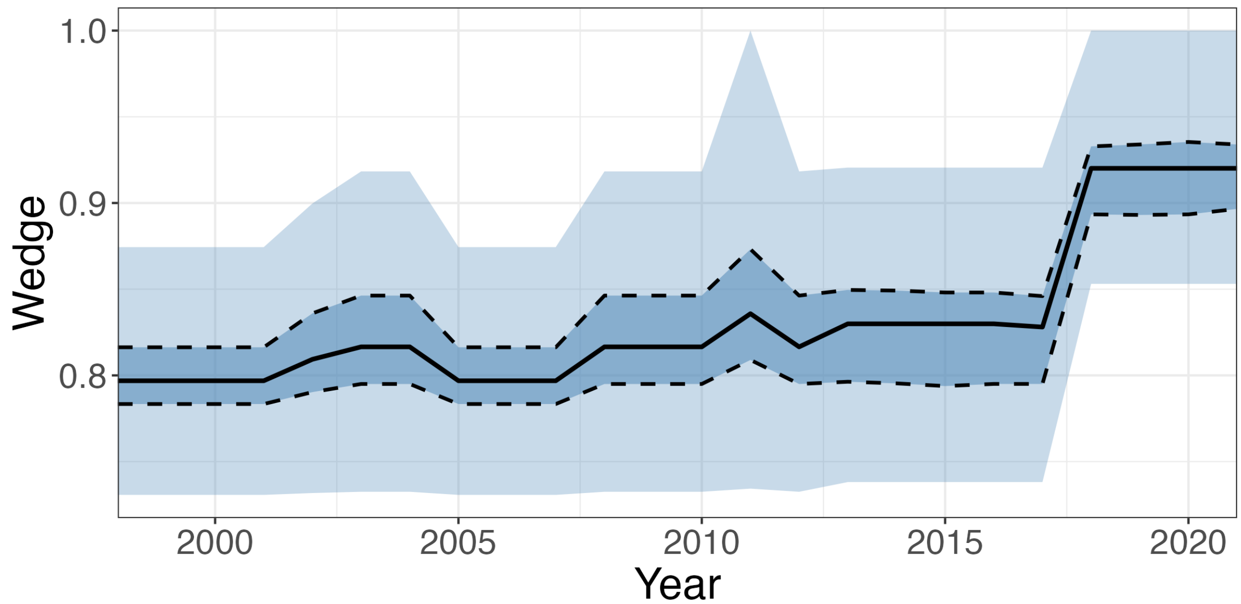
10. The labor cost share in maintenance is typically 30-45%. Appendix C.2 discusses a potential correction and shows that the main results of this paper are conservative as a result.

11. See Appendix Figure C.5 for a comparison.

12. Appendix C.2 provides details on asset-level wedge construction and capital weighting.

change until the TCJA in 2017. There is significant variation in the incentive to substitute between maintenance and investment over time, which is exactly what I require to estimate the elasticity of maintenance demand.

Figure 3: Distribution of industry-specific wedges over time



Notes: Every line is a quartile of the industry-specific wedge defined in the main text.

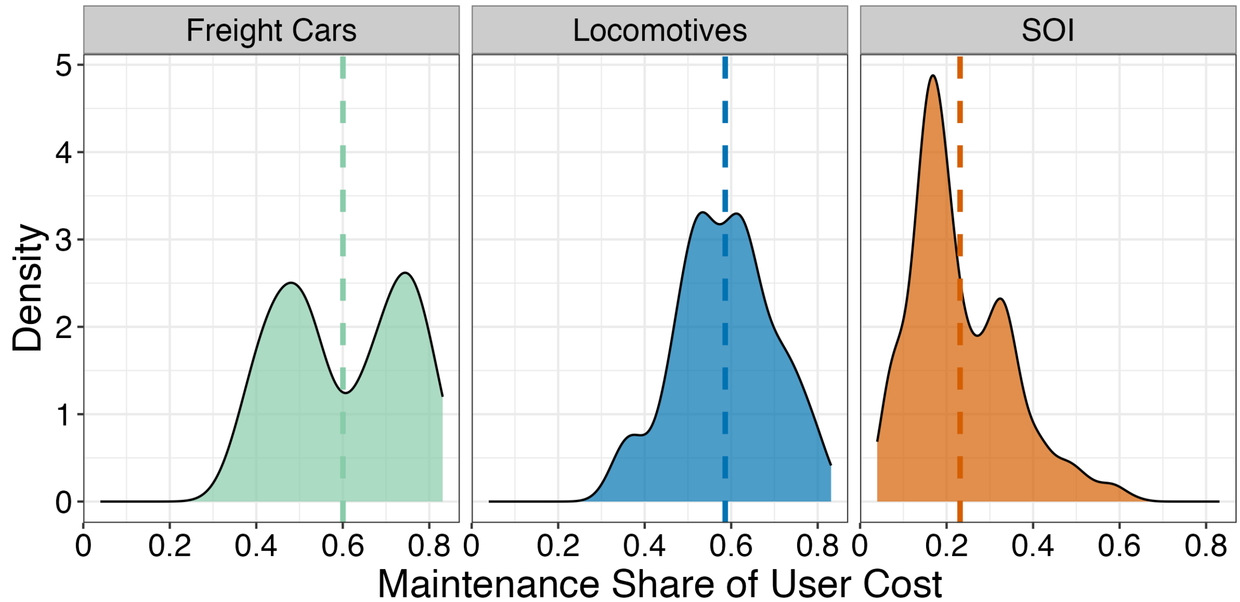
To construct the maintenance share, I use industry-specific depreciation rates from the BEA Fixed Asset Tables and industry-specific discount rates from Gormsen and Huber (2022). Appendix C.2 provides details.

With these two complementary data sources in hand—detailed firm-level variation from railroads and broad industry coverage from the SOI—I now turn to measuring the key parameters from the theory.

5 The Maintenance Share of User Cost

Standard models of capital taxation assume the entire user cost responds to tax incentives. In practice, only the investment component responds, while the maintenance component is already fully expensed. The maintenance share of user cost $s_m \equiv p^M m^* / \Psi$ therefore governs how much investment tax incentives actually reduce the cost of capital services. If maintenance accounts for half of user cost, tax incentives are half as effective as standard models predict. Figure 4 presents estimates from both the freight rail and SOI datasets.

Figure 4: The Maintenance Share of User Cost



Note: Density plots for the maintenance share of user cost. Dashed lines are means: 23% (SOI), 60% (Freight Cars), 58.6% (Locomotives). See Appendix C for construction details.

The railroad data show maintenance shares averaging nearly 60% for both locomotives and freight cars. Keeping rolling stock in service requires continuous upkeep, and these costs rival the financing and depreciation costs of the equipment itself. At $s_m = 0.6$, standard models overstate the responsiveness of user cost to tax policy by a factor of 2.5.[13]

The SOI data show a maintenance share of about 25%, though this is likely a lower bound. SOI maintenance expenditures exclude internal labor costs, which appear as wages on tax forms rather than repairs. Since labor accounts for 30–45% of total maintenance costs in the railroad data, the true economy-wide maintenance share may be closer to 35–40%. Even at the conservative SOI estimate, tax incentives reduce the cost of capital by only three-quarters as much as standard models imply.

The difference between the datasets reflects both measurement and economics. Railroad rolling stock is maintenance-intensive: locomotives and freight cars operate continuously, face harsh conditions, and require regular inspection and repair. The broader economy includes structures and other long-lived assets with lower maintenance requirements. For the quantitative analysis that follows, I use the SOI estimate as a conservative baseline, recognizing that the true attenuation is likely larger for equipment-intensive industries.

13. The user cost formula requires a depreciation rate, which I take from BEA estimates. These estimates assume constant depreciation, which is precisely what this paper relaxes. However, the BEA rates reflect average depreciation given average maintenance behavior over the sample period, making them appropriate for roughly measuring the level of s_m at current policy.

6 Estimating The Maintenance Demand Elasticity

The elasticity of maintenance demand governs the investment-capital gap. I estimate it from two complementary sources of variation. My primary estimate is economy-wide: in industry-level tax data, differential exposure to changes in investment subsidies and the corporate tax rate generates policy-driven variation in the relative price of maintenance across the corporate sector, which is then sufficient to recover the maintenance elasticity of demand. I then corroborate the economy-wide estimate with more granular data by evaluating how exogenous shocks to the relative price of maintenance affect the maintenance intensity of locomotives and freight cars. The two designs deliver large, positive elasticities of comparable magnitude. A closing subsection combines them and frames how the firm-level and economy-wide estimates relate.

6.1 Evidence from the Statistics of Income

My primary estimate of the maintenance elasticity comes from the Statistics of Income (SOI), the only economy-wide record of maintenance expenditures in the United States, reported annually at the industry level. Two features make it the right place to recover the parameter the model needs. First, the data span the whole corporate sector. Second, aggregation to the industry level nets out the reallocation of capital within industries. For both reasons the SOI elasticity is closer to a general-equilibrium object than to a firm- or asset-level one, and I read it as the economy-wide parameter. Appendix F.1 treats the netting-out formally.

Identification Strategy. The SOI carries no direct measure of the relative price of maintenance, so I identify the elasticity from policy-driven variation in the tax wedge of equation 1. Industries differ in capital composition: some rely heavily on equipment eligible for accelerated depreciation, others on structures. This creates differential exposure to two major types of reforms. First, bonus depreciation phased in and out repeatedly, with larger effects on equipment-intensive industries: it was enacted at 30% in 2002, allowed to lapse over 2005–2007, and reinstated in 2008. Second, the 2017 Tax Cuts and Jobs Act raised bonus to 100% and cut the corporate rate from 35% to 21%, creating further variation across industries. There is substantial evidence that both kinds of reforms boosted investment (House and Shapiro 2008; Zwick and Mahon 2017; Garrett, Ohrn, and Suárez Serrato 2020; Kennedy et al. 2023; Chodorow-Reich et al. 2025). Following Zwick and Mahon (2017), I estimate the maintenance demand elasticity using a continuous-intensity difference-in-differences design. The key idea is that equipment-intensity supplies the cross-sectional dose and the bonus schedule the time variation. Because the schedule is national, industries are treated simultaneously, so all variation comes in intensity of exposure rather than timing.

In particular, I estimate:

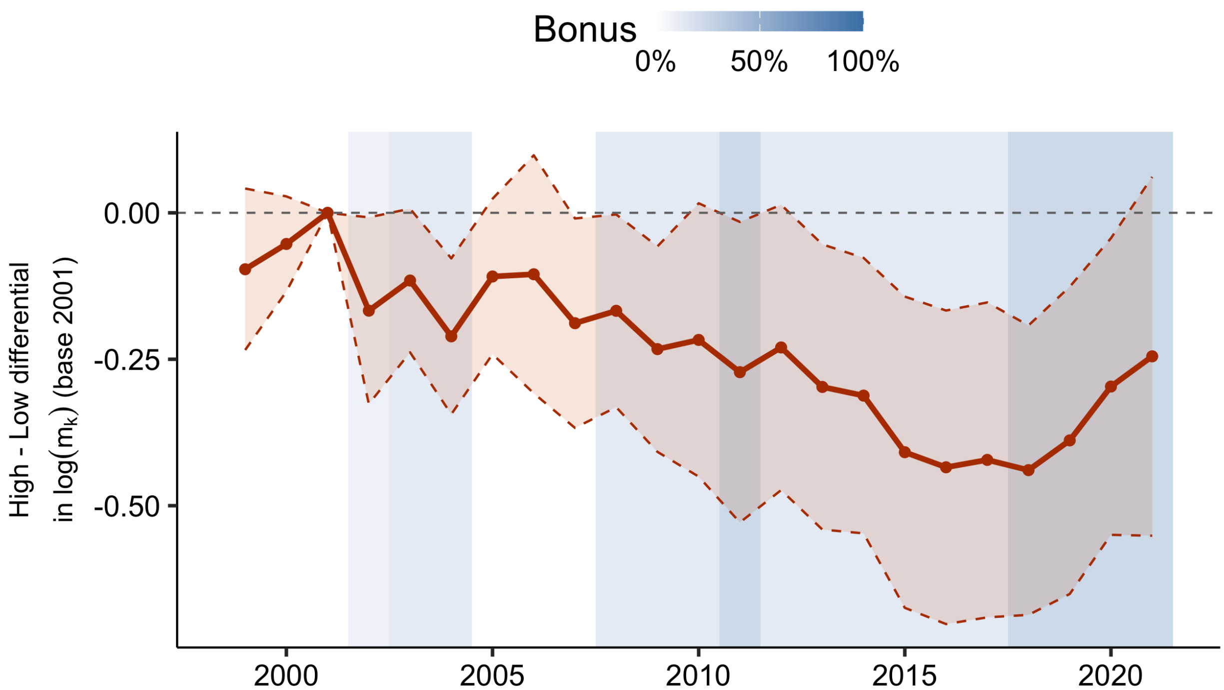
$$\log m_{i,t} = \alpha_i + \lambda_t - \omega \log \left(\frac{1 - \tau_t^c}{1 - \tau_t^c z_{i,t}} \right) + \text{Controls} + \eta_{i,t}, \tag{19}$$

where α_i is an industry fixed effect and λ_t is a time fixed effect, and the tax term is the wedge from equation 1. The present value of depreciation deductions $z_{i,t}$ determines how exposed each industry is to the reforms: bonus depreciation raises $z_{i,t}$, and by more for equipment-intensive industries whose capital is eligible. I construct $z_{i,t}$ using pre-reform capital weights, so its variation over time reflects the bonus schedule rather than contemporaneous changes in capital composition. This captures the economically relevant margin: firms are differentially exposed based on their existing stock of capital, which is what they maintain.[14]

Identification requires that this policy-driven variation is uncorrelated with other industry-year shocks to maintenance. Equivalently, the most- and least-exposed industries would have followed parallel paths absent the reforms. Figure 5 provides direct support. It plots the difference in log maintenance between the top and bottom terciles of exposure, year by year and relative to 2001, with cluster-robust confidence bands. Before bonus the difference is statistically indistinguishable from zero, with the two groups moving in parallel, and turns negative and significant only after 2002, as the most-exposed industries cut maintenance. The decisive feature is the lapse: when bonus expires over 2005–2007 the difference retraces toward zero, then reopens when bonus returns in 2008. A secular trend in equipment-intensive industries cannot switch off and on with the statute, so this on-again-off-again response is difficult to attribute to anything but the policy. To absorb any residual time-varying confounders, such as Covid-era disruptions, I include linear and quadratic trends at the two-digit NAICS level when estimating the main regression.

14. One could also consider an investment-weighted scheme, as is standard in investment elasticity studies (e.g., (Zwick and Mahon 2017)). In those studies, z is constructed by taking a weighted average of recent investment flows. This captures the elasticity for recently acquired assets. However, maintenance applies to the entire installed capital base, not just recent purchases, making investment weights conceptually misaligned. Appendix Table D.8 bears this out. Under investment weights the positive maintenance elasticity disappears, insignificant for taxable firms and wrong-signed in the pooled sample, confirming that maintenance responds to the installed capital base rather than recent purchases.

Figure 5: Calendar Difference-in-Differences



Note: Each point is the year-t gap in log maintenance between the top and bottom terciles of baseline (2001) exposure z_i (high-tercile-by-year coefficients, bottom tercile omitted), from a regression with a capital-age control and industry and year fixed effects. Bands are 95% cluster-robust (by industry) confidence intervals, measured relative to 2001. Shaded regions mark years bonus depreciation was in effect. The unshaded 2005–2007 window is the lapse.

Results. Equation (19), estimated on the continuous wedge, quantifies the pattern in Figure 5. Table 1 reports estimates for three samples (all firms, tax-paying firms, and untaxed firms), each with and without a control for capital age (the ratio of gross to net book capital, since older capital may require more maintenance). All columns include linear and quadratic two-digit NAICS trends. In the full sample the elasticity is $\omega = 2.61$ (column 1) and is little changed by the addition of a control for age.

I provide further validation of the mechanism by splitting the sample into firms with positive net income ("taxable") and firms without positive income ("untaxable").[15] In principle, maintenance should not move for untaxable firms and decline for taxable firms following a tax cut if the model is correct. Columns 3–6 of Table 1 show the estimates for each subsample. The taxable firm elasticity rises to 3.07 and becomes more statistically significant compared to the full sample. On the other hand, the untaxable elasticities are centered at zero. Thus, as a placebo test for the

15. The mapping is rough because some firms in the taxable sample carry net operating losses forward, so they are not actually taxable.

theory, the sample split provides sound evidence in favor of the hypotheses advanced in Section 3.

Table 1: The SOI Maintenance Elasticity

	All		Tax-paying		Untaxed	
	(1)	(2)	(3)	(4)	(5)	(6)
$\hat{\omega}$	2.61^{**}	2.71^{**}	3.07^{**}	3.16^{***}	-0.03	-0.09
	(1.01)	(1.04)	(1.15)	(1.13)	(1.69)	(1.70)
N (industry-years)	1,116	1,113	1,051	1,051	1,049	1,049
Industry and year FE	✓	✓	✓	✓	✓	✓
Capital age control		✓		✓		✓
Two-digit NAICS trends	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Estimates of (19). $\hat{\omega}$ is the price elasticity of maintenance demand. All columns include two-way (industry and year) fixed effects and linear and quadratic trends in two-digit NAICS codes; standard errors, in parentheses, are clustered by industry. Columns (2), (4), and (6) add a control for capital age. Columns (1)–(2) use the full sample; (3)–(6) split it into tax-paying and untaxed firms and re-estimate (19) on each. The all-firm columns scale maintenance by lagged capital; the tax-paying and untaxed columns use contemporaneous capital, since a firm's current tax status need not match its prior-period stock. Untaxed firms are not profitable enough to owe tax, so the depreciation incentive is worthless to them—a placebo.

Robustness and Validation. A continuous-intensity design invites three concerns about the estimate, which I take up in turn. The first is that the result is an artifact of how the regression is specified. Across alternative trend specifications, sample windows, and capital measures, the elasticity stays between roughly 1.3 and 3.2, sizable in every case, a range that also spans the leave-one-industry-out interval (Appendix Table D.6). The second is that the difference-in-differences is invalid, either because exposed and unexposed industries were already diverging or because the two-way fixed-effects estimator is biased by treatment-effect heterogeneity. The design diagnostics in Appendix Table D.7 answer it: pre-trends are flat, a heterogeneity-robust estimator returns the same elasticity, and the response scales with the size of the dose.

The third concern is the most specific. Koby and Wolf (2020) point out that low-depreciation equipment is both the most exposed to bonus and the most price-sensitive, so cross-sectional exposure could pick up a pre-existing elasticity gradient rather than the average response. Appendix Figure D.4 rules this out. The interaction between equipment intensity and the wedge is small and statistically insignificant, so the elasticity is homogeneous across industries and does not ride on differential exposure.

One feature of the response matters for how we later approach the quantitative model: its timing. Local projections in Appendix Figure D.5 show that maintenance responds essentially contemporaneously, with industries cutting maintenance in the same year the wedge falls and little persistence afterward. Discrete policy reforms can be acted on at once, and the model accordingly uses the immediate response these estimates imply.

6.2 Evidence from Freight Railroads

The SOI estimate is economy-wide, but it aggregates across industries. To see whether the same elasticity holds at the level of individual assets, I turn to a hand-collected panel of the locomotives and freight cars owned by seven Class I railroads from 1999 to 2023. The source of identifying variation changes with the setting: where the SOI relied on tax policy, the railroad design relies on the cost of maintenance labor, a shock to the relative price of upkeep that is exogenous to any individual firm. Let $P_{i,j,t}$ denote the firm i by asset j by year t after-tax relative price of maintenance to investment defined in Section 4. I estimate the maintenance elasticity of demand ω with

$$\log m_{i,j,t} = \alpha_{ij} + \lambda_t - \omega \log P_{i,j,t} + \text{Controls} + u_{i,j,t}, \tag{20}$$

where α_{ij} is a firm-by-capital type fixed effect and λ_t is a time fixed effect. The coefficient ω is identified by leveraging variation in relative prices within each firm-capital type over time, with fixed effects controlling for all time-invariant characteristics and common temporal shocks. I cluster standard errors by firm to account for correlation in maintenance decisions within firms.

Identification Strategy. The relative price $P_{i,j,t}$ is likely endogenous. If firms experiencing higher maintenance demand also have systematically higher input costs or choose different factor mixes, then OLS estimates will be biased. Unobserved firm-level supply shocks could simultaneously affect maintenance intensity and input prices. Similarly, regional economic expansions might raise both wages and freight demand, leading firms to defer maintenance to maximize utilization.

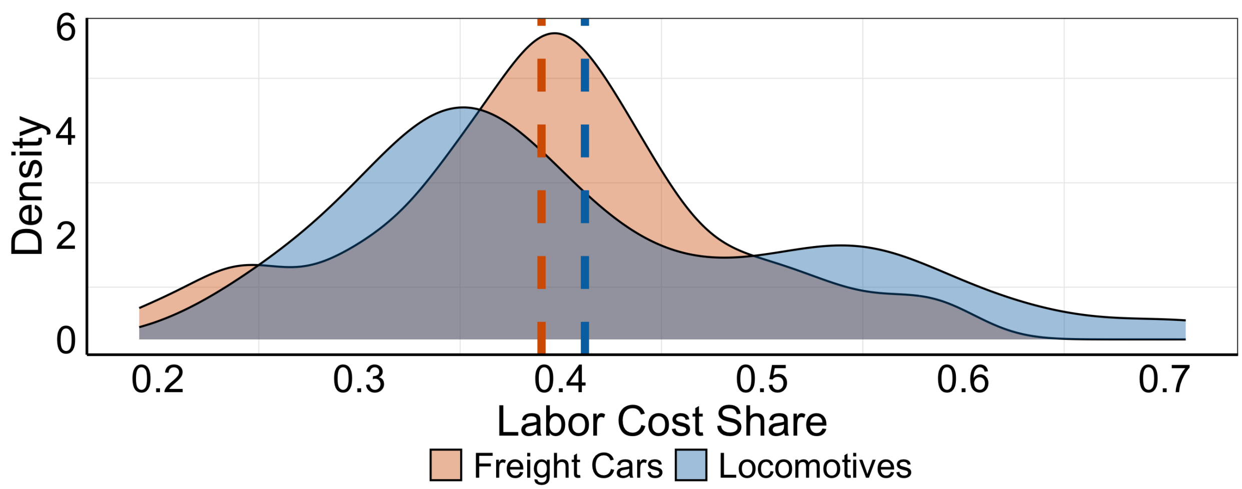
To address this endogeneity, I construct an instrument that isolates exogenous variation in maintenance costs:

$$Z_{i,j,t} = \frac{\text{Labor}_{i,j,t-2}}{\text{Internal Maintenance}_{i,j,t-2}} \sum_{s=1}^{S} \frac{\text{Rail Miles}_{i,t,s}}{\text{Rail Miles}_{i,t}} W_{s,t}. \tag{21}$$

The instrument has two components. The *shares* capture each firm's cost structure: the labor cost share of internal maintenance, lagged two years to ensure pre-determination. Firms that rely more heavily on internal labor are more exposed to wage movements. The *shifts* are state-level wage indices for maintenance occupations (BLS SOC code 49-0000), weighted by each firm's geographic footprint. These state wages reflect broad labor market conditions such as tightness, cost of living, and unionization, all plausibly orthogonal to individual railroads' maintenance needs.[16]

Identification rests on the exclusion restriction: the wage shock alters maintenance only through the relative price, and not through any channel that independently moves maintenance demand. The shares make this credible, since geographic footprints and cost structures are persistent and pre-determined, so the exposure-weighted wage tracks labor-market conditions rather than a firm's current upkeep decisions. The internal-versus-external split, which I turn to below, tests its leading violation directly.

Figure 6: Internal Labor Cost Share for Rolling Stock



Note: The figure plots the winsorized density of labor cost shares for locomotives and freight cars from R-1 reports.

Figure 6 plots the distribution of labor cost shares by asset type, showing substantial variation both within and between firms over time. The first stage has the correct sign and the effective F of Montiel-Olea and Pflueger (2013) is 12.6 in the headline specification, above 10 across every instrument (Appendix Table D.1), but the inference that follows is robust to weak instruments. Additionally, the instrument satisfies standard balance tests. Appendix Figure D.1 shows that lagged maintenance rates are not predicted by $Z_{i,j,t}$, indicating no anticipation effects. Appendix Figure D.2 shows that labor cost shares are orthogonal to lagged maintenance

16. Why focus on internal rather than total maintenance? External maintenance is typically contracted in advance, often negotiated at equipment purchase, which makes it inflexible in the short run. Any price-driven adjustment therefore occurs primarily through the internal margin, where firms retain discretion.

rates, relative prices, and asset age, confirming that the shares are not correlated with pre-period predictors of maintenance.

Table 2: The Railroad Maintenance Elasticity

	All		Internal		External	
	OLS	IV	OLS	IV	OLS	IV
$\hat{\omega}$	1.74^{***}	4.76^{***}	2.81^{***}	8.43^{***}	-0.52	0.12
	(0.26)	(1.07)	(0.74)	(1.76)	(0.59)	(3.74)
F-statistic	12.63		12.63		12.63	
t_F 95% CI	[1.48, 8.04]		[3.02, 13.84]		[-11.35, 11.58]	
AR 95% CI	[3.03, 8.37]		[3.84, 12.13]		[-6.99, 10.96]	
N	340		340		340	
Twoway FE	✓		✓		✓	
Controls	✓		✓		✓	

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Estimates of (20) for all maintenance and for its in-house (Internal) and outsourced (External) components, each by OLS and 2SLS. The 2SLS (IV) columns instrument the log relative maintenance price with the firm's exposure-weighted state maintenance wage ((21)). All columns add capital-age and GDP-exposure controls, include firm-asset and year (twoway) fixed effects, and cluster standard errors on the seven firms. $\hat{\omega}$ is $-1\times$ the log-relative-price coefficient. The F-statistic is the effective first-stage F of Montiel-Olea and Pflueger (2013); the t_F and AR rows are weak-IV-robust 95% intervals (Lee et al. 2022; Anderson and Rubin 1949).

To address remaining confounders I add two controls. First, firms may be differentially exposed to local demand conditions, which could alter the demand for maintenance through wages. I account for that by using a track-mileage-weighted average of state GDP growth to absorb any differential exposure to local business cycles that could otherwise move wages and maintenance together. Second, older equipment may require more maintenance. Hence, like in the SOI, I proxy for capital age by taking the ratio of gross to net book capital. With the firm-asset and year fixed effects, identification then requires only that the instrument provides variation in maintenance costs orthogonal to unobserved maintenance needs, conditional on these.

Results and Robustness. Table 2 presents estimates of equation (20) for all maintenance and, separately, for its internal and external components, which I examine below. For all maintenance, the baseline OLS estimate yields a point estimate near 2. Instrumenting for relative prices produces a substantially larger elasticity: the TSLS point estimate is approximately 4.8, above the economy-wide elasticity from the SOI. I return to that gap when I reconcile the two designs. That instrumenting raises the elasticity rather than shrinking it points to attenuation: measurement error in the relative price biases the uninstrumented estimate toward zero. Because the instrument is only moderately strong, I follow Lee et al. (2022) and Lal et al. (2024) and compute 95% confidence intervals by inverting the Anderson and Rubin (1949) and tF statistics. While wider than analytic intervals, both reject $\omega = 0$ and $\omega = 1$. It is important to reject $\omega = 1$ so that returns to scale are decreasing in maintenance. For comparison, the tax elasticity of the investment rate is generally between 0.5 and 1 (Hassett and Hubbard 2002), while other studies have found values about twice as large (Zwick and Mahon 2017; Hartley, Hassett, and Rauh 2025).

I now present two types of evidence supporting the baseline results: first, a test of the exclusion restriction using the internal-external maintenance split; second, a summary of additional robustness checks detailed in Appendix D.1.

Internal vs. External Maintenance: Testing the Exclusion Restriction. The preceding results aggregate all maintenance into a single input.[17] Examining internal and external maintenance separately provides a powerful test of whether the instrument isolates cost-side variation or captures correlated demand shocks. The instrument shifts the cost of internal maintenance performed by the firm's own employees through state-level wage movements. If this successfully isolates supply-side shocks, it leads to sharp predictions. Internal maintenance is flexible: firms can reallocate labor, adjust schedules, or defer maintenance when costs rise. External maintenance is typically contracted in advance with predetermined service agreements, typically negotiated at equipment purchase, and involves vendors with limited capacity. A valid cost-side shock therefore predicts two things: (i) large quantity responses for internal maintenance, where firms can change their maintenance mix, and (ii) minimal responses for external maintenance, where contracts are sticky. These predictions would not hold under alternative interpretations. If the instrument captured demand shocks, for example, regional booms simultaneously raising maintenance needs and wages, both internal and external maintenance would move together. The asymmetry predicted by the cost-side interpretation provides a falsifiable test.

The remaining columns of Table 2 re-estimate equation (20) separately for internal and external maintenance. The contrast is stark. Internal maintenance exhibits an elasticity nearly double the pooled estimate, reaching approximately 8.4, with confidence intervals easily rejecting both zero and unit elasticity. External maintenance sits near zero in every specification, OLS and IV alike. Instrumenting a margin governed by pre-set contracts is inherently low-powered, so the

17. Appendix F.2 extends the model to multiple maintenance inputs, separating internal maintenance (labor and materials) from contracted external maintenance, and verifies the maintenance channel carries through.

external interval is wide. But no estimate places the contracted margin anywhere near the responsiveness of the in-house one.

This pattern strongly supports the exclusion restriction. The instrument generates the asymmetric response a cost shifter predicts: a large adjustment on the flexible internal margin and a flat one on the sticky external margin. The pattern is difficult to reconcile with demand confounds. As the most price-responsive margin, internal maintenance bounds the firm's aggregate elasticity from above. The lower pooled estimate, which blends in the sticky external margin, better summarizes how a firm adjusts on net. The internal figure is nonetheless conservative. Any residual demand-side correlation would bias it toward zero, so maintenance on the flexible margin is at least as elastic as $\omega \approx 8.4$ indicates.

Additional Robustness Checks. Appendix D.1 provides extensive additional validation. I show that the first stage is strong across multiple instrument specifications (Table D.1). The appendix also re-estimates the elasticity using physical capital stocks rather than book values, a national wage shifter in place of the state-mix instrument, and a three-year-lagged rather than the two-year-lagged labor share (Table D.4). Across every specification the elasticity remains positive and large, ranging from roughly 3.7 to 6.8, and the headline estimates are larger still without any controls (Table D.3). The result is not an artifact of any particular modeling choice.

I also examine the dynamic response. Local projections in Figure D.3 indicate that the response of maintenance is immediate but also displays persistence. Firms reduce upkeep on impact, and the effect deepens for two to three years before attenuating. This pattern is consistent with some adjustment frictions, such as planning lags or a stock-of-disrepair channel, that make the full reallocation of resources away from maintenance gradual rather than instantaneous.

Identification at the frontier. Because the instrument is a shift-share, built from state wage shocks and firm exposure weights, I subject it to the inference now standard for such designs, reported for the no-controls 2SLS (Table D.2). Under firm clustering, the weak-instrument-robust intervals there keep the elasticity bounded away from zero and one. The harder test is exposure-robust inference, which credits the instrument only with its effective number of independent shocks, here roughly two dozen, since the state wage series move largely together. Under that most demanding standard the confidence interval is unbounded, with or without the controls. This is a statement about precision, not about sign or scale: every interval that binds excludes both a zero and a unit elasticity.

Taken together, the railroad evidence is striking. A shock to the wage of maintenance labor, cost-side by construction and plausibly orthogonal to any single firm's upkeep, moves the relative price of maintenance, and firms respond on exactly the margin they control, sharply and at scale. The same large elasticity emerges whether identification comes from national tax changes or from firm-level wage variation. Two designs this different in their identifying variation, and with weaknesses that do not overlap, point to the same conclusion: maintenance is highly sensitive to its price.

6.3 Interpreting the Estimates

Together, the two designs largely agree that maintenance is responsive to changes in relative prices. Moreover, their estimates form a gradient that rises as the unit of observation narrows: 2.6 across industries in the SOI, 4.8 across the firm-asset units of the railroad panel, and 8.4 on the flexible internal margin within them. This is likely not noise. Each step toward a finer unit strips away a layer of stickiness. The internal margin is the one firms control directly, the external margin is contracted, and the industry aggregate nets out the reallocation of capital within industries. When a tax change leads one carrier to let an asset deteriorate, another takes it on and keeps maintaining it, so the response visible in the aggregate is smaller than the response of any single firm. The elasticity is largest where adjustment is freest and most muted in the aggregate, a dampening Appendix F.1 rationalizes.

However, there is an interpretive issue with the coefficients. A regression of maintenance on the relative price recovers the structural demand elasticity ω only if the price is exogenous to quantity, and it is not, because maintenance supply slopes up. Maintenance and investment differ here: investment prices barely move, since globalized equipment markets held them flat through the bonus episodes of the 2000s (House and Shapiro 2008; Basu, Kim, and Singh 2021), but maintenance is labor-intensive, and wages rise when tax cuts stimulate investment (Fuest, Peichl, and Siegloch 2018; Kennedy et al. 2023). A boom therefore raises the relative price of maintenance, and the tax variation moves price as well as quantity, so the regressions recover not ω but an equilibrium elasticity $\omega_{\text{eq}} = \omega/(1 + \omega/\varepsilon_s) < \omega$, where ε_s is the supply elasticity.[18] The same upward slope, traced through the model in Appendix B.2, amplifies the dampening the maintenance margin already produces: a tax cut that raises capital raises the price of maintaining it, pushing firms further from upkeep.

How much the supply channel matters turns on its steepness, which I do not estimate, so its magnitude is an assumption rather than a result. If maintenance labor and parts are drawn from broad markets, ε_s is plausibly large relative to ω, which would leave the attenuation and the amplification both modest, but a smaller ε_s would make them more sizable. Modest or not, the two channels point the same way. The supply response puts the structural elasticity above the measured 2.6, and reallocation puts the firm-level response above it as well. The quantitative model, which I turn to in the following section, takes it as a floor, and the gap it opens between

18. Let demand be $m = a(Qp^M)^{-\omega}$ with $Q \equiv (1-\tau)/(1-\tau z)$, and inverse supply $p^M = bm^{1/\varepsilon_s}$. Substituting and differentiating with respect to $\ln Q$ yields $\omega_{\text{eq}} = \omega/(1+\omega/\varepsilon_s)$.

investment and capital responses grows in ω, so a fuller accounting of prices and of reallocation would likely only widen it.

7 Quantitative Implications of Maintenance Demand

The two mechanisms identified in Section 3 both attenuate the effects of tax incentives, and each carries an empirical consequence. The capital preservation effect—only the non-maintenance share of user cost responds to tax incentives—attenuates steady-state responses and, in standard regressions, biases the estimated investment elasticity downward. The investment-capital gap—tax cuts endogenously raise depreciation, so observed investment overstates capital accumulation—drives investment and capital elasticities apart and slows convergence. The empirical estimates from Sections 5 and 6 provide the parameters needed to quantify both. In this section, I first show how the investment-capital gap maps investment elasticities into capital elasticities. I then apply the OVB correction to estimates from the literature. Finally, I embed the estimated maintenance demand function into a general equilibrium model to assess the 2017 Tax Cuts and Jobs Act, where both mechanisms operate jointly.

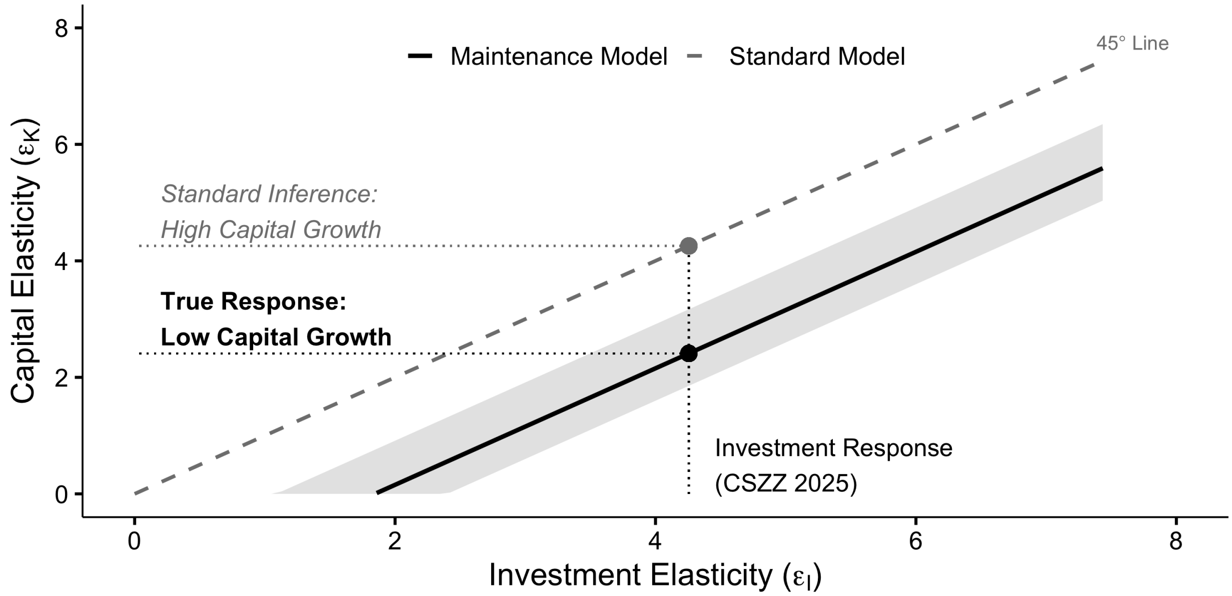
7.1 The Investment-Capital Gap

Proposition 3 establishes that $\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau}$: the investment response to tax policy exceeds the capital response by the depreciation elasticity. The standard identity $\varepsilon_{I,\tau} = \varepsilon_{K,\tau}$ holds only when $\varepsilon_{\delta,\tau} = 0$, which requires either perfectly inelastic maintenance demand or zero maintenance. The estimates from Section 6 reject both cases.

To quantify the gap, I use the SOI estimates of (γ, ω) in an otherwise standard calibration, pinning the depreciation rate to the pre-TCJA investment-output ratio of 10.9%. I construct the confidence band by drawing ω from its estimated distribution and recalibrating for each draw.

Figure 7 plots the result. The 45° line is the standard model where $\varepsilon_{K,\tau} = \varepsilon_{I,\tau}$. The solid line is the maintenance model. For a given investment elasticity on the horizontal axis, the vertical gap shows how much capital accumulation is overstated by standard inference. The shaded band reflects uncertainty in the maintenance parameters only. I take the investment elasticity as given rather than propagating uncertainty from that estimate, since the goal here is to show what maintenance implies for any measured investment response.

Figure 7: The Investment-Capital Elasticity Gap



Notes: The dashed line is the 45° line where $\varepsilon_{K,\tau} = \varepsilon_{I,\tau}$. The solid line plots $\varepsilon_{K,\tau} = \varepsilon_{I,\tau} - \varepsilon_{\delta,\tau}$ at the median estimates from Section 6. The shaded band spans the 5th to 95th percentiles across 5,000 draws of ω, holding the investment elasticity fixed. The vertical line marks the investment elasticity from Chodorow-Reich et al. (2025).

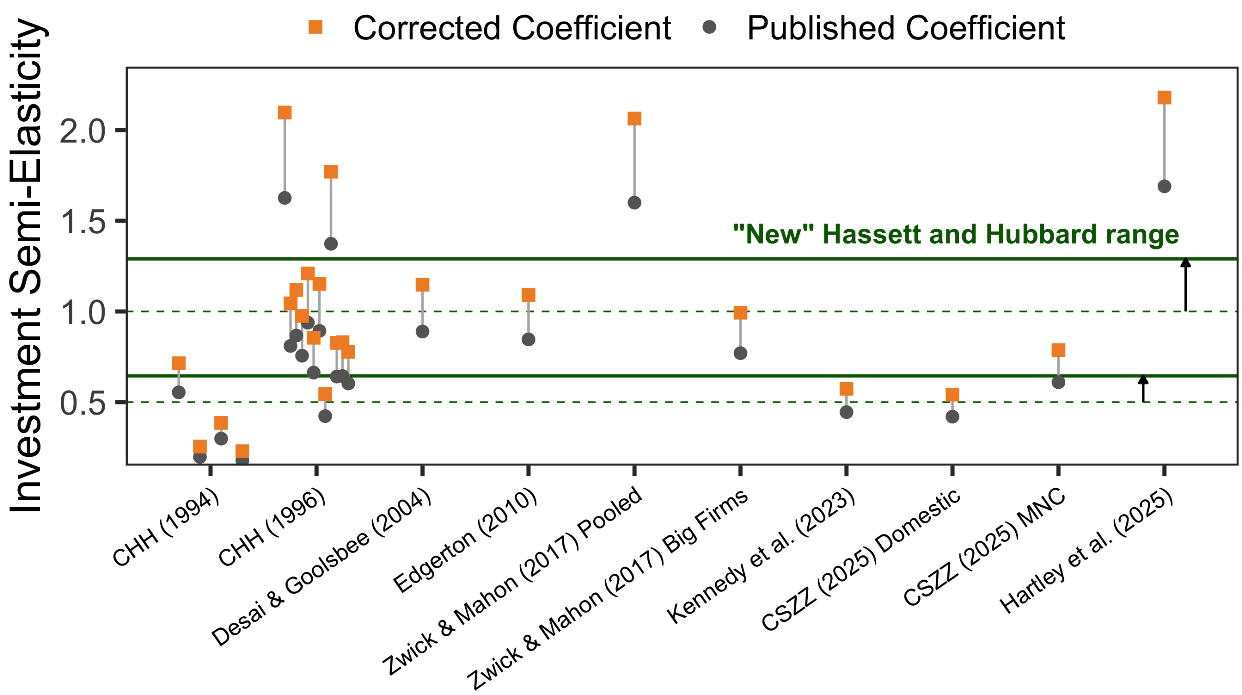
At the Chodorow-Reich et al. (2025) investment elasticity of roughly 4.3, the implied capital elasticity is about 2.4, or 57% of the investment response. Standard models that equate the two would overstate capital accumulation, and the associated effects on output and wages, by a factor of about 1.8. Maintenance also pushes the investment elasticities the literature reports below their true values, and I take up that correction before bringing both channels into general equilibrium.

7.2 Correcting Existing Estimates

Proposition 2 shows that standard investment regressions understate the short-run response because measured user cost overstates the true price change. The correction factor is $1/(1 - s_m)$: if maintenance accounts for share s_m of the user cost, the true coefficient is $1/(1 - s_m)$ times the estimated coefficient.

Under the estimates from Section 6, the correction factor is approximately 1.3. This adjustment is first-order for interpreting the empirical investment literature. Figure 8 applies Proposition 2 to a broad swath of studies, all of which have the investment rate as the left-hand side variable. The corrected estimates are uniformly larger than published estimates. The green lines plot what Hassett and Hubbard (2002) refer to as the "consensus range" for estimates in the literature, and the correction lifts it from roughly 0.5–1.0 to about 0.65–1.3.[19]

Figure 8: Corrected Investment Coefficients from the Literature



Notes: This figure applies Proposition 2 to studies from the literature using the maintenance demand function estimated in Section 6. Most estimates are taken from Chodorow-Reich, Zidar, and Zwick (2024). CHH1994 refers to Cummins, Hassett, and Hubbard (1994). CSZZ2025 refers to Chodorow-Reich et al. (2025). The correction factor is approximately 1.3.

The OVB correction is necessary but not sufficient for translating estimated coefficients into capital effects. Even after correction, three additional adjustments remain. First, the corrected coefficient is a short-run estimate. Transition dynamics relate short-run to long-run investment responses, and the maintenance model converges more slowly than standard models imply. Second, even long-run investment responses overstate capital accumulation because of the investment-capital gap (Section 3.3). Third, the capital preservation effect implies that the user cost elasticity itself is attenuated by $(1 - s_m)$, reducing steady-state capital responses (Section 3.2). These adjustments work in opposite directions: the correction raises the short-run investment elasticity, while the gap and the capital preservation effect lower the steady-state capital response it implies. The following section traces them through a full model of the 2017 Tax Cuts and Jobs Act.

19. As Chodorow-Reich (2025) argues, the coefficients in this figure are not strictly investment elasticities. The correction nonetheless applies because the bias operates through the same channel regardless of how the dependent variable is specified.

7.3 Capital Maintenance and the 2017 Tax Cuts and Jobs Act

Having established that investment responses overstate capital accumulation, I now embed the maintenance channel in a full model to assess TCJA. Throughout this subsection, I compare the maintenance model (NGMM) to the standard model with zero maintenance (NGM, the $s_m \to 0$ limit). The model extends Section 3 to multiple sectors with adjustment costs and general equilibrium in labor markets.

Model and Calibration

There are three sectors: corporate, non-corporate, and government. The representative firm in each private sector $i \in \{c, nc\}$ produces output with identical Cobb-Douglas technology and discounts the future at rate r^k. Capital and labor shares are α_K and α_L with $\alpha_K + \alpha_L \leq 1$. One unit of labor is supplied inelastically and allocated across sectors to equalize wages. I normalize corporate productivity to one and calibrate non-corporate productivity below. To facilitate comparison with Chodorow-Reich et al. (2025), I adopt their values for the discount rate, capital share, and labor share. Table E.1 reports the calibration in full.

To capture dynamics, both the NGM and the NGMM include capital adjustment costs:

$$\Phi(I_t, K_t) = \frac{\phi}{2}\left(\frac{I_t}{K_t} - \delta(m_t)\right)^2 K_t. \tag{22}$$

In the NGM, $\delta(m_t) = \delta$ since maintenance is zero.[20] I solve each model for the perfect-foresight transition between its initial and post-reform steady states. I set parameters so both models start from the same steady-state capital-labor ratio and sectoral capital shares, but because the models differ, some parameters like ϕ must differ as well.

Two corrections are required to take the model to TCJA. First, standard investment regressions omit maintenance from user cost, biasing coefficients toward zero. I correct the Chodorow-Reich et al. (2025) investment elasticity for this bias following Section 7.2, which at the model's calibrated maintenance share raises it by roughly 1.2. Second, reconciling a large investment response with a smaller capital response requires steeper adjustment costs in the NGMM than in the NGM. Appendix B.5 derives the implied adjustment cost parameter. The key intuition is that part of the observed investment surge replaces capital that now depreciates faster rather than expanding the net stock, so larger frictions are needed to rationalize the same gross response.

NGMM Calibration. The maintenance demand estimates from Section 6 pin down (γ, ω). For each of 5,000 draws of the estimated parameters, I calibrate δ_0 and non-corporate productivity A_{nc}

20. This specification implies maintenance adjusts instantaneously. I use capital adjustment costs for comparability with Chodorow-Reich et al. (2025).

to jointly match two targets: the ratio of corporate to total capital ($K_c/K = 0.7$, from Chodorow-Reich et al. 2025) and the pre-TCJA investment-output ratio (0.109, from NIPA).[21] Given these parameters, I compute the depreciation elasticity $\varepsilon_{\delta,\tau}$ using the first-order approximation in Appendix B.5. For the investment elasticity, I take the estimate from Chodorow-Reich et al. (2025), apply the OVB correction, and draw it independently of the maintenance parameters to propagate uncertainty from both sources. The implied adjustment cost ϕ_{NGMM} then follows from the derivations in Appendix B.5. The resulting calibration implies a maintenance share of user cost $s_m \approx 0.18$, consistent with the SOI data, so that steady-state responses are marked down by approximately $(1 - s_m)$.

NGM Calibration. As a benchmark, I calibrate an otherwise identical model with zero maintenance. I set the constant depreciation rate δ to match the same initial steady state as the NGMM. I then calculate ϕ_{NGM} using the *unadjusted* investment elasticity from Chodorow-Reich et al. (2025). This calibrates adjustment costs as if the NGM were true, yielding a conservative difference in convergence rates between models.

Policy Reform. I set pre- and post-TCJA tax rates following Chodorow-Reich et al. (2025), assuming permanent bonus depreciation and no change in non-corporate marginal rates.[22] Corporate tax rates come from a capital-weighted average of their domestic estimates. Given the reform, aggregate output growth is

$$\frac{\Delta Y_{0,t}}{Y_0} = \sum_{i \in \{c,nc,g\}} c_i \frac{\Delta Y_{i,0,t}}{Y_{i,0}},$$

where c_i is the sectoral weight and $\Delta Y_{i,0,t}/Y_{i,0}$ is cumulative growth from year zero to t.

I propagate uncertainty from two sources: the maintenance demand function (via its estimated sampling distribution from Section 6) and the investment elasticity (via the standard errors from Chodorow-Reich et al. (2025)). I hold all other parameters fixed. This differs from Chodorow-Reich et al. (2025), who propagate uncertainty across all structural parameters. I adopt the narrower approach for two reasons. First, it isolates uncertainty attributable to the maintenance channel. Second, it facilitates direct comparison: the NGM results exactly replicate theirs when uncertainty is limited to adjustment costs alone. The trade-off is that reported confidence intervals understate total uncertainty and should be interpreted as conditional on the broader calibration.

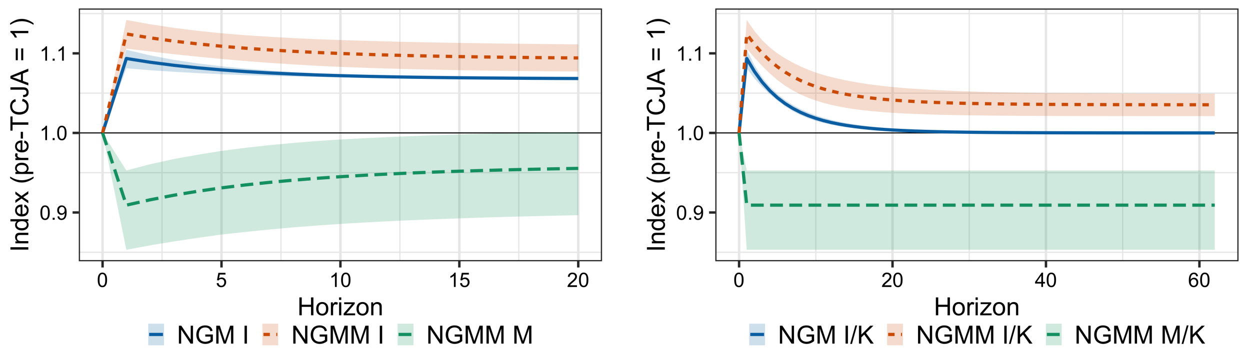
21. I divide the sum of non-residential investment in equipment and structures by GDP less government expenditures in NIPA Table 1.1.5.

22. This is not what actually happened: bonus was temporary and non-corporate taxes fell substantially. The goal is to isolate the maintenance channel rather than score TCJA comprehensively. The exercise is best viewed as a comparison to Chodorow-Reich et al. (2025).

Aggregate Effects of TCJA

The Tax Cuts and Jobs Act cut the effective tax on corporate capital by around 3.4% on average. Figure 9 plots how investment and maintenance evolved in response. The left panel shows levels and the right shows rates relative to capital.

Figure 9: The Effect of TCJA on Capital Inputs



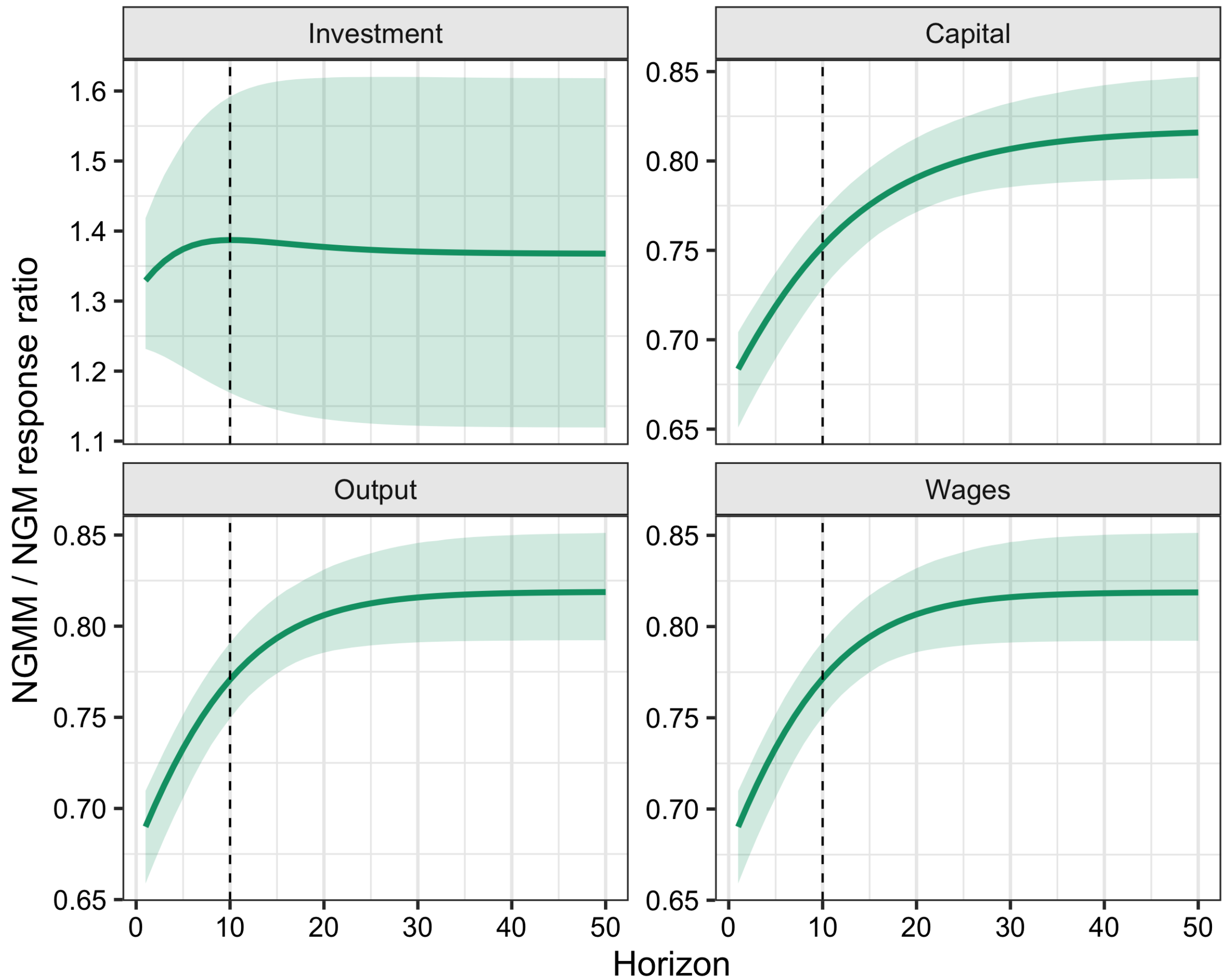
(a) Investment and Maintenance Levels

(b) Investment and Maintenance Rates

Notes: Panel (a) shows the response of domestic corporate investment and maintenance to TCJA in the NGMM (orange, with 95% CI) and the NGM (blue). Panel (b) shows corresponding changes in rates relative to capital.

In the NGM, investment rises by 6.8%. In the NGMM, the OVB-corrected investment elasticity implies a larger response: investment rises by 9.3%. But this larger gross investment response does not translate into proportionally larger capital accumulation. Maintenance falls, depreciation rises, and more of the investment flow goes toward replacement rather than expansion. The right panel shows the key distinction: in the NGMM, the investment rate rises permanently, to about 3.5% above baseline, offsetting the permanently lower maintenance rate. In the NGM, the investment rate increase is temporary, converging back to baseline as the capital stock adjusts. This is the churn-versus-growth distinction from Section 3.3: tax cuts raise the investment rate permanently, but this reflects faster replacement of a younger capital stock, not a larger capital stock.

Figure 10: Aggregate Effects of TCJA: NGMM Relative to NGM



Notes: Each panel plots the ratio of the NGMM response to the NGM response for one aggregate (investment, capital, output, or wages) along the transition. The dashed vertical line marks the ten-year horizon. Shaded bands are 95% confidence intervals from 5,000 draws.

Figure 10 plots the ratio of the NGMM response to the NGM response for investment, capital, output, and wages along the transition. Appendix Figure E.1 shows the underlying level paths. Three patterns stand out. First, steady-state effects on capital, output, and wages are roughly 81% as large in the NGMM as in the NGM, in line with the $(1 - s_m)$ markdown. This reflects the capital preservation effect: only the non-maintenance share of user cost responds to tax incentives. Second, investment and capital responses diverge sharply in the NGMM. In the NGM, both rise by 6.8% in steady state, replicating Chodorow-Reich et al. (2025). In the NGMM, investment rises by 9.3% while capital rises by only 5.5% in steady state, about three-fifths as much. The gap reflects Proposition 3: gross investment must rise to both expand the capital stock and offset faster depreciation. Third, convergence is markedly slower in the NGMM. At the ten-year horizon, which is the statutory scoring window, the NGMM capital stock has reached only about 75% of its steady-state change, compared to 81% in the NGM. This follows from the higher adjustment costs required to reconcile a large gross investment response with a smaller net capital impulse.

Together, these patterns show that all four theoretical channels operate in the quantitative model. The OVB correction recovers the true investment response from biased estimates. The investment-capital gap means gross investment overstates capital accumulation. The capital preservation effect dampens steady-state responses. And slower convergence reduces growth along the transition path.

Scoring TCJA

The quantitative difference in convergence rates matters for how we score tax reform. Government bodies like the Congressional Budget Office and the Joint Committee on Taxation publish budgetary and macroeconomic effects at a ten-year horizon. Static scores assume no behavioral changes. Dynamic scores account for the extra output, and hence the extra revenue, from growth effects.[23] Because convergence typically takes decades, the relevant metric is not steady-state effects but the ten-year mark along the transition path.

To score the reform, I follow Barro and Furman (2018). For both models, I adjust the CBO's static output projections using the model-implied growth path. The extra output generates additional tax revenue given the CBO's projected corporate revenue share. I express results as the percent of the bill's static cost offset by this additional revenue. If the offset is 100%, the bill pays for itself.

Table 3 presents ten-year outcomes for corporate capital accumulation, aggregate output growth, and the static-score offset.

Baseline Results (GE). In the baseline general equilibrium model, the NGM predicts corporate capital rises by 5.5% and aggregate output by 0.56% over ten years, offsetting 7.0% of the static cost. The NGMM predicts capital rises by 4.1% and output by 0.43%, offsetting only 5.2%. The bill does not pay for itself under either model. The ratios are stable across outcomes: NGMM responses are approximately 74–76% as large as NGM responses, reflecting the theoretical structure in which the maintenance share marks down all aggregates proportionally. Against an $18.3 trillion corporate capital base, the 1.4 percentage-point capital gap implies roughly $248 billion less corporate capital under the maintenance model.

Endogenous Maintenance Prices (NGMMP). The third row within each panel allows the pre-tax price of maintenance to rise with wages, as discussed in Appendix B.2. I assume labor

23. See Elmendorf, Hubbard, and Williams (2024) for discussion of when and how to dynamically score tax reforms.

comprises half of maintenance costs and compute the induced wage increase from output expansion. The quantitative difference from the baseline NGMM is minimal, and if anything the wage feedback deepens the attenuation slightly: the GE ratios edge down from 0.74–0.76 to 0.73–0.75. Wage endogeneity matters for interpreting elasticity estimates but has limited impact on policy counterfactuals.

Table 3: Ten-Year Corporate Capital, Aggregate Output, and Scores

		$\Delta K_c/K_c$ (%)		$\Delta Y/Y$ (%)		Static-Score Offset	
		10Y	Ratio	10Y	Ratio	10Y	Ratio
GE	NGM	5.5	-	0.56	-	7.0	-
	NGMM	4.1	0.75	0.43	0.76	5.2	0.74
	NGMMP	4.1	0.74	0.42	0.75	5.1	0.73
GE w/CO	NGM	3.6	-	0.21	-	2.6	-
	NGMM	2.3	0.65	0.10	0.46	1.1	0.44
	NGMMP	2.3	0.65	0.09	0.45	1.1	0.43
PE	NGM	8.8	-	3.25	-	37.5	-
	NGMM	6.4	0.73	2.37	0.73	26.6	0.71
PE w/CO	NGM	6.0	-	1.81	-	20.8	-
	NGMM	3.6	0.60	0.89	0.49	10.0	0.48

Notes: GE denotes general equilibrium, PE partial equilibrium, and w/CO with fiscal crowding out. Within each panel, the ratio is relative to the corresponding NGM model. NGMMP denotes the closure in which labor costs comprise half of maintenance costs, so tax cuts raise the relative price of maintenance beyond the direct tax effect. The ten-year corporate-capital gap between NGM and NGMM (GE) is ≈ \$248B (BEA Table 4.1, Line 17, 2017 base \$18,287.3 B). The δ_0-sensitivity range is \$170–516B.

Crowding Out (GE w/CO). The second panel incorporates fiscal crowding out. I compute the 2027 debt-GDP ratio implied by TCJA's fiscal cost and endogenous revenue feedback, then use Plante, Richter, and Zubairy (2026) to calculate the resulting increase in real interest rates. Higher rates reduce private investment. Crowding out affects both models but exacerbates the difference: the NGMM offset falls to 1.1% while the NGM falls to 2.6%. The larger proportional reduction in the NGMM reflects its slower initial growth. Because capital accumulates more gradually, debt accumulates faster over the scoring window. The ratio of NGMM to NGM responses falls from 0.74–0.76 to 0.44–0.65 with crowding out. With crowding out, both models contract, so the absolute capital gap narrows to about $228 billion even as the proportional markdown deepens.

Partial Equilibrium (PE). The final panels assume perfectly elastic labor supply. Aggregate outcomes are substantially larger. The NGM offset rises to 37.5%, but the ratio of NGMM to NGM responses remains similar, ranging from 0.48 to 0.73. This stability reinforces that the maintenance channel operates through the capital preservation effect and the investment-capital gap, both of which persist regardless of labor supply assumptions.

Interpretation. Given how little of the static cost is offset by either model, investment-driven growth provides minimal self-financing for capital tax cuts. The maintenance channel reduces this already-small revenue feedback by roughly a quarter, and by more than half once crowding out is included. This does not render dynamic scoring irrelevant. Dynamic scores inform distributional analysis, validate mechanisms, and guide fiscal timing. But for reforms targeting investment incentives, the growth-induced revenue feedback may be too small to substantially alter fiscal projections.[24]

The broader lesson extends beyond TCJA. The NGM results replicate Chodorow-Reich et al. (2025), so the haircuts I obtain apply to their analysis. But the mechanism generalizes: in richer models where capital accumulation responds to user cost reductions (Sedlacek and Sterk 2019; Zeida 2022), maintenance would similarly dampen responses. The channel operates through capital preservation and the investment-capital gap regardless of other frictions.

8 Concluding Remarks

In this paper, I establish that capital maintenance is a first-order margin for understanding how tax incentives transmit to aggregate outcomes. Using novel data from freight railroad filings and corporate tax returns, I show that maintenance demand is large and elastic. A calibrated model of the 2017 Tax Cuts and Jobs Act implies that accounting for maintenance reduces ten-year capital, output, and revenue effects by roughly one-quarter to one-third relative to standard models. Two mechanisms drive this result. The capital preservation effect operates through the level of maintenance, attenuating user cost elasticities and biasing standard regression estimates downward. The investment-capital gap operates through its elasticity, breaking the equivalence between investment and capital responses and slowing convergence to steady state. These forces matter most for policy through dynamic scoring: because growth effects are both smaller and slower to arrive, dynamic scores of investment-oriented tax reforms lie closer to static scores than the

24. The scoring exercise covers only domestic corporate provisions. Other margins may matter. The static score for permanent bonus comes from Barro and Furman (2018).

standard framework implies. This conclusion is reinforced from an independent direction by Chodorow-Reich (2025), who, reassessing the neoclassical benchmark itself, likewise finds that the neoclassical channel can rationalize only modest dynamic revenue offsets in the current low-tax regime. Maintenance is a distinct and complementary reason for the same bottom line. The core intuition is churn versus growth: because tax cuts raise depreciation through reduced maintenance, gross investment overstates net capital accumulation. Investment subsidies promote capital turnover, not necessarily capital expansion.

These findings extend beyond TCJA to any policy that changes the relative price of new capital versus maintenance. They also raise questions for national accounting: maintenance is treated as intermediate consumption while investment is final demand, but the substitution between them is endogenous to policy. Similarly, depreciation rates are treated as fixed, but if maintenance affects capital longevity, measured net capital stocks may overstate true accumulation when tax policy discourages upkeep. Given the groundwork laid here and in prior work by McGrattan and Schmitz Jr. (1999) and Goolsbee (2004), the case for accounting for maintenance in public finance and macroeconomics is too strong to ignore.

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Appendices

A Institutional Background

Once classified as an expenditure that must be capitalized, assets are slotted into one of eight lives under rules governed by the Modified Accelerated Cost Recovery System (MACRS)—three, five, seven, ten, fifteen, twenty, 27.5, or 39 years—which govern how quickly they may be depreciated; shorter class lives yield faster deductions and larger present-value tax benefits. Given a class life of T years and a discount rate r^k, the net present value of a dollar of deductions for a capital investment is

$$z = \sum_{t=0}^{T} \left(\frac{1}{1+r^k} \right)^t d_t,$$

where d_t is the allowable deduction by the IRS in period t. Table A.1 shows typical assets and associated present value of deductions z for each class category, and Table A.2 works out the year-by-year tax consequences of spending a marginal dollar investing in a new seven-year asset versus maintaining an existing one.

Table A.1: MACRS Asset Lives and NPV of Depreciation Allowances z at a 6% discount rate

MACRS Asset Life (Years)	**Representative Examples**	z
3	Racehorses; special tools	0.9467
5	Automobiles; computers; office machinery	0.9038
7	Furniture; fixtures; general-purpose equipment; locomotives[25]	0.8645
10	Appliances; vessels; barges	0.8108
15	Land improvements; sewage treatment facilities; telephone poles	0.6942
20	Farm buildings; municipal sewers	0.6219
27.5	Residential rental property	0.5001
39	Nonresidential real property (commercial buildings)	0.3958

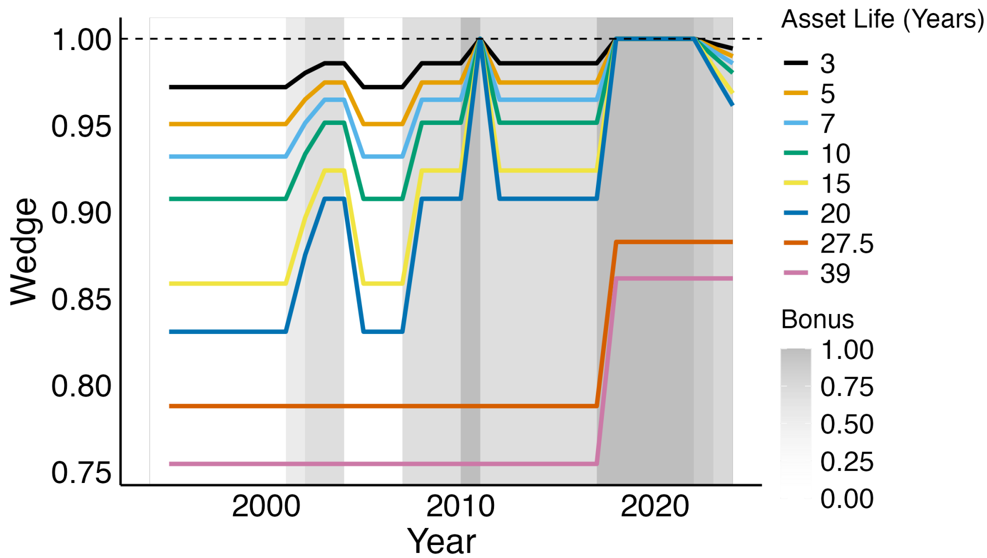
Table A.2: The Tax Treatment of Investment and Maintenance for a Seven-Year Asset

Year:	0	1	2	3	4	5	6	7	Total	z
Investment										
Deductions (000s)	142.9	244.9	174.9	124.9	89.3	89.2	89.3	44.6	1,000	
Tax benefit (τ = 35%)	50.0	85.7	61.2	43.7	31.3	31.2	31.3	15.6	350	0.86
Tax Benefit (PV)	50.0	80.9	54.5	36.7	24.8	23.3	22.0	10.4	302.6	
Maintenance										
Deductions (000s)	1,000	0	0	0	0	0	0	0	1,000	
Tax benefit (τ = 35%)	350	0	0	0	0	0	0	0	350	1.00
Tax Benefit (PV)	350	0	0	0	0	0	0	0	350	

Notes: This table adapts Table 1 from Zwick and Mahon (2017) to a seven-year MACRS schedule (8-year life with half-year convention and 200% declining balance until straightline becomes optimal). It includes year-by-year deductions, tax benefits, and present values using a 6% discount rate, and reports z, the PV factor per dollar of tax benefit.

Figure A.1 plots the wedge since 1995 for all MACRS assets. The wedge is largest for long-lived assets and with 100% bonus, the wedge vanishes for qualifying assets. Of course, some firms may elect not to claim bonus if they are not taxable because the deductions are worthless to them (Kitchen and Knittel 2011). Regardless, the wedge has varied considerably between asset types over time.

Figure A.1: The Maintenance-Investment Wedge for MACRS Assets



Notes: The wedge is defined in the main text as the ratio $(1-\tau)/(1-\tau\tilde{z})$, where $\tilde{z}$ is the net present value of depreciation allowances after accounting for bonus depreciation. I set the discount rate as $r = 0.06$.

B Model Derivations

This appendix contains proofs and additional results and discussion for the theoretical results in Section 3. Throughout, the NGM is the $s_m \to 0$ limit of the NGMM, with $\Psi^{NGM} = \lim_{s_m \to 0} \Psi^{NGMM}$.

B.1 Proofs

Proof of Proposition 3

In steady state, $K_{t+1} = K_t = K$. From the accumulation equation, $K = (1-\delta(m^*))K + I$, which implies $I = \delta(m^*)K$. Taking logs and differentiating with respect to $\log(1-\tau)$ yields the decomposition $\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau}$.

For the signs, note that by Lemma ??(iii), investment incentives reduce maintenance: $\partial m^*/\partial(1-\tau) < 0$. Since $\delta'(m) < 0$ by Assumption 1(i), lower m^* raises $\delta(m^*)$. A tax cut raises $(1-\tau)$, so δ increases, meaning $\varepsilon_{\delta,\tau} > 0$. Lower user cost also raises capital, so $\varepsilon_{K,\tau} > 0$. Since both elasticities are positive, $\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} > \varepsilon_{K,\tau}$. $\square$

Proof of Proposition 1

The proof relies on the envelope theorem. The optimized user cost is $\Psi^{NGMM} = \Psi(m^*(x), x)$ where x represents any parameter and $m^*(x)$ is optimal maintenance. By the chain rule:

$$\frac{d\Psi^{NGMM}}{dx} = \frac{\partial \Psi}{\partial m}\bigg|_{m=m^*} \cdot \frac{\partial m^*}{\partial x} + \frac{\partial \Psi}{\partial x}\bigg|_{m=m^*}$$

At the optimum, $\frac{\partial \Psi}{\partial m}|_{m=m^*} = 0$ by the first-order condition, so $\frac{d\Psi^{NGMM}}{dx} = \frac{\partial \Psi}{\partial x}|_{m=m^*}$.

Computing each partial derivative of $\Psi(m) = \frac{p^I(r+\delta(m))}{1-\tau} + p^M m$: for τ, $\frac{\partial \Psi}{\partial \tau}|_{m=m^*} = \frac{p^I(r+\delta(m^*))}{(1-\tau)^2}$, which equals $\frac{\partial \Psi^{NGM}}{\partial \tau}$ when $\bar{\delta} = \delta(m^*)$. The same equality holds for derivatives with respect to r and p^I. For p^M, $\frac{\partial \Psi}{\partial p^M}|_{m=m^*} = m^*$.

To convert to elasticities, note that for any $x \in \{\tau, r, p^I\}$, $\frac{\partial \Psi^{NGMM}}{\partial x} = \frac{\partial \Psi^{NGM}}{\partial x}$. Since $\varepsilon_{\Psi,x} = \frac{\partial \Psi}{\partial x} \cdot \frac{x}{\Psi}$:

$$\frac{\varepsilon_{\Psi^{NGMM},x}}{\varepsilon_{\Psi^{NGM},x}} = \frac{\Psi^{NGM}}{\Psi^{NGMM}} = 1 - s_m$$

Therefore $\varepsilon_{\Psi^{NGMM},x} = (1 - s_m)\varepsilon_{\Psi^{NGM},x}$. For the maintenance price elasticity: $\varepsilon_{\Psi^{NGMM},p^M} = m^* \cdot \frac{p^M}{\Psi^{NGMM}} = \frac{p^M m^*}{\Psi^{NGMM}} = s_m$. $\square$

B.2 General Equilibrium

To analyze general equilibrium, I extend the partial equilibrium model to include labor. Production is $F(K_t, L_t)$ where L_t is labor.

Assumption 2 (Production Technology). *The production function $F : \mathbb{R}^2_+ \to \mathbb{R}_+$ is twice continuously differentiable with $F_K > 0$, $F_L > 0$, $F_{KK} < 0$, and $F_{LL} < 0$.*

The firm's after-tax dividends become:

$$d_t = (1 - \tau^c)\left[F(K_t, L_t) - p_t^M M_t - w_t L_t\right] - (1 - \tau^c z - c)p_t^I I_t \tag{A.1}$$

where w_t is the wage. The firm's first-order conditions now include labor demand:

$$F_L(K_t, L_t) = w_t \tag{A.2}$$

$$\begin{aligned}(1 - \tau_t^c z_t - c_t)p_t^I = \frac{1}{1 + r_t}\Big\{&(1 - \tau_{t+1}^c)F_K(K_{t+1}, L_{t+1}) \\ &+ (1 - \tau_{t+1}^c z_{t+1} - c_{t+1})p_{t+1}^I\left[1 - \delta(m_{t+1}) + \delta'(m_{t+1})m_{t+1}\right]\Big\}\end{aligned} \tag{A.3}$$

$$(1 - \tau_t^c)p_t^M = (1 - \tau_t^c z_t - c_t)p_t^I\left[-\delta'(m_t)\right] \tag{A.4}$$

The maintenance and capital conditions are unchanged from the partial equilibrium analysis; labor demand (A.2) governs the wage response to capital accumulation.

Throughout this appendix, I compare the maintenance model (NGMM) to a standard neoclassical model without maintenance (NGM). The NGM is the $s_m \to 0$ limit of the NGMM: maintenance costs are negligible, so user cost reduces to the Hall-Jorgenson formula $\Psi^{NGM} \equiv \frac{p^I(r+\delta)}{1-\tau} = \lim_{s_m \to 0} \Psi^{NGMM}$.

The capital preservation effect is a partial equilibrium result: it characterizes how user cost responds to tax policy holding prices fixed. But the effects that matter for policy occur in general equilibrium, where capital accumulation feeds back through factor markets. When capital demand rises, interest rates may adjust, wages increase, and input prices change. This appendix shows the $(1 - s_m)$ attenuation survives—and under the most empirically relevant conditions, general equilibrium amplifies it.

Assumption 3 (GE Closure)**.** *The general equilibrium is characterized by:*

(i) A GE-adjusted marginal product $\tilde{F}_K(K)$ with elasticity $\tilde{\varepsilon}_{F_K} \equiv \frac{d \log \tilde{F}_K}{d \log K} < 0$, incorporating diminishing returns and labor market clearing.

(ii) Reduced-form price-capital elasticities $\varepsilon_{r,K}$, $\varepsilon_{p^I,K}$, and $\varepsilon_{p^M,K}$.

(iii) The function $\tilde{F}_K$ is identical across models: maintenance affects the cost of capital services but not the production technology.

(iv) Stability: $\tilde{\varepsilon}_{F_K} - \Lambda < 0$ where Λ is the relevant price feedback.

To state the results, I first derive the user cost elasticities with respect to prices. From the Hall-Jorgenson formula $\Psi^{NGM} = \frac{p^I(r+\bar{\delta})}{1-\tau}$:

$$\varepsilon_{\Psi^{NGM},r} = \frac{r}{r + \bar{\delta}}, \qquad \varepsilon_{\Psi^{NGM},p^I} = 1$$

Define the GE price feedback in the standard model as the effect of capital accumulation on user cost through prices:

$$\Lambda_0 \equiv \varepsilon_{\Psi^{NGM},r} \cdot \varepsilon_{r,K} + \varepsilon_{\Psi^{NGM},p^I} \cdot \varepsilon_{p^I,K} = \frac{r}{r + \bar{\delta}} \varepsilon_{r,K} + \varepsilon_{p^I,K}$$

The first term captures crowding out: when capital rises, interest rates may rise, increasing user cost. The second term captures investment goods price effects.

For the maintenance model, define:

$$\Lambda^M \equiv \varepsilon_{p^M,K}$$

This is the elasticity of maintenance prices with respect to capital. The user cost elasticity $\varepsilon_{\Psi^{NGMM},p^M} = s_m$ from Proposition 1 will appear separately in the formulas below. I compare steady states before and after a permanent tax change, with hats denoting log-deviations.

Proposition 6 (GE Response Ratio)**.** *Under Assumptions 2–3, the steady-state capital responses to a tax shock $\hat{\tau}$ are:*

$$\hat{K}^{NGM} = \frac{\varepsilon_{\Psi^{NGM},\tau}}{\tilde{\varepsilon}_{F_K} - \Lambda_0}\hat{\tau}, \qquad \hat{K}^{NGMM} = \frac{(1-s_m)\varepsilon_{\Psi^{NGM},\tau}}{\tilde{\varepsilon}_{F_K} - (1-s_m)\Lambda_0 - s_m\Lambda^M}\hat{\tau}$$

The ratio is:

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = (1-s_m)\cdot\frac{\tilde{\varepsilon}_{F_K} - \Lambda_0}{\tilde{\varepsilon}_{F_K} - (1-s_m)\Lambda_0 - s_m\Lambda^M} \tag{A.5}$$

Proof. The standard user cost is $\Psi^{NGM} = \frac{p^I(r+\bar{\delta})}{1-\tau}$. Taking logs gives $\log\Psi^{NGM} = \log p^I + \log(r+\bar{\delta}) - \log(1-\tau)$, so differentiating:

$$\hat{\Psi}^{NGM} = \hat{p}^I + \frac{r}{r+\bar{\delta}}\hat{r} + \frac{\tau}{1-\tau}\hat{\tau}$$

where I use $\frac{d\log(r+\bar{\delta})}{d\log r} = \frac{r}{r+\bar{\delta}}$ and $\frac{d\log(1-\tau)}{d\log\tau} = -\frac{\tau}{1-\tau}$. The user cost elasticities are therefore $\varepsilon_{\Psi^{NGM},p^I} = 1$, $\varepsilon_{\Psi^{NGM},r} = \frac{r}{r+\bar{\delta}}$, and $\varepsilon_{\Psi^{NGM},\tau} = \frac{\tau}{1-\tau}$.

Substituting the price responses $\hat{r} = \varepsilon_{r,K}\hat{K}$ and $\hat{p}^I = \varepsilon_{p^I,K}\hat{K}$ from Assumption 3:

$$\hat{\Psi}^{NGM} = \frac{\tau}{1-\tau}\hat{\tau} + \left(\frac{r}{r+\bar{\delta}}\varepsilon_{r,K} + \varepsilon_{p^I,K}\right)\hat{K} = \varepsilon_{\Psi^{NGM},\tau}\hat{\tau} + \Lambda_0\hat{K}$$

where $\Lambda_0 \equiv \frac{r}{r+\bar{\delta}}\varepsilon_{r,K} + \varepsilon_{p^I,K}$. The equilibrium condition $\tilde{F}_K(K) = \Psi$ log-linearizes to $\tilde{\varepsilon}_{F_K}\hat{K} = \hat{\Psi}$. Substituting and solving:

$$\hat{K}^{NGM} = \frac{\varepsilon_{\Psi^{NGM},\tau}}{\tilde{\varepsilon}_{F_K} - \Lambda_0}\hat{\tau}$$

For the maintenance model, Proposition 1 establishes that the derivatives of Ψ^{NGMM} with respect to (r, p^I, τ) equal those of Ψ^{NGM}. Since $\Psi^{NGM} = (1-s_m)\Psi^{NGMM}$, the elasticities satisfy $\varepsilon_{\Psi^{NGMM},x} = (1-s_m)\varepsilon_{\Psi^{NGM},x}$ for $x \in \{r, p^I, \tau\}$, and $\varepsilon_{\Psi^{NGMM},p^M} = s_m$. Log-linearizing:

$$\hat{\Psi}^{NGMM} = (1-s_m)\frac{\tau}{1-\tau}\hat{\tau} + (1-s_m)\frac{r}{r+\delta}\hat{r} + (1-s_m)\hat{p}^I + s_m\hat{p}^M$$

where $\delta = \delta(m^*) = \bar{\delta}$ at the common steady state. Substituting price responses $\hat{r} = \varepsilon_{r,K}\hat{K}$, $\hat{p}^I = \varepsilon_{p^I,K}\hat{K}$, and $\hat{p}^M = \varepsilon_{p^M,K}\hat{K}$:

$$\hat{\Psi}^{NGMM} = (1-s_m)\varepsilon_{\Psi^{NGM},\tau}\hat{\tau} + \left[(1-s_m)\Lambda_0 + s_m\Lambda^M\right]\hat{K}$$

where $\Lambda^M \equiv \varepsilon_{p^M,K}$. Substituting into the equilibrium condition and solving:

$$\hat{K}^{NGMM} = \frac{(1-s_m)\varepsilon_{\Psi^{NGM},\tau}}{\tilde{\varepsilon}_{F_K} - (1-s_m)\Lambda_0 - s_m\Lambda^M}\hat{\tau}$$

Taking the ratio yields (A.5). □

The numerator of the maintenance model is $(1-s_m)$ times the standard model—the capital preservation effect from Proposition 1 carries through to general equilibrium. The denominators reflect general equilibrium adjustment: $\tilde{\varepsilon}_{F_K}$ captures real crowding out through diminishing returns and labor markets, while the Λ terms capture price feedbacks. The coefficient $(1-s_m)$ on Λ_0 in the maintenance model's denominator reflects that only the investment component of user cost responds to interest rates and investment prices. The coefficient s_m on Λ^M reflects that maintenance prices affect only the maintenance component.

Why does $\tilde{\varepsilon}_{F_K}$ appear identically in both models? Because it works through the production technology, which maintenance does not affect. At any capital stock K, output is $F(K,L)$ regardless of how that capital was maintained. Diminishing returns and labor market adjustment therefore scale both models' responses equally, and $\tilde{\varepsilon}_{F_K}$ cancels when taking ratios. What remains is differential price exposure: the standard model faces Λ_0, while the maintenance model faces a weighted average $(1-s_m)\Lambda_0 + s_m\Lambda^M$.

Proposition 7 (GE Attenuation)**.** *Under Assumptions 2–3:*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = 1 - s_m \qquad \Longleftrightarrow \qquad \Lambda^M = \Lambda_0 \tag{A.6}$$

Proof. For the forward direction, suppose $\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = 1 - s_m$. By Proposition 6:

$$(1-s_m)\cdot\frac{\tilde{\varepsilon}_{F_K} - \Lambda_0}{\tilde{\varepsilon}_{F_K} - (1-s_m)\Lambda_0 - s_m\Lambda^M} = 1 - s_m$$

Dividing by $(1-s_m) > 0$ and cross-multiplying gives $\tilde{\varepsilon}_{F_K} - \Lambda_0 = \tilde{\varepsilon}_{F_K} - (1-s_m)\Lambda_0 - s_m\Lambda^M$, which simplifies to $-s_m\Lambda_0 = -s_m\Lambda^M$, hence $\Lambda_0 = \Lambda^M$.

For the reverse direction, suppose $\Lambda^M = \Lambda_0$. The denominator of Proposition 6 becomes $\tilde{\varepsilon}_{F_K} - (1-s_m)\Lambda_0 - s_m\Lambda_0 = \tilde{\varepsilon}_{F_K} - \Lambda_0$, so the ratio equals $(1-s_m)\cdot\frac{\tilde{\varepsilon}_{F_K}-\Lambda_0}{\tilde{\varepsilon}_{F_K}-\Lambda_0} = 1 - s_m$. □

When maintenance prices respond to capital accumulation the same way as the combination of interest rates and investment goods prices (weighted by their user cost elasticities), the ratio equals $(1-s_m)$ exactly.

Two economic forces determine whether this condition holds. The first is interest rate insensitivity. The maintenance component $\Psi^M = p^M m^*$ does not depend on the interest rate, so when capital accumulates and interest rates rise, only share $(1 - s_m)$ of the maintenance model's user cost is affected. The maintenance model is less exposed to crowding out than the standard model.

The second force is wage sensitivity. Maintenance is labor-intensive—repairs require technicians, upkeep requires workers. When capital accumulates and wages rise, maintenance prices rise more than investment goods prices if maintenance is more labor-intensive. The maintenance model is more exposed to wage-driven cost increases than the standard model.

The condition $\Lambda^M = \Lambda_0$ holds when these forces offset. In a small open economy where interest rates and input prices are pinned globally, neither force operates and the condition holds trivially. The empirically relevant case is likely an open economy where investment goods are traded globally but maintenance services are local. House and Shapiro (2008) and House, Mocanu, and Shapiro (2017) find that investment goods prices did not respond to tax policy changes during the 2000s, largely due to foreign competition—suggesting $\varepsilon_{p^I,K} \approx 0$. Interest rate crowding out depends on how tax cuts are financed: deficit-financed cuts may raise interest rates (Engen and Hubbard 2004; Neveu and Schafer 2024; Plante, Richter, and Zubairy 2026), though the magnitude is debated. But maintenance requires domestic labor and cannot be imported. Fuest, Peichl, and Siegloch (2018) and Kennedy et al. (2023) document that wages rise following corporate tax cuts, implying $\varepsilon_{p^M,K} > 0$. If interest rate crowding out is modest and maintenance prices are wage-sensitive, then $\Lambda^M > \Lambda_0$, and the ratio falls below $(1 - s_m)$: general equilibrium amplifies the attenuation. In most empirically plausible scenarios, $(1 - s_m)$ is either exact or a conservative upper bound. The partial equilibrium results are robust.

With a concave production function and complementarity between capital and labor, the responses of output and wages inherit the $(1 - s_m)$ attenuation. A smaller capital response means less output and lower wages in equilibrium. To illustrate, consider Cobb-Douglas production $Y = K^\alpha L^{1-\alpha}$. The elasticities of capital, output, and wages with respect to user cost are $\varepsilon_{K,\tau} = \frac{1}{1-\alpha}\varepsilon_\Psi$ and $\varepsilon_Y = \varepsilon_w = \frac{\alpha}{1-\alpha}\varepsilon_\Psi$. Since $\varepsilon_\Psi^{NGMM} = (1 - s_m)\varepsilon_\Psi^{NGM}$, the same $(1 - s_m)$ markdown applies to all three. The maintenance margin dampens the real effects of investment incentives across all aggregates.

Additional Results

The following results characterize how the response ratio deviates from $(1 - s_m)$ when $\Lambda^M \neq \Lambda_0$.

Corollary 1 (Special Cases)**.** *Under Assumptions 2–3: (a) if* $\Lambda_0 = \Lambda^M = 0$*, the ratio equals* $1 - s_m$*;*

(b) if $\varepsilon_{p^I,K} = \Lambda^M = 0$*, then*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = (1 - s_m) \cdot \frac{\tilde{\varepsilon}_{F_K} - \frac{r}{r+\bar{\delta}}\varepsilon_{r,K}}{\tilde{\varepsilon}_{F_K} - (1 - s_m)\frac{r}{r+\bar{\delta}}\varepsilon_{r,K}}$$

(c) if $\varepsilon_{r,K} = \varepsilon_{p^I,K} = 0$*, then*

$$\frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}} = (1 - s_m) \cdot \frac{\tilde{\varepsilon}_{F_K}}{\tilde{\varepsilon}_{F_K} - s_m\Lambda^M}$$

Proof. Part (a) follows from Proposition 7. Parts (b) and (c) follow by substitution into Proposition 6. □

Proposition 8 (Deviation from $(1 - s_m)$)**.** *Define* $R \equiv \frac{\hat{K}^{NGMM}}{\hat{K}^{NGM}}$*. Under Assumptions 2–3: (a) if* $\Lambda^M > \Lambda_0$*, then* $R < 1 - s_m$*; (b) if* $\Lambda^M < \Lambda_0$*, then* $R > 1 - s_m$*; (c) if* $\Lambda^M = \Lambda_0$*, then* $R = 1 - s_m$*.*

Proof. Define $N \equiv \tilde{\varepsilon}_{F_K} - \Lambda_0$ and $D \equiv \tilde{\varepsilon}_{F_K} - (1 - s_m)\Lambda_0 - s_m\Lambda^M$. By Assumption 3(iv), $N < 0$ and $D < 0$. By Proposition 6, $R = (1 - s_m) \cdot \frac{N}{D}$. Since $D - N = s_m(\Lambda_0 - \Lambda^M)$, if $\Lambda^M > \Lambda_0$ then $D < N < 0$, so $|D| > |N|$ and $R < 1 - s_m$. The other cases follow analogously. □

Proposition 9 (First-Order Approximation)**.** *For* $|\Lambda_0|, |\Lambda^M| \ll |\tilde{\varepsilon}_{F_K}|$*:*

$$R \approx (1 - s_m)\left(1 + \frac{s_m(\Lambda^M - \Lambda_0)}{\tilde{\varepsilon}_{F_K}}\right)$$

Proof. From Proposition 6, factor out $\tilde{\varepsilon}_{F_K}$ and apply the approximation $\frac{1-a}{1-b} \approx 1 - a + b$ for small a, b. □

B.3 The Tax Elasticity of Investment

In steady state, investment is given by $I = \delta(m)K$. Defining investment explicitly in terms of the tax wedge $(1 - \tau)$:

$$I(1 - \tau) = \delta(m(1 - \tau)) \cdot K(1 - \tau, m(1 - \tau), \delta(m(1 - \tau))) \tag{A.7}$$

Taking logs and differentiating:

$$\begin{aligned} \varepsilon_{I,\tau} &= \frac{d \ln I}{d \ln(1 - \tau)} = \frac{d \ln \delta}{d \ln(1 - \tau)} + \frac{d \ln K}{d \ln(1 - \tau)} \\ &= \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} \end{aligned} \tag{A.8}$$

I now derive explicit expressions for each component using the constant-elasticity depreciation technology:

$$\delta(m) = \delta_0 - \frac{\gamma^{1/\omega}}{1 - 1/\omega} m^{1-1/\omega}$$

and Cobb-Douglas production $F(K) = K^\alpha$.

Depreciation elasticity. Substituting optimal maintenance $m^* = \gamma(1-\tau)^{-\omega}$ into the depreciation technology yields:

$$\delta(m^*) = \delta_0 + \frac{\gamma\omega}{1-\omega}(1-\tau)^{1-\omega}$$

Taking the derivative with respect to $(1-\tau)$:

$$\frac{\partial\delta}{\partial(1-\tau)} = \gamma\omega(1-\tau)^{-\omega}$$

Evaluating at $\tau \approx 0$:

$$\varepsilon_{\delta,\tau} = \frac{\partial \ln \delta}{\partial \ln(1-\tau)} = \frac{\gamma\omega}{\delta_0 + \frac{\gamma\omega}{1-\omega}} \tag{A.9}$$

Capital elasticity. With Cobb-Douglas production, the capital stock satisfies:

$$K = \left(\frac{\alpha}{\Psi}\right)^{\frac{1}{1-\alpha}}$$

where $\Psi = \frac{r^k + \delta(m^*)}{1-\tau} + m^*$ is the user cost. Taking logs and differentiating:

$$\varepsilon_{K,\tau} = \frac{d \ln K}{d \ln(1-\tau)} = \frac{-1}{1-\alpha} \cdot \varepsilon_\Psi$$

where $\varepsilon_\Psi = \frac{d \ln \Psi}{d \ln(1-\tau)}$ is the user cost elasticity. By Proposition 1, $\varepsilon_\Psi = -(1-s_m)$, so:

$$\varepsilon_{K,\tau} = \frac{1-s_m}{1-\alpha} \tag{A.10}$$

where $s_m = \frac{m^*}{\Psi}$ is the maintenance share of user cost.

Investment elasticity. Combining (A.9) and (A.10):

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} = \frac{\gamma\omega}{\delta_0 + \frac{\gamma\omega}{1-\omega}} + \frac{1-s_m}{1-\alpha} \tag{A.11}$$

Both terms are positive: investment increases in response to higher $(1-\tau)$ both because capital expands ($\varepsilon_{K,\tau} > 0$) and because depreciation rises as firms substitute away from maintenance ($\varepsilon_{\delta,\tau} > 0$). The investment-capital gap from Proposition 3 is precisely $\varepsilon_{\delta,\tau}$.

Balanced Growth

Note that with balanced growth, the steady state investment condition is

$$I = (\delta(m) + n + g)K,$$

where n and g are exogenous rates of population and productivity growth. In that case, the gap would simply become

$$\varepsilon_{I,\tau} = \varepsilon_{\delta,\tau}\left(\frac{\delta(m)}{\delta(m)+n+g}\right) + \varepsilon_{K,\tau}$$

B.4 Omitted Variable Bias in Investment Regressions

This appendix proves Proposition 2 and applies the correction to estimates from the literature.

Proof of Proposition 2

The correctly specified regression uses the full user cost $\Psi = \Psi^I + \Psi^M$:

$$f(I_{i,t}, K_{i,t-1}) = \beta \times \log\left(\frac{r^k + \delta(m_{i,t})}{1-\tau_{i,t}} + m_{i,t}\right) + X_{i,t} + u_{i,t}. \tag{A.12}$$

Standard regressions use only $\Psi^I = \frac{r^k+\delta}{1-\tau}$, omitting the maintenance component. The coefficient β represents the short-run investment response to a change in user cost; the omission biases this short-run estimate toward zero.

Under the constant-elasticity functional form, optimal maintenance is $m^* = \gamma(1-\tau)^{-\omega}$ and depreciation is $\delta(m^*) = \delta_0 + \frac{\gamma\omega}{1-\omega}(1-\tau)^{1-\omega}$. The omitted term is the difference between the true and standard log user costs. Taking a first-order approximation around $\tau \approx 0$:

$$\begin{aligned}
\text{Omitted Term} &= \log\left(\frac{r^k + \delta(m^*)}{1-\tau} + m^*\right) - \log\left(\frac{r^k+\delta}{1-\tau}\right) \\
&= \log\left(\frac{r^k + \delta_0 + \frac{\gamma}{1-\omega}(1-\tau)^{1-\omega}}{1-\tau}\right) - \log\left(\frac{r^k+\delta}{1-\tau}\right) \\
&\approx \log\left(\frac{r^k + \delta_0 + \frac{\gamma}{1-\omega}(1-(1-\omega)\tau)}{r^k+\delta}\right) \\
&= \log\left(\frac{r^k + \delta_0 + \frac{\gamma}{1-\omega}}{r^k+\delta}\left(1 - \frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}}\right)\right) \\
&\approx -\frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}},
\end{aligned}$$

where constant terms are absorbed by fixed effects.

Applying the omitted variable bias formula:

$$\begin{aligned}
\text{Bias} &= \beta \cdot \frac{\text{Cov}\left(\log\left(\frac{r^k+\delta}{1-\tau}\right), -\frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}}\right)}{\text{Var}\left(\log\left(\frac{r^k+\delta}{1-\tau}\right)\right)} \\
&\approx -\beta \cdot \frac{\text{Cov}\left(\tau, \frac{\gamma\tau}{r^k+\delta_0+\frac{\gamma}{1-\omega}}\right)}{\text{Var}(\tau)} \\
&= -\beta \cdot \frac{\gamma}{r^k+\delta_0+\frac{\gamma}{1-\omega}}.
\end{aligned}$$

Since $\hat{\beta} = \beta + \text{Bias}$, rearranging yields:

$$\beta = \frac{\hat{\beta}}{1 - \frac{\gamma}{r^k+\delta_0+\frac{\gamma}{1-\omega}}}. \tag{A.13}$$

Under the functional form assumptions, the denominator equals $(1-s_m)$ evaluated at the steady state, yielding the expression in Proposition 2. The correction factor depends only on the relationship between the included and omitted terms on the right-hand side, not on the left-hand side variable $f(\cdot)$. □

B.5 Adjustment Costs and Dynamic Scoring

Throughout this appendix, the standard model (NGM) is the $s_m \to 0$ limit of the maintenance model (NGMM), with $\Psi^{NGM} = \lim_{s_m \to 0} \Psi^{NGMM}$.

The investment-capital gap established in Proposition 3 has direct implications for transition dynamics. Because the NGMM features a higher long-run investment elasticity than the NGM—investment must both expand capital and replace faster depreciation—matching the same observed short-run investment response requires slower convergence to steady state. This section proves that result in general and then quantifies the magnitude using the standard quadratic specification.

Proof of Proposition 4

The proof proceeds in three steps: (i) establish that long-run investment elasticities are higher in the NGMM, (ii) characterize transition paths under convex adjustment costs, and (iii) show that calibrating to the same short-run elasticity implies slower convergence.

Step 1: Long-run investment elasticities. In steady state, investment replaces depreciated capital: $I = \delta(m)K$. Taking logs and differentiating with respect to the tax term yields

$$\begin{aligned} \varepsilon_{I,\tau}^{LR,NGM} &= \varepsilon_{K,\tau} \\ \varepsilon_{I,\tau}^{LR,NGMM} &= \varepsilon_{\delta,\tau} + \varepsilon_{K,\tau} \end{aligned}$$

where I use the fact that δ is fixed in the NGM. By Proposition 3, $\varepsilon_{\delta,\tau} > 0$ when maintenance demand is elastic, so $\varepsilon_{I,\tau}^{LR,NGMM} > \varepsilon_{I,\tau}^{LR,NGM}$.

Step 2: Transition paths. With smooth, strictly convex adjustment costs, the economy converges monotonically to steady state. For the class of adjustment cost specifications commonly used in the tax literature—including quadratic costs in the investment rate—the transition path for the investment elasticity takes the form

$$\varepsilon_{I,\tau}(t) = \varepsilon_{I,\tau}^{LR}\left[1 - (1-\theta)e^{-\kappa t}\right] \tag{A.14}$$

where $\kappa > 0$ is the convergence rate and $\theta \in (0, 1)$ is the impact response as a fraction of the long-run response.[26] Both θ and κ depend on the adjustment cost parameter ϕ: higher adjustment costs reduce the impact response and slow convergence. The key property is that θ and κ are co-monotonic in ϕ—specifications with larger impact jumps also converge faster.

26. This functional form is exact for linearized dynamics around steady state with quadratic adjustment costs. More generally, any smooth convex specification generates monotonic convergence that can be approximated by (A.14) locally.

Step 3: Calibration and convergence rates. Suppose both models are calibrated to match the same observed short-run investment elasticity $\hat{\varepsilon}_I^{SR}$ at horizon T:

$$\hat{\varepsilon}_I^{SR} = \varepsilon_{I,\tau}^{LR,NGM}\left[1-(1-\theta^{NGM})e^{-\kappa^{NGM}T}\right]$$
$$\hat{\varepsilon}_I^{SR} = \varepsilon_{I,\tau}^{LR,NGMM}\left[1-(1-\theta^{NGMM})e^{-\kappa^{NGMM}T}\right]$$

Rearranging each expression:

$$(1-\theta)e^{-\kappa T} = 1-\frac{\hat{\varepsilon}_I^{SR}}{\varepsilon_{I,\tau}^{LR}} \tag{A.15}$$

Since $\varepsilon_{I,\tau}^{LR,NGMM} > \varepsilon_{I,\tau}^{LR,NGM}$, the right-hand side is larger for the NGMM:

$$(1-\theta^{NGMM})e^{-\kappa^{NGMM}T} > (1-\theta^{NGM})e^{-\kappa^{NGM}T} \tag{A.16}$$

The left-hand side $(1-\theta)e^{-\kappa T}$ is monotonically decreasing in both θ and κ. By co-monotonicity, a higher value of this expression implies lower θ and lower κ. Therefore $\kappa^{NGMM} < \kappa^{NGM}$: the NGMM converges more slowly. □

The intuition is straightforward. Both models start from the same initial steady state and display the same short-run investment response. But the NGMM must reach a higher long-run investment level (to cover both capital expansion and higher depreciation). With the same starting point, the same short-run response, and a higher destination, the NGMM must be traveling more slowly along the transition path.

Example with Quadratic Adjustment Costs

To quantify the difference in convergence rates, consider the standard quadratic adjustment cost specification from Summers (1981) and Koby and Wolf (2020). A firm chooses sequences of investment, maintenance, labor, and capital to maximize the present value of after-tax profits:

$$\begin{aligned}\max_{I_t,M_t,L_t,K_{t+1}} \sum_{t=0}^{\infty}\left(\frac{1}{1+r^k}\right)^t \Bigg\{&(1-\tau_t^c)\left(F(K_t,L_t)-w_tL_t-\frac{\phi}{2}\left(\frac{I_t}{K_t}-\delta(m_t)\right)^2K_t-p_t^M M_t\right)\\ &-(1-c_t-z_t\tau_t^c)p_t^I I_t\Bigg\} \quad \text{s.t.} \quad K_{t+1}=(1-\delta(m_t))K_t+I_t.\end{aligned} \tag{A.17}$$

The parameter ϕ governs how costly it is to deviate from replacement investment $\delta(m_t)K_t$. Higher ϕ means slower adjustment. Note that this specification implies maintenance adjusts instantaneously—I rely on capital adjustment costs because it allows for easier comparison with Chodorow-Reich et al. (2025), who use the same specification.

Letting q_t denote the shadow value of capital, the first-order conditions for investment and maintenance are:

$$(1-\tau_t^c)\phi\left(\frac{I_t}{K_t}-\delta(m_t)\right)=q_t-(1-c_t-z_t\tau_t^c)p_t^I \tag{A.18}$$

$$-\delta'(m_t)=\frac{1-\tau_t^c}{1-c_t-z_t\tau_t^c}\frac{p_t^M}{p_t^I}. \tag{A.19}$$

The maintenance condition is static and identical to the model without adjustment costs. The investment condition shows that the gap between the investment rate and the depreciation rate is proportional to the gap between the shadow value q_t and the after-tax purchase price.

Define $i \equiv I_t/K_t$. Rearranging the investment FOC:

$$i=\frac{q-(1-c-z\tau^c)p^I}{(1-\tau^c)\phi}+\delta(m). \tag{A.20}$$

In steady state, $i=\delta(m)$, so the investment rate equals the depreciation rate. Taking derivatives and expressing in elasticity form:

$$|\varepsilon_{I,\tau}|\equiv\left|\frac{\partial\ln i}{\partial\ln q}\right|=\frac{(1-c-z\tau^c)p^I}{i(1-\tau^c)\phi}+|\varepsilon_{\delta,\tau}| \tag{A.21}$$

where $|\varepsilon_{\delta,\tau}|\equiv|\frac{\partial\ln\delta}{\partial\ln q}|$ is the elasticity of depreciation with respect to the shadow value of capital. The key insight is that the observed investment elasticity has two components: the direct effect through the adjustment cost channel, and the indirect effect through endogenous depreciation. Only the first component represents the adjustment friction; the second is a consequence of elastic maintenance demand.

Solving for the adjustment cost parameter, evaluated locally around a zero marginal tax rate ($(1-c-z\tau^c)p^I=q\approx 1$) and using the steady-state condition $i=\delta(m)$:

$$\phi^{NGMM}=\frac{1}{\delta(m)}\cdot\frac{1}{|\hat{\varepsilon}_I|-|\hat{\varepsilon}_\delta|}. \tag{A.22}$$

When maintenance demand is perfectly inelastic, $|\varepsilon_{\delta,\tau}|=0$ and this collapses to

$$\phi^{NGM}=\frac{1}{\delta}\cdot\frac{1}{|\hat{\varepsilon}_I|}. \tag{A.23}$$

Since $|\hat{\varepsilon}_\delta|>0$ whenever maintenance demand is elastic, we have $\phi^{NGMM}>\phi^{NGM}$: the maintenance model requires larger adjustment costs to rationalize the same observed investment response.

To quantify the difference, suppose the empirical investment elasticity is $|\hat{\varepsilon}_I| = 5$ (consistent with estimates in Section 7.3) and the depreciation elasticity is $|\hat{\varepsilon}_\delta| = 1.5$ (implied by the maintenance demand estimates). Then:

$$\frac{\phi^{NGMM}}{\phi^{NGM}} = \frac{|\hat{\varepsilon}_I|}{|\hat{\varepsilon}_I| - |\hat{\varepsilon}_\delta|} = \frac{5}{5 - 1.5} \approx 1.43.$$

Adjustment costs are roughly 40% larger in the maintenance model. In the calibrated model of Section 7.3, this translates to meaningfully slower convergence: at the ten-year horizon, the NGMM capital stock reaches only about 75% of its steady-state change, compared to roughly 85% in the NGM. The combination of a smaller destination and a slower path produces dynamic scores substantially closer to static scores than conventional models suggest.

Implications for Dynamic Scoring

Slower convergence to a smaller steady state has direct implications for scoring tax reform. By statute, the Congressional Budget Office (CBO) scores tax reforms over ten-year windows, computing both a static score (assuming no behavioral changes) and a dynamic score (accounting for revenue feedback from growth effects). Since model convergence typically takes far longer than ten years, the transition path critically affects dynamic scores.

Corollary 2 (Dynamic Scoring)**.** *Consider a permanent tax cut scored over a ten-year window. Because the NGMM features slower convergence and a smaller steady-state capital stock than the NGM, the NGMM generates strictly less additional output and tax revenue over the scoring window. The dynamic score is therefore closer to the static score.*

This establishes that the maintenance channel substantially reduces estimated revenue feedback from investment incentives. Although economists emphasize the importance of dynamic scoring (Barro and Furman 2018; Elmendorf, Hubbard, and Williams 2024), incorporating the maintenance margin produces revenue projections closer to static scoring.

C Data

C.1 R-1 Data Construction

Data Sources and Digitization

Class I freight railroads must file annual R-1 reports with the Surface Transportation Board (STB) under 49 CFR §1241. These reports follow the Uniform System of Accounts for Railroad Companies, which differs meaningfully from Generally Accepted Accounting Principles (GAAP). For example, railroads often use composite depreciation rates—group rates applied to classes of similar assets—which must be approved ex ante by the STB.

I hand-collected and digitized R-1 reports for all Class I railroads from 1999-2023. Reports are available directly from the STB website for years after 2012. For earlier years (1999-2011), I used Amazon Textract to extract the relevant data from PDF scans of physical filings. Each report is independently audited by major accounting firms (e.g., KPMG, PwC, Deloitte) and then reviewed by the STB, providing high data quality.

The reports contain dozens of detailed schedules. For this paper, I extract data from:

- **Schedule 410:** All components of maintenance expenditures come from Line 202 (Locomotives) and Line 221 (Freight Cars). This schedule also breaks down maintenance costs by materials, labor, and purchased services.

- **Schedules 330 and 335:** Investment expenditures and capital stocks for locomotives and freight cars

- **Schedule 702:** Miles of track by state (used for constructing geographic exposure weights)

- **Schedule 710:** Detailed capital inventories by asset type

- **Wage Form A&B:** Hourly wage rates for maintenance workers by occupation and firm

Sample Construction

My sample includes seven Class I railroads:

1. Burlington Northern & Santa Fe Railway (BNSF)

2. CSX Transportation

3. Norfolk Southern Railway (NS)

4. Union Pacific Railroad (UP)

5. Kansas City Southern Railway (KCS)

6. Soo Line Railroad (SOO)

7. Grand Trunk Western Railroad (GT)

These firms are geographically dispersed. Burlington Northern and Union Pacific dominate the western United States, with extensive networks covering the Great Plains and West Coast. CSX and Norfolk Southern operate primarily on the eastern seaboard and in the South. The Soo

Line, Kansas City Southern, and Grand Trunk (operated by Canadian National Railway) have networks concentrated in the Midwest, with Kansas City Southern also serving the Southwest and Mexico.

The industry was highly fragmented prior to the Staggers Rail Act of 1980, which deregulated much of the industry. Throughout the 1980s and 1990s, extensive consolidation occurred through mergers and acquisitions. By the late 1990s, the industry had stabilized into the current seven-firm structure. I begin my sample in 1999 because: (1) the industry structure was stable by then, (2) some data elements in Schedule 410 are unavailable or inconsistent before this period, and (3) maintenance behavior may differ systematically during periods of anticipated merger activity. I end the sample in 2023 because Canadian Pacific Railway formally took control of both the Soo Line and Kansas City Southern by 2024, fundamentally altering the competitive structure. Including post-merger years would conflate firm-level maintenance decisions with merger-related reorganization.

Institutional Background: The R-1 Reporting Requirement. The R-1 reporting requirement originates from the Interstate Commerce Commission (ICC), the predecessor to the STB. Prior to the 1990s, the ICC extensively used R-1 reports to regulate rate-setting through cost-of-service regulation. While the STB still maintains regulatory oversight and occasionally intervenes in rate disputes, its role has diminished substantially since deregulation. The detailed reporting requirements remain in place primarily for regulatory monitoring and dispute resolution, though the data are rarely used for academic research.

Maintenance Rate Construction

The primary maintenance rate is:

$$m_{i,j,t} = \frac{\text{Maintenance Expenditures}_{i,j,t}}{\text{Beginning Book Capital}_{i,j,t}}, \tag{A.24}$$

where i indexes firms, j indexes asset types (locomotives or freight cars), and t indexes years.

Maintenance expenditures come directly from Schedule 410, Line 202 for locomotives and Line 221 for freight cars. These lines capture all repair and maintenance activities and are broken down into whether the expenditures were for labor, materials, or external services. Beginning book capital is the prior year's ending value from Schedules 330 and 335. This timing ensures maintenance in year t responds to the capital stock at the start of year t, consistent with the theoretical model. I construct three alternative measures to check robustness:

Physical Maintenance Rate. To control for inflation and asset quality changes, I construct:

$$m_{i,j,t}^{\text{phys}} = \begin{cases} \frac{\text{Maintenance Expenditures}_{i,j,t}}{\text{Total Horsepower}_{i,j,t}} & \text{if } j = \text{locomotives} \\ \frac{\text{Maintenance Expenditures}_{i,j,t}}{\text{Total Freight Ton Capacity}_{i,j,t}} & \text{if } j = \text{freight cars} \end{cases} \tag{A.25}$$

Horsepower and freight ton capacity come from Schedule 710, which reports quantities and physical characteristics of equipment.

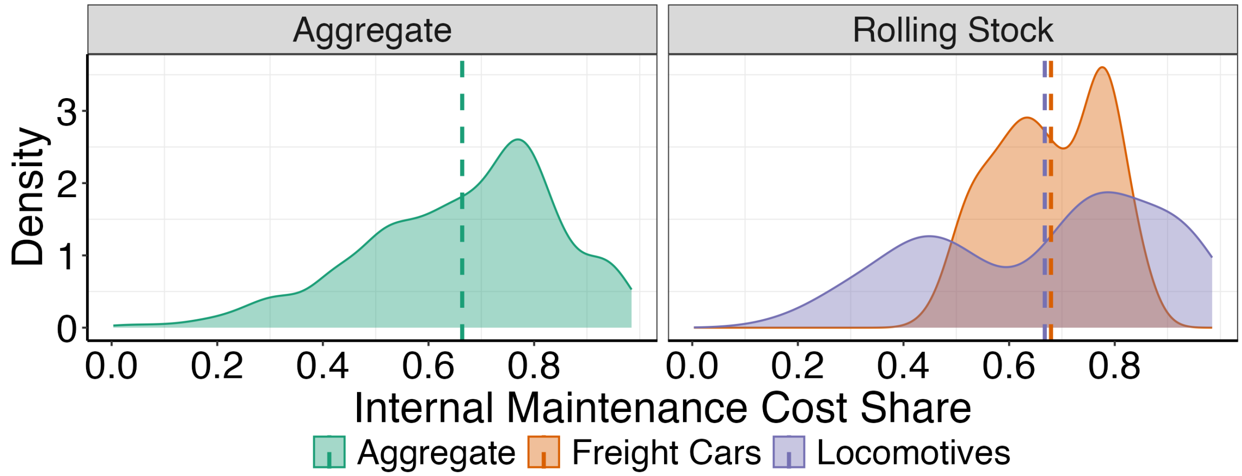
Internal vs. External Maintenance. Schedule 410 breaks down maintenance costs into three components:

- Materials costs (e.g., replacement parts, lubricants, consumables)
- Labor costs (wages and benefits for railroad employees performing maintenance)
- Purchased services (maintenance contracted to outside vendors)

I define **internal maintenance** as the sum of materials and labor costs, and **external maintenance** as purchased services. This distinction is crucial for identification. Internal maintenance costs vary with local labor market conditions and firm-specific wage policies, while external maintenance costs reflect national market prices for specialized services. If maintenance demand is elastic, the adjustment will occur primarily through the internal margin in the short run, as external maintenance often involves predetermined contracts.

There is significant variability across firms in their internal maintenance practices. On average, approximately 65% of total maintenance is performed internally. However, this share varies considerably: some firms like Norfolk Southern perform nearly all locomotive maintenance in-house, while similarly sized competitors like CSX outsource around 70% of locomotive maintenance. Freight car maintenance tends to have less variability, perhaps because locomotives are more technically complex and require specialized expertise.

Figure C.1: Ratio of Internal Maintenance to Total Maintenance



Notes: The distribution of aggregate internal maintenance rates is constructed using Table 1 from the Statistics of Income, which encompasses all firms regardless of legal type, together with input-output tables from the BEA on purchases of equipment repairs from NAICS code 811 except housing services. For 2007, 2012, and 2017, I subtract payments to NAICS 811 from total maintenance expenditures after applying a labor cost correction. The distribution of the share of internal railroad equipment maintenance comes from dividing internal maintenance expenditures by total maintenance expenditures for all Class I freight railroads for locomotives and freight cars.

Relative Price Construction

The after-tax relative price is:

$$P_{i,j,t} = \frac{p^M_{i,j,t}}{p^I_{j,t}} \frac{1-\tau_{i,t}}{1-\tau_{i,t} z_t}. \tag{A.26}$$

I construct each component as follows:

Investment Price ($p^I_{j,t}$). The investment price varies by asset type but not by firm. I use BLS Producer Price Indices:

Freight Cars: Through 2016, I use BLS series PCU3365103365103Z ("Producer Price Index by Industry: Railroad Rolling Stock Manufacturing: Passenger and Freight Train Cars, New (Excluding Parts)"). For subsequent years, I splice this series with WPU14440102 ("Wholesale Price Index: Passenger and Freight Train Cars, New (Excluding Parts)"). The two series have a correlation of 0.98 in the overlapping period.

Locomotives: Through 2018, I use BLS series WDU1441 ("Wholesale Price Index: Locomotives"). The BLS discontinued this series in 2019 without a direct replacement. For 2019-2023, I construct a locomotive-specific price index by decomposing the broader rolling stock price index (WPU144) using investment shares from the R-1 data. Specifically, I calculate the locomotive share of total rolling stock investment for each year and construct a Tornqvist index that separates locomotive price movements from freight car price movements.

Maintenance Price ($p^M_{i,j,t}$). The pre-tax maintenance price is a firm-asset-time specific weighted average of the two internal input prices, labor and materials:

$$p^M_{i,j,t} = \alpha^{\text{labor}}_{i,j,t-1} \cdot w_{i,t} + \alpha^{\text{materials}}_{i,j,t-1} \cdot p^{\text{mat}}_t, \tag{A.27}$$

where:

- $\alpha^{\text{labor}}_{i,j,t-1}$ and $\alpha^{\text{materials}}_{i,j,t-1}$ are lagged labor and materials cost shares from Schedule 410, normalized to sum to one. I use lagged shares to avoid mechanical correlation between prices and quantities.
- $w_{i,t}$ is the firm-specific average hourly wage for rolling stock maintenance workers from Wage Form A&B filed with the STB. This form reports detailed wage information by occupation and skill level.
- p^{mat}_t is BLS series PCU33651033651054 ("Producer Price Index by Industry: Railroad Rolling Stock Manufacturing: Railway Maintenance of Way Equipment and Parts, Parts for All Railcars, and Other Railway Vehicles").

Tax Parameters. I construct firm-specific statutory tax rates ($\tau_{i,t}$) as:

$$\tau_{i,t} = \tau^{\text{federal}}_t + \tau^{\text{state}}_{i,t} - \tau^{\text{federal}}_t \cdot \tau^{\text{state}}_{i,t}, \tag{A.28}$$

where τ^{federal}_t is the federal corporate tax rate (35% before 2018, 21% thereafter) and $\tau^{\text{state}}_{i,t}$ is a revenue-weighted average state corporate tax rate.

The state component is calculated using the geographic distribution of each railroad's operations. Schedule 702 reports track miles by state for each firm. I use these track miles as weights to construct a firm-specific exposure to state corporate tax rates:

$$\tau^{\text{state}}_{i,t} = \sum_s \left(\frac{\text{Track Miles}_{i,s,t}}{\text{Total Track Miles}_{i,t}} \right) \cdot \tau^{\text{state}}_{s,t}, \tag{A.29}$$

where $\tau^{\text{state}}_{s,t}$ is the statutory corporate tax rate in state s at time t. I obtain state tax rates by extending the dataset of Suárez Serrato and Zidar (2016) through 2023.

The present value of depreciation allowances (z_t) varies over time due to changes in bonus depreciation and MACRS schedules. Following House and Shapiro (2008), I calculate:

- Both locomotives and freight cars are classified as MACRS 7-year property
- Bonus depreciation rates
- Discount rate: $r^k = 0.0713$ (industry average for NAICS 48 from Gormsen and Huber (2022))

Because the IRS places both asset types in the same depreciation class, z_t does not vary between locomotives and freight cars. However, there is firm-level variation in the effective tax rate through the state tax component.

Additional Control Variables

Local GDP Exposure. I construct a firm-specific measure of exposure to local demand shocks using:

$$\Delta \log Y_{i,t} = \sum_s \left(\frac{\text{Track Miles}_{i,s,t}}{\text{Total Track Miles}_{i,t}} \right) \cdot \Delta \log \text{GDP}_{s,t}, \tag{A.30}$$

where $\Delta \log \text{GDP}_{s,t}$ is the log change in gross state product from the Bureau of Economic Analysis. This measure captures how aggregate demand shocks in a railroad's service territory might affect maintenance decisions.

Local Wage Index. I construct a firm-specific maintenance wage index using Bureau of Labor Statistics data on wages by state in the Installation, Maintenance, and Repair Occupation (SOC Code 49-0000):

$$W_{i,t} = \sum_s \left(\frac{\text{Track Miles}_{i,s,t}}{\text{Total Track Miles}_{i,t}} \right) \cdot \frac{w_{s,t}}{w_{s,1999}}, \tag{A.31}$$

where $w_{s,t}$ is the average wage in state s at time t, normalized to 1999 = 1. This provides an alternative measure of maintenance labor costs that is not firm-specific and can serve as an instrument for firm-level wage variation.

Maintenance Share of User Cost. Following the definition in Section 3, I calculate:

$$s_{m,i,j,t} = \frac{(1 - \tau_{i,t}) \cdot p^M_{i,j,t} \cdot m_{i,j,t}}{(1 - \tau_{i,t} z_t) \cdot p^I_{j,t} \cdot (r^k + \delta_j) + (1 - \tau_{i,t}) \cdot p^M_{i,j,t} \cdot m_{i,j,t}}. \tag{A.32}$$

The depreciation rate δ_j is assumed constant across firms but varies by asset type:

- Locomotives: $\delta = 0.04$ (4% annual depreciation)
- Freight cars: $\delta = 0.03$ (3% annual depreciation)

These rates are industry standards for freight railroads and are consistent with the depreciation rates railroads report to the STB for composite depreciation calculations. The discount rate is $r^k = 0.0713$ as noted above.

Summary Statistics

Tables C.1 and C.2 present summary statistics for the R-1 sample, separately for locomotives and freight cars.

Table C.1: Summary Statistics: R-1 Data (Locomotives)

Variable	Mean	10th Pctl	Median	90th Pctl	N
Year	2011.19	2001	2011	2021	172
$m_{i,j,t}$ (Total)	0.17	0.07	0.14	0.30	172
$m_{i,j,t}$ (Internal)	0.11	0.04	0.09	0.19	172
$m_{i,j,t}$ (External)	0.06	0.00	0.05	0.11	172
$m_{i,j,t}$ (Physical)	0.03	0.02	0.02	0.04	172
$\log M_{i,j,t}$	11.98	10.35	12.36	13.41	172
$\log I_{i,j,t}$	11.56	9.40	11.94	13.32	172
Investment Rate	0.15	0.02	0.10	0.28	172
$P_{i,j,t}$	1.11	1.00	1.10	1.25	172
Capital Age	1.48	1.24	1.51	1.69	172
Local GDP Exposure	1.99	-0.21	2.13	4.09	172
$z_{i,j,t}$	0.56	0.36	0.50	0.83	172
Labor Cost Share (lag)	0.41	0.29	0.37	0.57	172
Local Wage Index	1.33	1.08	1.32	1.64	172
Effective Tax Rate	1.04	1.00	1.04	1.09	172
Maintenance Share (s_m)	0.59	0.45	0.59	0.74	172

Notes: Sample covers 7 firms over 25 years (1999-2023), with some missing observations. Maintenance and investment rates are scaled as shares of beginning-of-period book capital. Capital Age is the ratio of gross to net book capital.

Table C.2: Summary Statistics: R-1 Data (Freight Cars)

Variable	Mean	10th Pctl	Median	90th Pctl	N
Year	2011.19	2001	2011	2021	172
$m_{i,j,t}$ (Total)	0.22	0.08	0.16	0.45	172
$m_{i,j,t}$ (Internal)	0.15	0.05	0.10	0.34	172
$m_{i,j,t}$ (External)	0.07	0.02	0.05	0.12	172
$m_{i,j,t}$ (Physical)	0.03	0.02	0.03	0.06	172
$\log M_{i,j,t}$	11.70	10.35	11.92	13.03	172
$\log I_{i,j,t}$	10.33	8.12	10.72	12.24	172
Investment Rate	0.08	0.00	0.05	0.19	172
$P_{i,j,t}$	0.90	0.77	0.88	1.04	172
Capital Age	1.59	1.18	1.62	1.93	172
Local GDP Exposure	1.99	-0.21	2.13	4.09	172
$z_{i,j,t}$	0.52	0.36	0.53	0.67	172
Labor Cost Share (lag)	0.39	0.26	0.39	0.51	172
Local Wage Index	1.33	1.08	1.32	1.64	172
Effective Tax Rate	1.04	1.00	1.04	1.09	172
Maintenance Share (s_m)	0.60	0.41	0.60	0.78	172

Notes: Sample covers 7 firms over 25 years (1999-2023), with some missing observations. Maintenance and investment rates are scaled as shares of beginning-of-period book capital. Capital Age is the ratio of gross to net book capital.

C.2 SOI Data Construction

Data Sources and Sample Construction

The Statistics of Income (SOI) is published annually by the Internal Revenue Service and aggregates corporate tax return data into industry-level samples. The SOI uses a stratified sampling procedure that oversamples large corporations and then applies weights to make the sample representative of the full corporate population. Corporations report maintenance expenditures (labeled "repairs and maintenance") and book capital as line items on their tax forms (Form 1120), which the SOI aggregates across firms within industries.

I use the SOI Corporate Sample files from 1999-2021. The data are publicly available at the industry level but not at the firm level. Each observation is an industry-year cell, with all dollar amounts representing weighted aggregates across firms in that industry. The SOI reports data at varying levels of aggregation, roughly corresponding to 2-3 digit NAICS codes. However, the number of SOI industries fluctuates over time as the IRS adjusts its classification scheme. In 2014, for example, the SOI changed from a "major" to a "minor" industry scheme.

To maintain consistency across years, I map SOI industries to Bureau of Economic Analysis (BEA) industry definitions, which are more stable over time. Specifically, I use the 49-industry classification from the BEA's Fixed Asset Tables. This mapping is not one-to-one—some SOI industries must be aggregated—but it ensures definitional consistency across my sample period. I exclude several categories, the financial industry and passthrough entities, from the analysis, focusing only on corporate filings. The resulting sample consists of 49 industries observed annually from 1999-2021, yielding up to 1,127 industry-year observations. Some industry-year cells have missing data, resulting in a final sample of approximately 1,116 observations.

I use the BEA industry definition rather than the raw SOI definition for three reasons. First, I combine SOI data with BEA data in some specifications (e.g., using BEA depreciation rates, capital composition weights, and price deflators). Using a consistent industry definition facilitates this merging. Second, the BEA definition is more aggregated than the SOI definition. This aggregation arguably better captures general equilibrium effects: when national tax policy changes, firms can reallocate capital within broadly defined industries, and the BEA aggregation level implicitly nets out this within-industry reallocation. Third, the BEA classification is stable over time, whereas the SOI classification changes periodically, creating artificial breaks in the panel.

Taxable vs. Untaxable Firms. For some analyses, I separate the sample into taxable and untaxable firms. I define **taxable firms** as those with positive net income in a given year, and **untaxable firms** as those without positive net income. The SOI reports both categories separately.

This classification is imperfect. Some firms in the "taxable" category may not actually face positive tax liability because they have accumulated net operating losses (NOLs) from prior years that they can carry forward to offset current income. Conversely, some "untaxable" firms may face some tax liability through alternative minimum tax provisions or other mechanisms. Despite these limitations, the taxable/untaxable split provides a useful approximation: taxable firms face stronger incentives to respond to tax policy changes than untaxable firms.

Variable Construction

Maintenance Rate. The maintenance rate is defined as the ratio of the maintenance and repairs line item to lagged book capital:

$$m_{i,t} = \frac{\text{Repairs and Maintenance}_{i,t}}{\text{Book Capital}_{i,t-1}}, \tag{A.33}$$

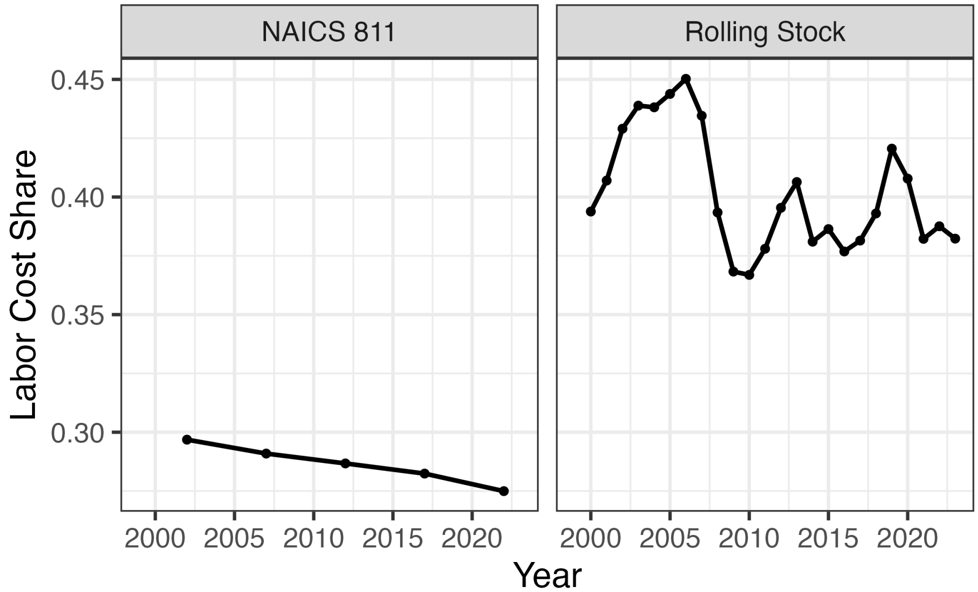
where i indexes industries and t indexes years.

The SOI maintenance measure is noisier than the R-1 measure for two important reasons:

First, labor costs are likely understated. On corporate tax forms, labor expenditures are reported as a separate line item (wages and salaries). As a result, the repairs and maintenance line primarily reflects spending on materials and external maintenance services purchased from vendors. Internal maintenance performed by the firm's own employees—which can be substantial—is not fully captured. Additionally, some materials costs may be classified under cost of goods sold rather than the maintenance line item, further understating true maintenance expenditures.

In the R-1 railroad panel, internal labor accounts for about 40 percent of total maintenance expenditures on both freight cars and locomotives (Figure 6). There is not a corresponding aggregate labor cost share, but Appendix Figure C.2 plots the ratio of labor costs to total receipts for the maintenance and repair sector using the Economic Census from 2002-2022. The labor share is consistently 30%. Thus, materials and parts are usually around 60-70% of maintenance expenditures. Given the rough agreement between the railroad data and the equipment repair sector, we could reasonably boost the typical SOI maintenance rate to between 7% and 8%, or about 2/3 as large as the usual investment rate.

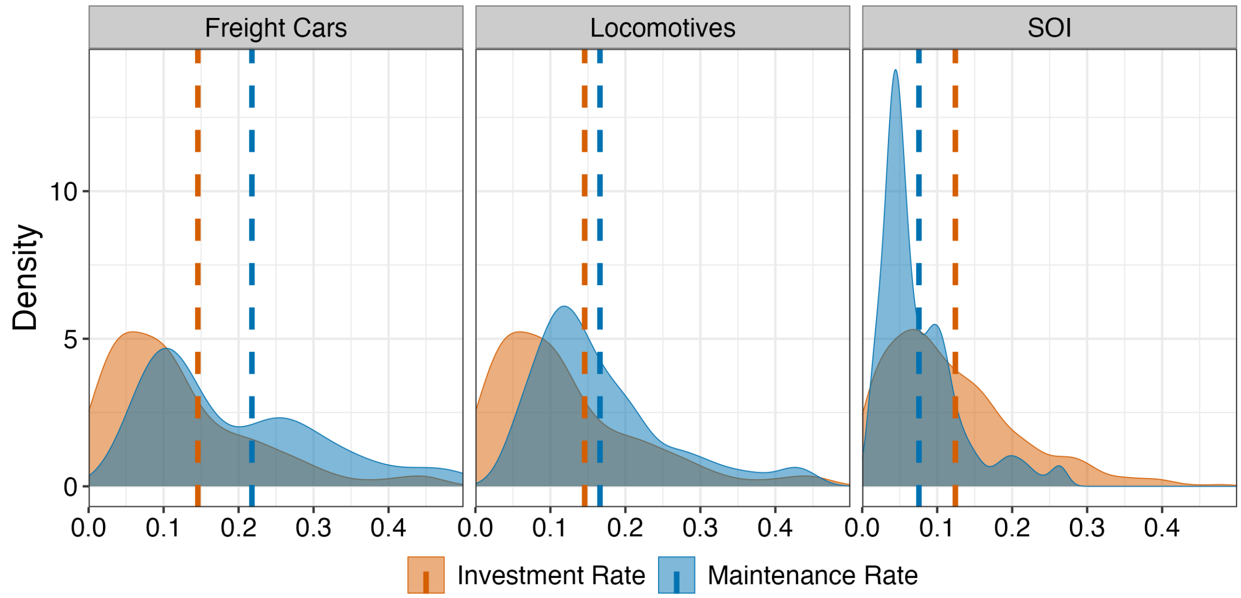
Figure C.2: Aggregate Internal Labor Cost Share



Notes: The share is computed by dividing labor costs by total internal maintenance costs. The left panel plots the ratio of labor costs to total receipts for NAICS code 811, which is the maintenance and repair sector, from 2002-2022. Each data point comes from the Economic Census. The right panel is the average for all rolling stock in the R-1 data.

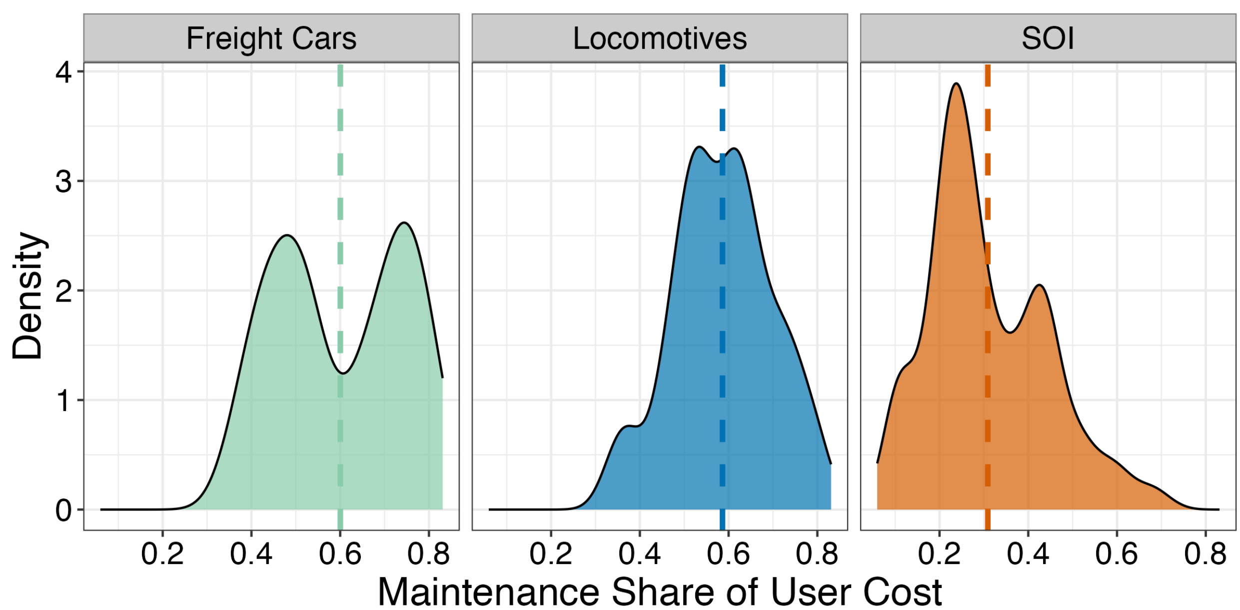
The Equipment Maintenance and Repair sector (NAICS 811) suggests that labor accounts for 30-45% of total maintenance costs. If this labor share applies broadly, then reported SOI maintenance may understate true maintenance by a factor of roughly 1.4. In robustness checks, I construct an "adjusted" maintenance rate by dividing reported maintenance by 0.65 to approximate total maintenance inclusive of labor. The elasticity results are invariant to this measure, but the *level* results in the main text are necessarily conservative because they are a lower bound.

Figure C.3: Density plots for maintenance and gross investment rates (adjusted)



Notes: Each density plot is constructed with beginning of period book capital in the denominator. The dashed lines are mean maintenance and investment rates. From left to right, the mean maintenance rates are 7.9%, 21.8%, and 16.6%. The corresponding investment rates are 13.8%, 7.9%, and 14.6%. The SOI maintenance rate is adjusted for missing labor and materials costs.

Figure C.4: Density plots for the maintenance share of user cost (adjusted)



Notes: Each density plot is constructed with beginning of period book capital in the denominator. The dashed lines are mean maintenance shares. Across the SOI, freight cars, and locomotives, the mean maintenance shares are 30%, 60%, and 58.6%. The maintenance rate is adjusted for labor costs.

Second, the capital stock denominator uses tax depreciation. The SOI constructs book capital using firms' reported accumulated depreciation, which follows tax depreciation schedules rather than economic depreciation. Tax depreciation can be accelerated (especially during bonus depreciation periods), causing book capital to understate true economic capital. This introduces measurement error in the denominator, which may attenuate estimated elasticities. Despite these measurement issues, SOI maintenance rates are comparable to those observed in aggregate Canadian *survey* data, which is constructed by Statistics Canada.

Investment Rate. The investment rate is the year-over-year change in gross property, plant, and equipment, scaled by lagged book capital:

$$\frac{I_{i,t}}{K_{i,t-1}} = \frac{\text{Gross PP\&E}_{i,t} - \text{Gross PP\&E}_{i,t-1}}{\text{Book Capital}_{i,t-1}}. \tag{A.34}$$

Capital Age. I proxy for capital age using the ratio of gross to net book capital:

$$\text{Capital Age}_{i,t} = \frac{\text{Gross Book Capital}_{i,t}}{\text{Net Book Capital}_{i,t}}. \tag{A.35}$$

Higher values indicate older capital (more accumulated depreciation relative to gross value). This measure is imperfect because it reflects tax depreciation schedules rather than true physical age, but it provides a useful control for capital vintage effects.

I winsorize maintenance rates, investment rates, and capital age at the 2nd and 98th percentiles to reduce the influence of outliers and measurement error.

Policy Wedge Construction

The key source of identification in the SOI data is variation in the policy wedge:

$$\text{Wedge}_{i,t} = \frac{1 - \tau_t}{1 - \tau_t z_{i,t}}, \tag{A.36}$$

where τ_t is the statutory federal corporate tax rate (35% before 2018, 21% thereafter) and $z_{i,t}$ is the net present value of depreciation allowances for industry i at time t.

I construct $z_{i,t}$ in three steps:

Step 1: Asset-Level Present Value of Depreciation. For every asset type j with MACRS class life T, I compute the baseline present value of depreciation allowances as:

$$z_{i,j} = \sum_{t=0}^{T} \left(\frac{1}{1 + r_i^k} \right)^t d_t, \tag{A.37}$$

where d_t is the depreciation rate in year t under the applicable MACRS schedule (e.g., 200% declining balance for 5-year property, 150% declining balance for 15-year property, straight-line for structures), and r_i^k is the industry-specific discount rate.

I obtain industry-specific discount rates from Gormsen and Huber (2022). I map their firm-level estimates to BEA industries and take a time-series average within each industry.

Step 2: Adjusting for Bonus Depreciation. Time-series variation in $z_{i,j,t}$ comes from changes in bonus depreciation policy. Let θ_t denote the bonus depreciation rate (the percentage of asset cost that can be immediately expensed). Then:

$$z_{i,j,t} = \theta_t + (1 - \theta_t) z_{i,j}, \tag{A.38}$$

where θ_t only applies to eligible assets (primarily equipment). Structures are generally ineligible for bonus depreciation. My sample ends before the TCJA bonus provisions began phasing out.

Step 3: Aggregating to Industry-Level Wedges. To construct the industry-specific $z_{i,t}$, I aggregate across the 39 detailed asset categories in the BEA's Fixed Asset Tables using capital-weighted shares:

$$z_{i,t} = \sum_{j=1}^{39} \alpha_{i,j} \cdot z_{i,j,t}, \tag{A.39}$$

where $\alpha_{i,j}$ is the share of asset type j in industry i's total capital stock. I map the 39 BEA assets to their corresponding MACRS depreciation classes using the concordance from House and Shapiro (2008).

Capital Weights and Exogeneity. A key consideration is which years to use for constructing the capital weights $\alpha_{i,j}$. Using contemporaneous weights would introduce endogeneity: if tax policy affects investment, it also affects capital composition, which would feed back into the construction of $z_{i,t}$.

To maintain exogeneity while preserving relevance, I use two different weighting periods:

- **1998-2001 weights** for all years 1999-2012: These years had no variation in tax policy (no bonus depreciation, stable MACRS schedules), so capital weights are pre-determined relative to subsequent policy changes.

- **2013-2017 weights** for years 2018-2019: These years also had stable policy (50% bonus ramping down to zero), making them pre-determined relative to the TCJA changes in 2018.

This approach balances two objectives: maintaining exogeneity (using pre-policy weights) while ensuring relevance (using weights not too far in the past from the policy change). The 2017 TCJA represented a major structural break (corporate rate cut from 35% to 21% plus 100% bonus), so using 2014-2017 weights better captures the capital composition relevant to that reform.

Industries with more equipment-intensive capital structures (higher shares of short-lived assets eligible for bonus depreciation) experience larger changes in $z_{i,t}$ when bonus depreciation rates change. Figure 3 in the main text shows substantial cross-industry variation in the wedge over time, with equipment-intensive industries like manufacturing experiencing larger policy-driven changes than structures-intensive industries like utilities or real estate.

Maintenance Share of User Cost

Following the definition in Section 3, I calculate the maintenance share of user cost as:

$$s_{m,i,t} = \frac{(1-\tau_t^c)\cdot p_t^M \cdot m_{i,t}}{(1-\tau_t^c z_{i,t})\cdot p_t^I \cdot (r_i^k+\delta_i) + (1-\tau_t^c)\cdot p_t^M \cdot m_{i,t}}, \tag{A.40}$$

where:

- r_i^k is the industry-average discount rate from Gormsen and Huber (2022)

- δ_i is a capital-weighted industry-average depreciation rate constructed from the BEA Fixed Asset Tables. For each industry, I compute $\delta_i = \sum_j \alpha_{i,j}\delta_j$, where δ_j is the BEA's estimate of economic depreciation for asset type j.

- p_t^I is the implicit price deflator for fixed investment from the BEA National Income and Product Accounts (NIPA Table 1.1.4)

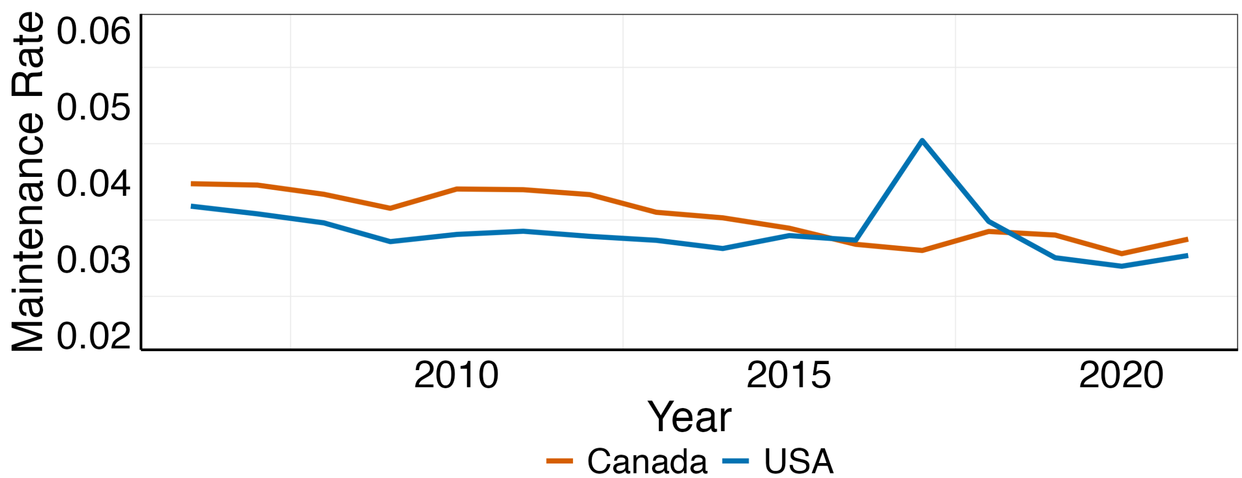
- p_t^M is the Producer Price Index for equipment maintenance and repair services (BLS series PCU81138113). For years before 2007 when this series is unavailable, I use the CPI for motor-vehicle maintenance and repair (BLS series CUSR0000SETD) as a proxy.

Table C.3 also reports an "adjusted" maintenance share that accounts for the labor understatement issue discussed above. This is calculated by dividing reported maintenance by 0.65 before computing the share, under the assumption that labor accounts for approximately 35% of total maintenance costs.

Comparison to Canadian Data

To validate the SOI maintenance rates, I compare them to aggregate maintenance data from Canada. The Canadian data compiled by Statistics Canada are from survey data on maintenance expenditures combined with an estimated aggregate economic capital stock. Figure C.5 plots the aggregate maintenance rate (total maintenance divided by total book capital) for both the U.S. SOI sample and Canadian data.

Figure C.5: American vs. Canadian Maintenance Rates



Notes: This figure plots the maintenance rate (net of government capital and maintenance) from Statistics Canada against the SOI maintenance rate.

The two series track each other closely, with mean maintenance rates of 4.8% (U.S.) and 5.1% (Canada). Both exhibit similar cyclical patterns, rising during recessions (when investment falls but maintenance continues) and declining during booms. This similarity suggests that despite measurement issues, the SOI provides a reasonable proxy for economy-wide maintenance behavior.

Summary Statistics

Table C.3 presents summary statistics for the SOI sample, separately for all firms, taxable firms, and untaxable firms.

Table C.3: Summary Statistics: SOI Data

Variable	Mean	10th Pctl	Median	90th Pctl	N
All Firms					
Year	2009.95	2001	2010	2019	1116
$m_{i,t}$	0.05	0.02	0.04	0.09	1116
Investment Rate	0.06	-0.14	0.06	0.23	1116
Wedge	0.83	0.76	0.82	0.92	1116
Maintenance Share (s_m)	0.23	0.10	0.20	0.38	1116
s_m (Adjusted)	0.31	0.15	0.28	0.49	1116
Capital Age	2.21	1.66	2.15	2.80	1116
Taxable Firms					
Year	2009.04	2000	2009	2019	1114
$m_{i,t}$	0.05	0.02	0.04	0.10	1114
Investment Rate	0.11	-0.47	0.05	0.61	1114
Wedge	0.83	0.76	0.82	0.92	1114
Maintenance Share (s_m)	0.24	0.10	0.21	0.42	1114
s_m (Adjusted)	0.32	0.14	0.29	0.52	1114
Capital Age	2.25	1.69	2.19	2.93	1114
Untaxable Firms					
Year	2009.05	2000	2009	2019	1113
$m_{i,t}$	0.05	0.01	0.04	0.09	1113
Investment Rate	0.13	-0.40	0.02	0.83	1113
Wedge	0.83	0.76	0.82	0.92	1113
Maintenance Share (s_m)	0.24	0.08	0.21	0.44	1113
s_m (Adjusted)	0.31	0.11	0.29	0.55	1113
Capital Age	2.12	1.52	2.06	2.75	1113

Notes: Sample covers 49 industries over 23 years (1999-2021). Maintenance and investment rates are winsorized at 2nd and 98th percentiles. "Adjusted" maintenance share divides reported maintenance by 0.65 to approximate total maintenance inclusive of labor costs. Wedge is $\frac{1-\tau_t}{1-\tau_t z_{i,t}}$. Capital Age is the ratio of gross to net book capital.

D Empirical Robustness and Identification Details

This appendix provides detailed validation of identification strategies, robustness checks, and supplementary analyses for the maintenance demand estimates in Section 6.

D.1 R-1 Estimates

Identifying assumption. Identification requires that the instrument affect maintenance only through the relative price—the exclusion restriction. The state wage indices $W_{s,t}$ (BLS Installation, Maintenance, and Repair occupations, SOC 49-0000) reflect broad labor-market conditions: tightness, cost of living, unionization, and prevailing-wage rules. Railroad maintenance facilities are geographically dispersed by network structure and historical infrastructure. Union Pacific's major facilities sit in Omaha (its historical headquarters), North Platte (the system's geographic midpoint), and division points set by crew-change and fueling needs—locations established, in many cases, in the nineteenth century and not chosen to minimize current labor costs. The exposure-weighted wage is therefore plausibly orthogonal to firm-specific maintenance demand, and the internal-versus-external split in the main text tests the exclusion restriction directly.

First stage and balance. The first stage is strong and correctly signed: across instruments the effective F runs from about 10 to 15, and the instrument raises the relative price as predicted (Table D.1). Two balance tests support the design. Figure D.1 regresses the instrument on lagged maintenance rates—no lag is significant, so firms do not anticipate wage shocks. Figure D.2 regresses the labor-share component on lagged maintenance, prices, and capital age—none is significant, so the shares are not correlated with pre-period determinants of maintenance.

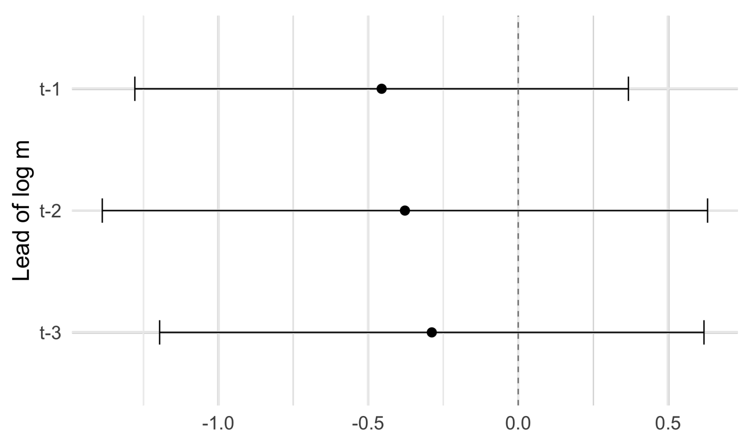
Table D.1: First-stage regressions

Instrument	Coef. on $Z_{i,j,t}$	Cluster SE	Effective F	N
Baseline (no controls)	0.165^{***}	(0.049)	11.54	340
Baseline (all controls)	0.168^{***}	(0.047)	12.63	340
Three-year-lagged shares	0.156^{***}	(0.041)	14.68	326
National wage	0.001^{***}	(0.000)	10.68	314

* p < 0.1, ** p < 0.05, *** p < 0.01

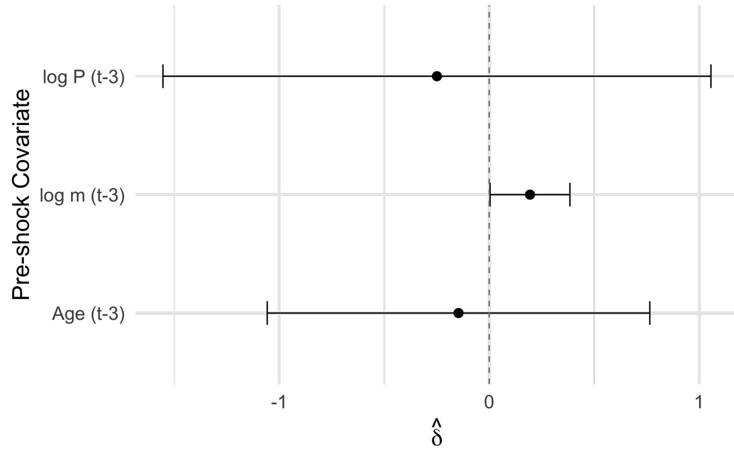
Notes: First stage of (20): the log relative maintenance price regressed on the instrument $Z_{i,j,t}$ of (21), with firm-asset and year fixed effects and standard errors clustered on the seven firms. The controlled rows add capital age and GDP exposure. Instruments are the baseline two-year-lagged labor share, the three-year-lagged labor share, and a national (rather than state-mix) wage shifter. The effective F is from Montiel-Olea and Pflueger (2013).

Figure D.1: Lead tests of the log maintenance rate on $Z_{i,j,t}$



Notes: Coefficients from regressing a pre-shock maintenance rate on the instrument $Z_{i,j,t}$ of (21), controlling for firm trends, year fixed effects, and firm-asset fixed effects. Standard errors clustered by firm.

Figure D.2: Balance tests on pre-shock outcomes



Notes: Coefficients from regressing each pre-shock outcome on the instrument $Z_{i,j,t}$ of (21), controlling for firm-asset and year fixed effects and firm trends. Standard errors clustered by firm.

Inference at the shift-share frontier. Because the instrument aggregates state wages with firm exposure weights, it has a shift-share structure. Table D.2 reports the inference appropriate to such designs: weak-IV-robust intervals, exposure-robust standard errors, and the effective number of independent wage shocks.

Table D.2: Railroad: exposure-robust and weak-IV inference

Inference	With controls ($\hat{\omega} = 4.76$)	No controls ($\hat{\omega} = 5.41$)
Firm-clustered (analytic)	[2.66, 6.85]	[3.34, 7.48]
weak-IV-robust (t_F)	[1.48, 8.04]	[2.04, 8.78]
weak-IV-robust (AR)	[3.03, 8.37]	[2.50, 7.63]
Shift-level, clustered by state	[2.09, 7.43]	[2.24, 8.58]
Shift-level, clustered by year	[2.52, 7.00]	[2.93, 7.89]
AKM exposure-robust	[-384.95, 394.47]	[-204.36, 215.18]
AKM0 (weak-IV-robust)	$(-\infty, \infty)$	$(-\infty, \infty)$
Effective number of shifts (raw)	627	
Effective number of shifts (state-cluster)	26	

Notes: Inference on the 2SLS of (20) with the instrument (21), for the headline with-controls specification and the no-controls specification. Firm-clustered rows give the conventional and weak-IV-robust (t_F, AR) intervals (Lee et al. 2022; Anderson and Rubin 1949). Shift-level rows aggregate the instrument to the state×year shock and cluster by state and by year (Borusyak, Hull, and Jaravel 2024), and AKM and AKM0 are exposure-robust intervals (Adão, Kolesár, and Morales 2019). The effective number of shifts is $1/\sum_s \bar{s}_s^2$, with $\bar{s}_s$ the mean exposure share on shift s, a property of the route-mileage shares and the lagged labor share that is identical across specifications. The with-controls column adds capital age and GDP exposure (the Table 2 specification), the no-controls column the twoway fixed effects only.

Robustness. Table D.3 re-estimates the main results without controls. Table D.4 re-estimates the elasticity across instruments—three-year-lagged versus two-year-lagged shares, and a national versus the state-mix wage shifter—and across outcomes (physical versus book maintenance). Table D.5 reports the reduced form and the implied elasticity.

Table D.3: The Railroad Maintenance Elasticity (No Controls)

	All		Internal		External	
	OLS	IV	OLS	IV	OLS	IV
$\hat{\omega}$	1.84^{***}	5.41^{***}	2.91^{**}	9.21^{***}	-0.41	0.85
	(0.55)	(1.05)	(1.15)	(2.55)	(0.47)	(3.40)
F-statistic	11.54		11.54		11.54	
t_F 95% CI	[2.04, 8.78]		[1.07, 17.35]		[-10.01, 11.70]	
AR 95% CI	[2.50, 7.63]		[0.77, 13.60]		[-6.70, 9.69]	
N	340		340		340	
Twoway FE	✓		✓		✓	
Controls						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: As in Table 2, but with no controls beyond the twoway fixed effects. Estimates of (20) for all maintenance and its in-house (Internal) and outsourced (External) components, by OLS and by 2SLS with the instrument of (21); firm-asset and year fixed effects throughout, standard errors clustered on the seven firms. $\hat{\omega}$, the F-statistic, and the weak-IV-robust intervals are defined as in Table 2.

Table D.4: Railroad: robustness of $\hat{\omega}$ across specifications

Specification	$\hat{\omega}$	Cluster SE	First-stage F
Headline (all controls)	4.76^{***}	(1.07)	12.63
No controls	5.41^{***}	(1.05)	11.54
Three-year-lagged-share instrument	6.06^{***}	(1.07)	14.68
National-wage instrument	6.77^{***}	(0.91)	10.68
Physical maintenance outcome	3.70^{**}	(1.67)	12.63

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The 2SLS elasticity of (20), re-estimated under alternative instruments (21) and outcomes. Except for the no-controls row, all add capital-age and GDP-exposure controls; all include firm-asset and year fixed effects, with standard errors clustered on the seven firms. Rows vary the labor-share weighting (three-year vs. two-year lag), the wage shifter (national vs. state-mix), and the maintenance measure (physical vs. book). The first-stage F is the effective F of Montiel-Olea and Pflueger (2013).

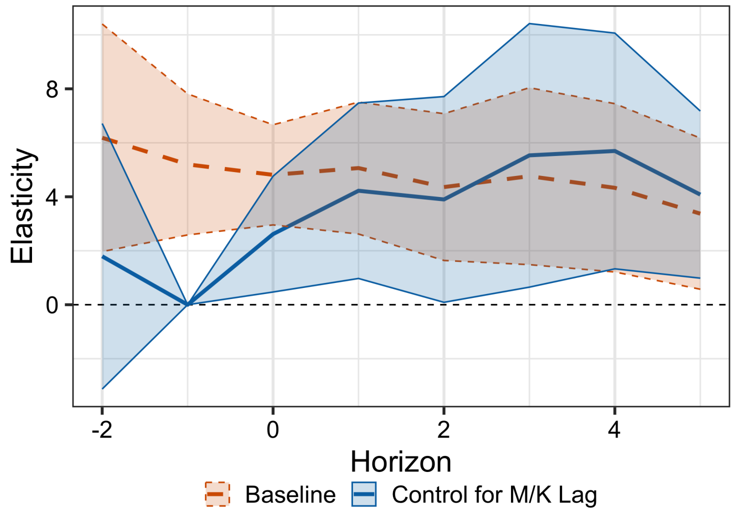
Table D.5: Railroad: reduced form and implied elasticity

	Main	Internal	External
Reduced form (M/K on exposure)	-0.801^{***}	-1.418^{***}	-0.020
	(0.194)	(0.534)	(0.627)
Implied $\hat{\omega}$ ($= -\mathrm{RF}/\mathrm{FS}$)	4.76	8.43	0.12

Notes: Each outcome regressed directly on the intensity-weighted wage exposure of (21) (capital age, GDP exposure, firmtype and year fixed effects; standard errors clustered on the seven firms). The first stage (relative price on exposure) is 0.168 (0.047), $F = 12.63$. The implied elasticity is $\hat{\omega} = -\mathrm{RF}/\mathrm{FS}$. $^{*}/^{**}/^{***}$: 10/5/1%.

Dynamics. Figure D.3 traces the local-projection response of maintenance to the relative price by horizon.

Figure D.3: Local Projection of Relative Price Shocks on the Maintenance Rate (R-1)



Notes: IV coefficients ω_h from the local projection $\log m_{i,j,t+h} = \alpha_{ij} + \lambda_t - \omega_h \log P_{i,j,t} + \text{Controls} + u_{i,j,t+h}$. The solid line controls for the lagged maintenance rate, and the dashed line does not.

D.2 SOI Robustness Checks

Specification band. Table D.6 collects the headline elasticity across alternative specifications: the trend ladder varies the industry time trends; the pre-2014 window ends the sample before the 2013 change in SOI industry sampling, which made some industries too small to report maintenance separately; the alternative capital stock replaces SOI book capital, built from tax depreciation, with BEA capital built from economic depreciation; and the leave-one-out row deletes each industry in turn. The elasticity stays positive and between 1.3 and 2.6 throughout.

Table D.6: Robustness of the SOI Maintenance Elasticity

Specification (All sample)	$\hat{\omega}$	Cluster SE
FE only (no trends)	1.83*	(1.03)
Linear NAICS trends	2.45**	(0.94)
Quadratic NAICS trends *(headline)*	2.61**	(1.01)
Industry-specific linear trends	1.33*	(0.79)
Pre-2014 window	1.43**	(0.65)
Alternative capital stock	1.50	(1.11)
Leave-one-industry-out (range)	[2.18, 3.17]	—

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The headline maintenance elasticity ($-1\times$ the coefficient on $\log(\text{wedge}_k)$ in (19), All sample, clustered on the 49 BEA industries) across trend specifications, the pre-2014 (bonus-era) window, an alternative BEA capital stock, and the 49 single-industry deletions. The elasticity is positive in every specification and economically large (the 1.3–3.2 band, which spans the leave-one-out interval), and significant at conventional levels in five of the six point specifications—the lone exception being the alternative capital-stock measure. The wide standard errors reflect the limited within-industry variation in the wedge, so the SOI alone pins down ω imprecisely.

Design diagnostics. The specification is a continuous-intensity difference-in-differences: each industry's asset-composition exposure supplies the cross-sectional dose and the national bonus-depreciation schedule supplies the timing, following Zwick and Mahon (2017). Because the schedule is national, all industries are treated simultaneously and no not-yet-treated comparison group exists, so the heterogeneity-robust difference-in-differences estimators of Callaway, Goodman-Bacon, and Sant'anna (2025) and Chaisemartin and D'Haultfœuille (2026) are not identified in this design. I therefore validate the two-way fixed-effects slope directly. Such a design must satisfy three conditions, each of which Table D.7 tests. First, the two-way fixed-effects slope must recover a well-behaved average of heterogeneous dose responses. A heterogeneous-adoption consistency test (*did_had*, Chaisemartin et al. 2026), implemented as a Yatchew (1997) nonparametric test of mean-independence of the effect from the dose, does not reject (p between 0.40 and 0.85), so the linear coefficient is consistent for the weighted-average slope. Second, exposed and unexposed industries must not have been diverging beforehand. An exposure event study finds flat pre-trends across exposure terciles (joint $F = 0.66$, $p = 0.62$), and a balance test finds no differential pre-2002 trend in exposure (-0.07, $p = 0.68$). Third, the response must scale with the dose. A within-fixed-effects dose-response check finds the maintenance response monotone in the dose ($p < 0.001$), confirming that the linear form is not masking heterogeneity. The specification passes each.

Table D.7: SOI design: identification diagnostics

Diagnostic	Result	Reading
Heterogeneity-robust consistency (*did_had*)	not rej., $p = 0.40$–0.85	TWFE consistent for the weighted-average slope
Pre-trends across exposure (event study)	$F = 0.66$ ($p = 0.62$)	not rejected
Selection on exposure (balance)	-0.07 ($p = 0.68$)	no differential pre-2002 trend
Dose-response monotonicity (within-FE bins)	increasing, $p < 0.001$	linear form not masking heterogeneity

Notes: The SOI specification is a continuous-intensity difference-in-differences, with each industry's asset-composition exposure as the dose and the national bonus-depreciation schedule as the timing, following Zwick and Mahon (2017). Because adoption is synchronized across industries (no not-yet-treated control), the heterogeneity-robust difference-in-differences estimators of Callaway, Goodman-Bacon, and Sant'anna (2025) and Chaisemartin and D'Haultfœuille (2026) are not identified here, so the two-way fixed-effects slope is validated directly. *did_had* (Chaisemartin et al. 2026) tests mean-independence of the effect from the dose with a Yatchew (1997) nonparametric test, the consistency condition for the slope. The event study tests parallel pre-trends across exposure terciles, the balance test checks for a differential pre-2002 trend in exposure, and the dose-response check confirms the response is monotone in the dose.

Investment Weights (Placebo Test). Table D.8 re-estimates equation (19) with z_i built from pre-reform investment flows rather than the capital stock. The positive maintenance elasticity disappears—insignificant for tax-paying firms and wrong-signed in the pooled sample—confirming that maintenance responds to the installed capital base, not recent investment flows.

Table D.8: The SOI Maintenance Elasticity: Investment-Weighted Wedge

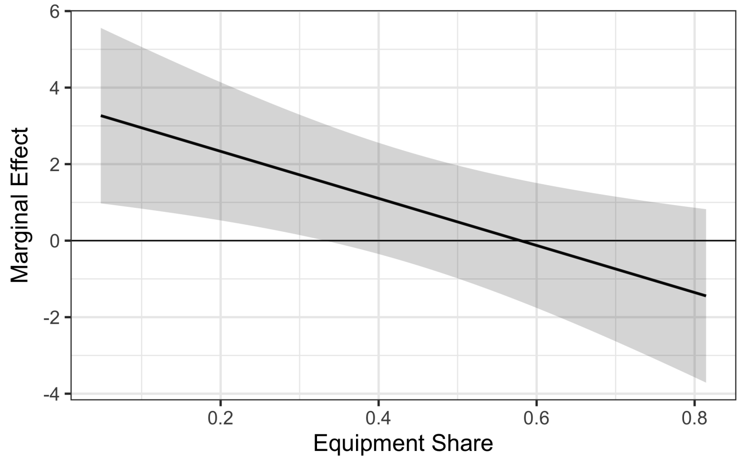
	All		Tax-paying		Untaxed	
	(1)	(2)	(3)	(4)	(5)	(6)
$\hat{\omega}$	-2.29^{*}	-2.24^{*}	-0.33	-0.15	0.45	0.19
	(1.29)	(1.23)	(1.32)	(1.30)	(1.72)	(1.70)
N (industry-years)	1,116	1,113	1,051	1,051	1,049	1,049
Industry and year FE	✓	✓	✓	✓	✓	✓
Capital age control		✓		✓		✓
Two-digit NAICS trends	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: As in Table 1, but the industry exposure z_i in (19) is built from pre-reform investment flows rather than the capital stock. All columns include two-way (industry and year) fixed effects and linear and quadratic two-digit NAICS trends, clustered by industry. Columns (2), (4), and (6) add a capital-age control; (1)–(2) use all firms, (3)–(6) split into tax-paying and untaxed firms.

No Selection by Equipment Intensity. Koby and Wolf (2020) raise concerns that bonus depreciation identification may be confounded by selection: equipment-intensive industries may have inherently different elasticities than structures-intensive industries. If true, differential exposure to bonus would be correlated with pre-existing elasticity heterogeneity, biasing estimates. To test this, I add an interaction between the equipment-capital ratio and the wedge term to equation (19). If the elasticity varies by equipment intensity, the interaction should be significant. Figure D.4 shows the interaction is small and statistically insignificant, indicating the elasticity is homogeneous across industries. This rules out selection effects.

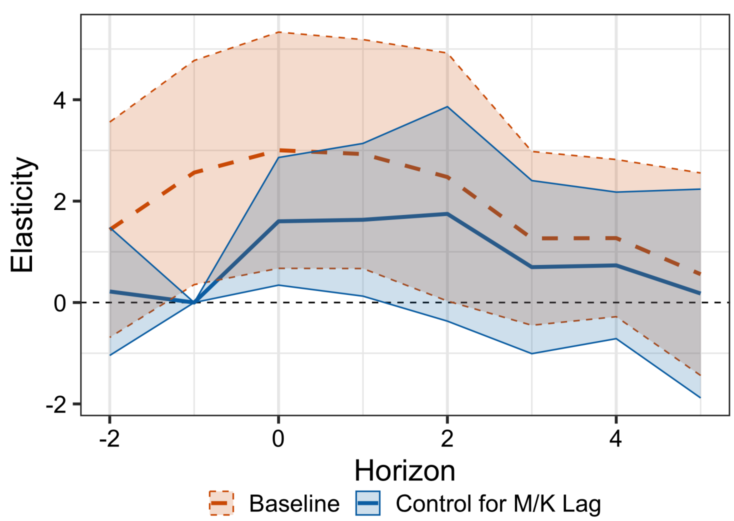
Figure D.4: Marginal Effect of Wedge on Maintenance Rate by Equipment Share



Notes: Marginal maintenance-wedge elasticity by equipment-capital ratio. The black line plots $\partial \log m / \partial \log(\text{wedge})$ against each industry's average equipment-capital ratio. Shaded bands are 95% confidence intervals.

Dynamics. Figure D.5 traces the response of maintenance to the wedge by horizon.

Figure D.5: Local Projection of Relative Price Shocks on the Maintenance Rate (SOI)



Notes: This figure plots the coefficients ω_h from the regression

$$\log m_{i,t+h} = \alpha_i + \lambda_t - \omega_h \log\left(\frac{1-\tau_t^c}{1-\tau_t^c z_{i,t}}\right) + \text{Controls} + \eta_{i,t+h}.$$

The solid path controls for the lagged maintenance rate. Both specifications control for industry trends and correspond to the SOI sample with all types of firms.

E Quantification

This appendix collects the full calibration of the quantitative model and the variants referenced in Section 7.3: the endogenous-maintenance-price closure (NGMMP), fiscal crowding out, and partial equilibrium. Table E.1 lists every assigned, estimated, and calibrated parameter.

Table E.1: Calibration of the Quantitative Model

Description	Symbol	Value	Source / target
A. Technology and aggregation			
Capital elasticity	α_K	0.245	Chodorow-Reich et al. (2025)
Labor elasticity	α_L	0.650	Chodorow-Reich et al. (2025)
Discount rate	r^k	0.060	Chodorow-Reich et al. (2025)
Sectoral output weights (c / nc)	c_c, c_{nc}	0.70 / 0.30	Chodorow-Reich et al. (2025)
Private share of output	—	0.81	NIPA
Maintenance-price wage weight	θ_w	0.50	assumed (NGMMP)
B. Tax rates (τ^c, τ^x)			
Corporate, pre-TCJA	—	0.249, 0.210	statutory
Corporate, post-TCJA	—	0.151, 0.138	Chodorow-Reich et al. (2025)
Non-corporate (held fixed)	—	0.280, 0.230	Chodorow-Reich et al. (2025)
C. Estimated parameters (standard error)			
Maintenance demand level	γ	0.0305 (0.0056)	§6
Maintenance demand elasticity	ω	2.608 (1.012)	§6
Investment elasticity (raw)	ε_I	4.27 (0.51)	Chodorow-Reich et al. (2025)
Crowding-out coefficient	—	3.20 (1.22)	Plante, Richter, and Zubairy (2026)
D. Calibrated to match targets			
Non-corporate productivity	A_{nc}	0.919	$K_c/K = 0.70$
Baseline depreciation	δ_0	0.137	$I/Y = 0.109$
E. Implied at the calibration (corporate sector)			
Steady-state depreciation	$\delta(m^*)$	0.083	
Maintenance rate	m^*	0.035	
Maintenance share of user cost	s_m	0.171	
Depreciation elasticity	ε_δ	0.91	
OVB-corrected investment elast.	$\hat{\varepsilon}_I$	5.15	$= \varepsilon_I/(1 - s_m)$
Adjustment cost, NGMM	ϕ_{NGMM}	2.83	
Adjustment cost, NGM	ϕ_{NGM}	2.01	

Notes: SOI main-spec estimates. Panels A–B are assigned, C estimated, D calibrated so the steady state matches $K_c/K = 0.70$ (Chodorow-Reich et al. 2025) and the pre-TCJA gross $I/Y = 0.109$ (NIPA). s_m is the maintenance share of user cost, and the leading-order OVB factor $1/(1 - s_m) \approx 1.21$ gives $\hat{\varepsilon}_I = 5.15$. The NGM shares the NGMM initial steady state, with $\delta_{\text{NGM}} = \delta(m^*) + m^* P = 0.116$ and ϕ_{NGM} from the uncorrected ε_I. Non-corporate: $\delta(m^*) = 0.082$, $m^* = 0.036$, $\phi_{\text{NGMM}} = 2.88$.

The baseline depreciation rate δ_0 is the one parameter calibrated with discretion, pinned to the pre-TCJA investment-output ratio. Table E.2 sweeps it across a ±50% range, re-calibrating non-corporate productivity at each node so the corporate capital share still hits 0.70. The maintenance markdown holds throughout, with the NGMM offset ratio below one at every node. Its magnitude does move with δ_0: the ten-year capital gap runs from \$170B to \$516B and the offset ratio from 0.21 to 0.84, with the calibrated values at \$249B and 0.74.[27]

Table E.2: Sensitivity of the Headline Result to the Baseline Depreciation Rate δ_0

				GE (No Crowd-Out)			Dollar (\$B)	
δ_0	% cal.	I/Y	s_m	Off. NGM	Off. NGMM	Ratio	Gr.-rev.	Cap. gap
0.068	50%	0.031	0.278	4.34	0.93	0.21	20	516
0.080	58%	0.050	0.252	4.96	2.14	0.43	47	418
0.103	75%	0.079	0.212	5.96	3.69	0.62	81	323
0.137	100%	0.109	0.171	7.08	5.25	0.74	116	249
0.171	125%	0.129	0.144	7.88	6.32	0.80	139	204
0.205	150%	0.143	0.124	8.46	7.11	0.84	157	173
0.210	153%	0.145	0.121	8.53	7.20	0.84	159	170

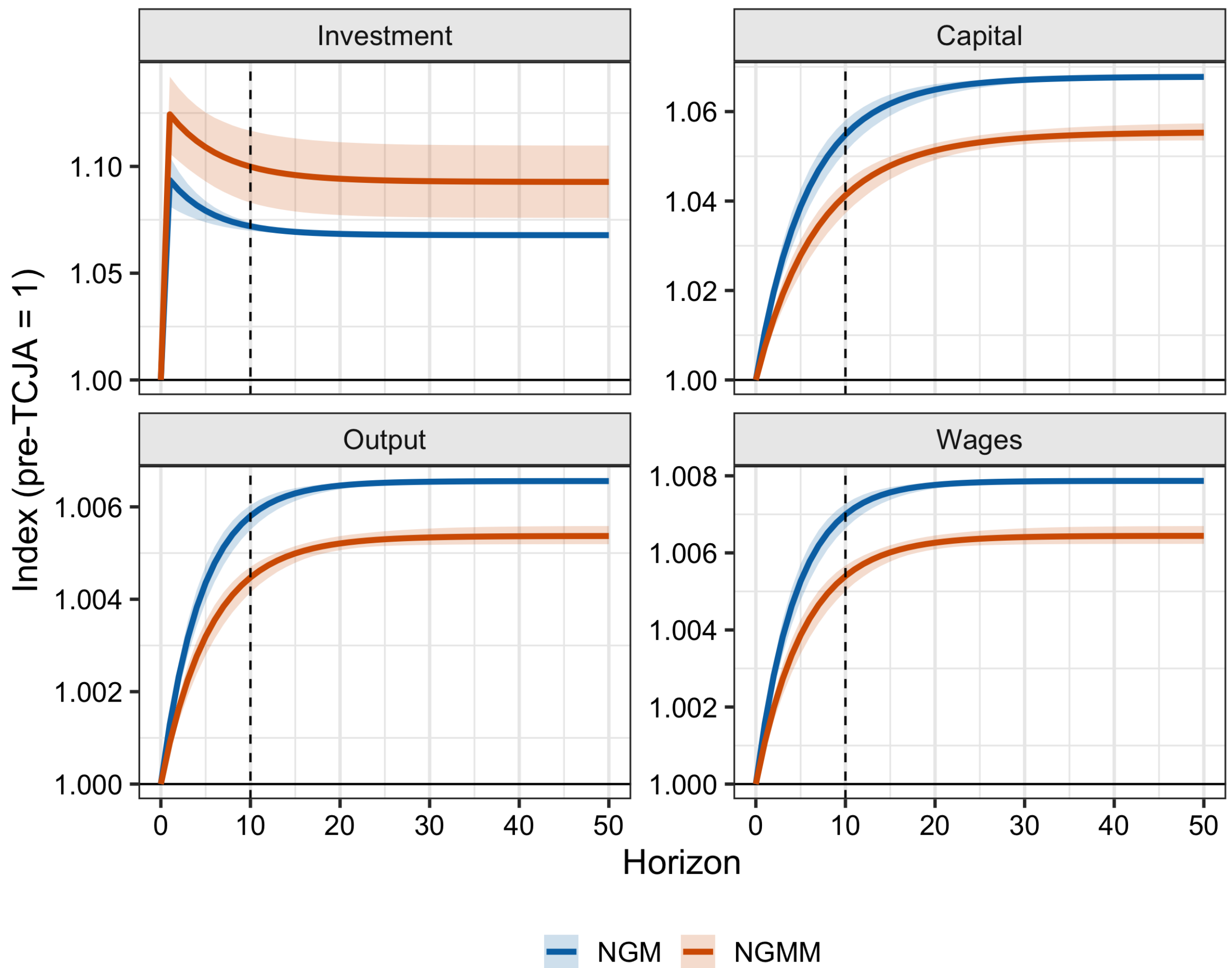
Notes: The §7.3 Monte Carlo propagates uncertainty from only two sources (the maintenance parameters γ, ω and the investment elasticity β) and holds the baseline depreciation rate δ_0, α, and the depreciation calibration fixed. This table instead sweeps the calibrated δ_0 deterministically, holding γ, ω, β and the crowd-out at their central values. At each node A_{nc} is re-calibrated to hold $K_c/K = 0.70$; the induced gross I/Y is shown (it equals the 0.109 target at the calibrated δ_0, in bold). The static cost is \$2206B; the capital gap is $(\Delta K_c^{NGM} - \Delta K_c^{NGMM})$ applied to the \$18,287.3B corporate-capital base (BEA Table 4.1, Line 17, 2017; GE no-crowd-out panel, the headline). The "Off." columns are static-score offsets (%); "Gr.-rev." is the NGMM growth-revenue = offset× static cost.

Figure E.1 plots the level paths of investment, capital, output, and wages that underlie the response ratios in Figure 10. The NGM and NGMM start from a common pre-TCJA steady state and converge to model-specific terminal levels, with the maintenance model reaching a lower

27. This is the deterministic gap at the point estimates. The \$248 billion headline is the Monte Carlo mean over the maintenance and investment-elasticity draws, within rounding of this value.

capital stock along a slower path.

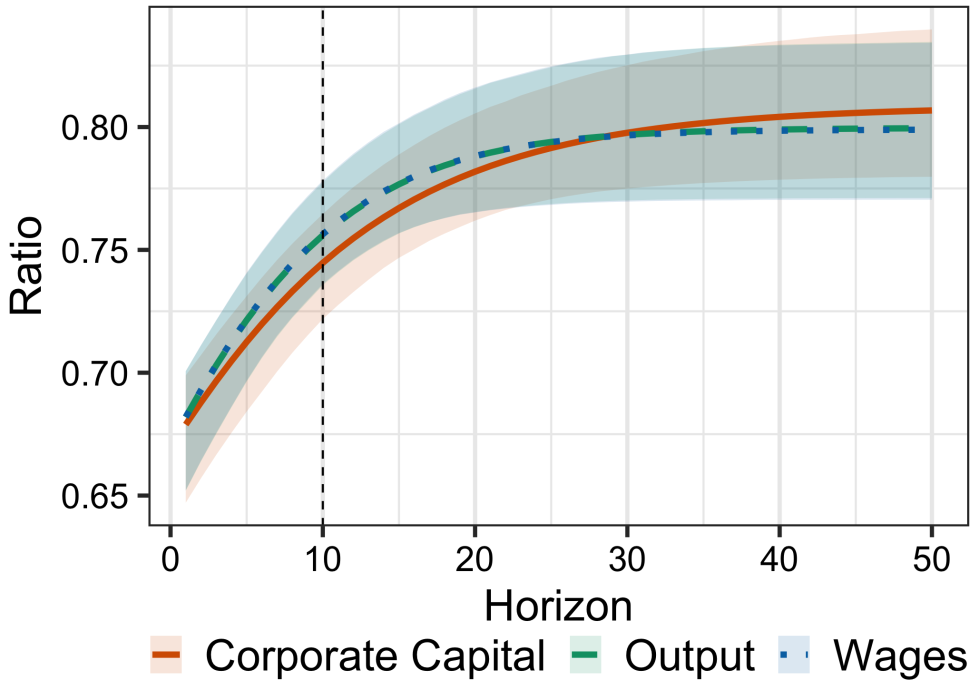
Figure E.1: Aggregate Effects of TCJA: Level Paths



Notes: Level paths (index, pre-TCJA = 1) of investment, capital, output, and wages under the NGM (blue) and NGMM (orange), each with 95% confidence intervals. The dashed vertical line marks the ten-year horizon.

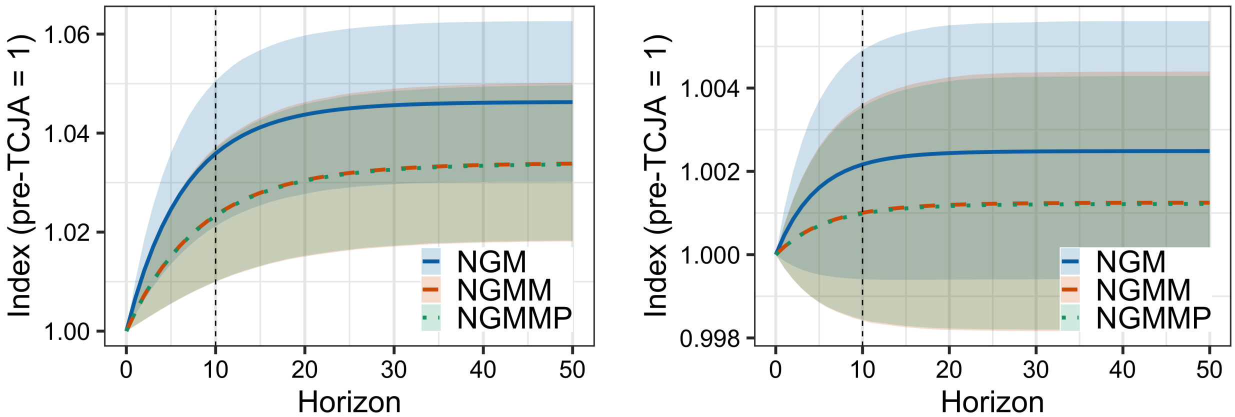
The remaining figures report the model variants. Figure E.2 adds the NGMMP closure, in which the pre-tax price of maintenance rises with the wage. Its responses are nearly identical to the baseline NGMM, confirming that wage feedback in maintenance prices has little quantitative bite. Figures E.3 and E.4 reintroduce fiscal crowding out and partial equilibrium, respectively.

Figure E.2: NGMMP-NGM Ratio of Aggregates



Notes: The NGM corporate capital, output, and wages are exactly as in the main text. The NGMMP is an extension of the baseline NGMM which specifies that the price of maintenance rises with wages. I assume that wages are half the cost of maintenance.

Figure E.3: The Effect of TCJA on Corporate Capital and Aggregate Output with Crowding Out

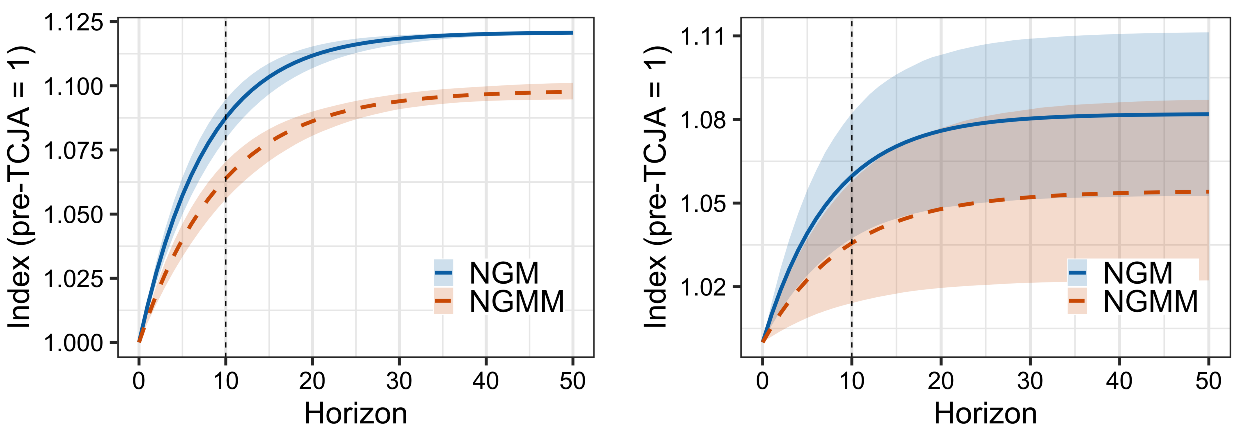


(a) Paths of Corporate Capital

(b) Paths of Output

Notes: Panel (a) shows the response of capital to the TCJA in the NGMM (orange line) and the NGM (blue line), and the NGMMP (green line), and Panel (b) plots the corresponding IRFs for aggregate output. To account for crowding out, we translate the 2027 debt-output ratio into an increase in discount rates using the static score of each model. The increase in debt-output corresponds to an increase in discount rates drawn from Plante, Richter, and Zubairy (2026). A one percentage point increase in the debt-GDP ratio corresponds to a 3.2 basis point increase in the discount rate (SE = 1.22). All lines are bootstrapped with a 95% confidence interval accounting for uncertainty in maintenance demand and crowding out.

Figure E.4: The Effect of TCJA on Corporate Capital in Partial Equilibrium



(a) Baseline

(b) Crowding Out

Notes: Panel (a) shows the response of corporate capital to the TCJA in the NGMM (orange line) and the NGM (blue line), and Panel (b) plots the corresponding IRFs with crowding out. To account for crowding out, we translate the 2027 debt-output ratio into an increase in discount rates using the static score of each model. drawn from Plante, Richter, and Zubairy (2026). A one percentage point increase in the debt-GDP ratio corresponds to a 3.2 basis point increase in the discount rate (SE = 1.22). All lines are bootstrapped with a 95% confidence interval accounting for uncertainty in maintenance demand and crowding out.

F Model Extensions

F.1 Capital Reallocation

This subsection extends the base model to heterogeneous firms which only differ in their tax status. That leads to reallocation through variation in the marginal product of capital. A fraction $\eta \in (0,1)$ of firms is taxable (Type T) and the remaining $1-\eta$ are untaxed (Type U). Variation in taxability comes from realization of an i.i.d. fixed cost $\bar{F}$, which is assumed to be sufficiently large that it exceeds profits. Untaxed firms also pay a higher cost of investment $p^I + b$, which is meant to account (in a reduced-form way) for the fact that untaxable firms often face some kind of financial constraint that hinders their ability to access new investment (Lian and Ma 2020). This formulation allows us to model the fact that investment is typically more expensive for unprofitable firms without explicitly modeling the borrowing constraint.

Each period, a firm observes its tax type $\theta \in \{T, U\}$:

- Type T (taxable) firms face corporate tax τ^c.
- Type U (untaxed) firms pay no corporate tax because they have a fixed cost $\bar{F}$. They also have a higher cost of investment $p^I + b$.

After observing its type, a firm chooses:

1. Maintenance expenditure M, with $m = \frac{M}{K}$ being the maintenance intensity.
2. Investment I (at new capital price p^I).
3. Net used–capital sales s, where $s > 0$ indicates selling a fraction of the capital and $s < 0$ indicates buying used capital. Firms pay a convex adjustment cost $G(s)$ for participating in the used capital market, which can be thought of as accounting for information frictions in a reduced–form way. For example, a firm selling a used car must furnish information about the vehicle, which is costly to do. The equilibrium used–capital price is denoted by q.

Putting together maintenance expenditures $m = M/K$, investment I, and capital sales s, the law of motion for capital (independently of tax status) is

$$K' = \Big[1 - \delta(m)\Big](1-s)K + I. \tag{A.41}$$

Because firms vary in their tax status, they also vary in their cash flows. Profitable firms have cash flows

$$\pi^T = (1-\tau^c)\Big[F(K) - p^m M + q\,s\,K - G(s)K\Big] - \big(1 - c - \tau^c z\big)p^I I,$$

where z incorporates the possibility that the firm may be untaxable in the future and hence will not always be able to take advantage of a tax depreciation allowance. Note that capital sales are taxed at rate τ^c, reflecting current policy practice. Given those cash flows, we get the following recursive formulation for a firm choosing K', s, M:

$$V^T(K) = \max_{m,s,K'} \Bigg\{ (1-\tau^c)\Big[F(K) - p^m\, mK + q\, s\, K - G(s)K\Big] \\ - \big(1 - c - \tau^c z\big) p^I \Big(K' - [1-\delta(m)](1-s)K\Big) + \frac{V^*(K')}{1+r^k} \Bigg\}, \tag{A.42}$$

where

$$V^*(K') = \eta\, V^T(K') + (1-\eta)\, V^U(K')$$

is the expected continuation value. Taxable firms therefore have the following FOCs:

$$\textbf{Investment:} \quad \frac{V^{*'}(K')}{1+r^k} = \big(1 - c - \tau^c z\big) p^I, \tag{A.43}$$

$$\textbf{Maintenance:} \quad -\delta'(m) = \frac{1-\tau^c}{1-c-\tau^c z} \frac{p^m}{p^I\,(1-s)}, \tag{A.44}$$

$$\textbf{Sales:} \quad (1-\tau^c)\Big[q - G'(s)\Big] = \big(1 - c - \tau^c z\big) p^I\, [1-\delta(m)]. \tag{A.45}$$

On the other hand, untaxed type U firms do not face any tax, so their cash flows are

$$\pi^U = F(K) - p^m M + q\, s\, K - G(s)K - (1-\tau^c \tilde{z})(p^I + b)I - \bar{F}.$$

Here $\tilde{z}$ is the net present value of depreciation allowances for a currently untaxed firm, the counterpart to z for taxable firms. Given those cash flows, we get the following recursive formulation for a firm choosing K', s, M:

$$V^U(K) = \max_{m,s,K'} \Bigg\{ F(K) - p^m\, mK + q\, s\, K - G(s)K \\ - (1-\tau^c\tilde{z})(p^I + b)\Big(K' - [1-\delta(m)](1-s)K\Big) - \bar{F} + \frac{V^*(K')}{1+r^k} \Bigg\}. \tag{A.46}$$

The type U FOCs are:

$$\textbf{Investment:} \quad \frac{V^{*'}(K')}{1+r^k} = (1-\tau^c\tilde{z})(p^I+b), \tag{A.47}$$

$$\textbf{Maintenance:} \quad -\delta'(m) = \frac{p^m}{(1-\tau^c\tilde{z})(p^I+b)(1-s)}, \tag{A.48}$$

$$\textbf{Sales:} \quad q - G'(s) = (1-\tau^c\tilde{z})(p^I+b)\,[1-\delta(m)]. \tag{A.49}$$

The envelope condition enables us to define the user cost of capital in this economy. For taxed firms,

$$V^{T\prime}(K) = (1-\tau^c)\Big[F'(K) - p^m\, m^T + q\, s^T - G(s^T)\Big] + \big(1-c-\tau^c z\big)p^I\,(1-\delta(m^T))(1-s^T),$$

while for untaxed firms we have

$$V^{U\prime}(K) = F'(K) - p^m\, m^U + q\, s^U - G(s^U) + (1-\tau^c\tilde{z})(p^I+b)\,(1-\delta(m^U))(1-s^U).$$

The combined envelope condition is

$$V^{*'}(K) = \eta\, V^{T\prime}(K) + (1-\eta)\, V^{U\prime}(K).$$

Equilibrium

An equilibrium in this economy is a collection of policies $\{V^T,\, V^U\}$, prices $\{q, p^I, p^M\}$ (the latter two exogenous) and allocations $\{\, m^T,\, s^T,\, K'^T,\, m^U,\, s^U,\, K'^U\}$ such that:

1. For all K, the Bellman equations for $V^T(K)$ and $V^U(K)$ are satisfied with the corresponding optimal policies.

2. The chosen policy functions satisfy the FOCs for investment, maintenance, and used–capital sales (with type–specific controls m^ι and s^ι) and the envelope conditions hold.

3. The law of motion for capital,

$$K' = \Big[1-\delta(m^\iota)\Big](1-s)K + I,$$

   is satisfied for each firm.

4. The used–capital market clears at the equilibrium price q; that is, the total net sales of used

capital by all firms (taxable and untaxed) sum to zero:

$$\int s^T(K) \cdot K \, d\mu(K) + \int s^U(K) \cdot K \, d\nu(K) = 0,$$

where $\mu(K)$ and $\nu(K)$ are the distributions of capital among taxable and untaxed firms.

5. The expected continuation value is given by

$$V^*(K') = \eta \, V^T(K') + (1-\eta) V^U(K'),$$

and firms form rational expectations consistent with the aggregate outcomes.

Trading Conditions

Trading of used capital will only arise under certain conditions. In particular, an active trading market is characterized by:

- Taxable firms choosing $s^T > 0$ (selling used capital) because their subsidized cost $\frac{1-c-\tau^c z}{1-\tau^c} p^I (1-\delta(m^T))$ is lower than that faced by untaxed firms.

- Untaxed firms choosing $s^U < 0$ (buying used capital) because the equilibrium used capital price q (adjusted by the marginal cost $G'(s^U)$) falls below the cost of acquiring new capital $p^I(1-\delta(m^U))$.

- An equilibrium price q that satisfies

$$q = \frac{1-c-\tau^c z}{1-\tau^c} p^I (1-\delta(m^T)) + G'(s^T) = (1-\tau^c \tilde{z})(p^I + b)(1-\delta(m^U)) + G'(s^U),$$

  along with the market clearing condition

$$\eta \, s^T + (1-\eta) \, s^U = 0.$$

Essentially, we require that the gap between the pre–tax price and after–tax price of new capital is sufficiently large that it is worth it for firms to sell used capital and for untaxed firms to buy it, i.e.,

$$\frac{(1-c-\tau^c z) p^I \, [1-\delta(m^T)]}{1-\tau^c} < q < (1-\tau^c \tilde{z})(p^I + b)[1-\delta(m^U)].$$

I focus on this equilibrium because we observe active used capital trading in practice.

Aggregate User Cost of Capital

Let $\Psi^{\iota} \equiv F'(K^{\iota})$ denote the steady-state user cost of capital for a type-θ firm, the marginal product of capital obtained by combining its investment FOC with its envelope condition. The aggregate user cost is the capital-weighted average of the two type-specific user costs:

$$\boxed{\Psi_{\text{agg}} = \frac{\eta\, K^T}{K}\, \Psi^T + \frac{(1-\eta)\, K^U}{K}\, \Psi^U, \quad \text{with } K = \eta\, K^T + (1-\eta)\, K^U.}$$

Indeed, one can observe that the proportional change in user cost will be strictly smaller to first order in this economy than in the representative firm economy in the main model because the share of untaxable firms will not react very much. Moreover, capital sales will prop up the maintenance rate (since the maintenance optimality condition implies a higher maintenance rate for untaxable firms), so maintenance and depreciation change less in this economy in the aggregate.

F.2 Extension to Multiple Maintenance Inputs

In this section, I extend the model to consider multiple inputs to maintenance production as well as a choice between using internal or external maintenance services. Production is carried out entirely using capital:

$$Y_t = F(K_t).$$

Capital evolves according to

$$K_{t+1} = \Big[1 - \delta(m_{I,t}, m_{E,t})\Big] K_t + I_t,$$

where $m_{I,t}$ is internal maintenance intensity and $m_{E,t}$ is external maintenance intensity. I assume that internal maintenance is the sum of labor and materials purchases

$$M_{I,t} = g(L_{m,t}, I_{m,t}).$$

The firm purchases internal maintenance labor at price w_t and materials at price p_t^X. External maintenance $m_{E,t} = M_{E,t}/K_t$ is purchased at price p_t^m and faces some convex cost of adjustment $C(M_{E,t}, M_{E,t-1})$, with $C(M_E, M_E) = 0$ in steady state. This is to reflect the fact that external maintenance contracts are typically quite sticky. Since internal maintenance also includes labor, then internal maintenance is also sticky, but presumably less so than external maintenance because internal resources can be reallocated more quickly. I could capture that by having an outside option for labor or introducing a separate adjustment cost for internal maintenance, but it would be unnecessarily complicated. Note that with multiple maintenance types, the price of maintenance is a weighted average of each.

Taking K_0 as given, the firm chooses the sequence $\{K_{t+1},\ L_{m,t},\ I_{m,t},\ M_{E,t}\}_{t\geq 0}$ to maximize

$$\mathcal{L} = \sum_{t=0}^{\infty} \left(\frac{1}{1+r^k}\right)^t \Bigg\{(1-\tau_t^c)\Big[F(K_t) - w_t\, L_{m,t} - p_t^X\, I_{m,t} - M_{E,t} - C\big(M_{E,t}, M_{E,t-1}\big)\Big]$$
$$- \ (1-\tau_t^c z_t)p_t^I\Big[K_{t+1} - \Big(1 - \delta\Big(m_{I,t}, m_{E,t}\Big)\Big)K_t\Big]\Bigg\},$$

subject to $m_{I,t} = \frac{g(L_{m,t}, I_{m,t})}{K_t}$. The first-order conditions are not substantially different from the baseline model. The capital Euler equation is:

$$\begin{aligned}(1-\tau_t^c z_t)p_t^I =& \frac{1}{1+r^k}\Bigg\{(1-\tau_{t+1}^c)\, F_K(K_{t+1}) + (1-\tau_{t+1}^c z_{t+1})\, p_{t+1}^I\Big[1 - \delta\big(m_{I,t+1}, m_{E,t+1}\big) \\ &+ \delta_1(t+1)\, m_{I,t+1} + \delta_2(t+1)\, m_{E,t+1}\Big]\Bigg\},\end{aligned} \tag{A.50}$$

which in steady state simplifies to

$$F_K = p^I(1-\tau^c z)\left(\frac{r^k + \delta(m_I, m_E)}{1-\tau^c}\right) + p^{M,I} m_I + p^{M,E} m_E, \tag{A.51}$$

where $p^{M,I}$ is the price of internal maintenance. If internal maintenance is a CES aggregator of labor and materials, then $p^{M,I}$ would just be the usual CES price index of p^X and w.

There are three maintenance choices. Throughout, $\delta_1 = \partial\delta/\partial m_I$ and $\delta_2 = \partial\delta/\partial m_E$ denote the partials of depreciation with respect to internal and external maintenance intensity, $g_L = \partial g/\partial L_m$ and $g_M = \partial g/\partial I_m$ the marginal products of labor and materials in internal maintenance, and C_1, C_2 the partials of the adjustment cost $C(M_{E,t}, M_{E,t-1})$ with respect to its current and lagged arguments. Beginning with the choice of internal labor $L_{m,t}$, the firm's optimal choice is satisfied when

$$-\delta_1(t)\, g_L(t) = \frac{(1-\tau_t^c)w_t}{(1-\tau_t^c z_t)p_t^I}. \tag{A.52}$$

Similarly, optimal materials is given by

$$-\delta_1(t)\, g_M(t) = \frac{(1-\tau_t^c)p_t^X}{(1-\tau_t^c z_t)p_t^I}, \tag{A.53}$$

which implies the marginal rate of substitution between materials and labor is

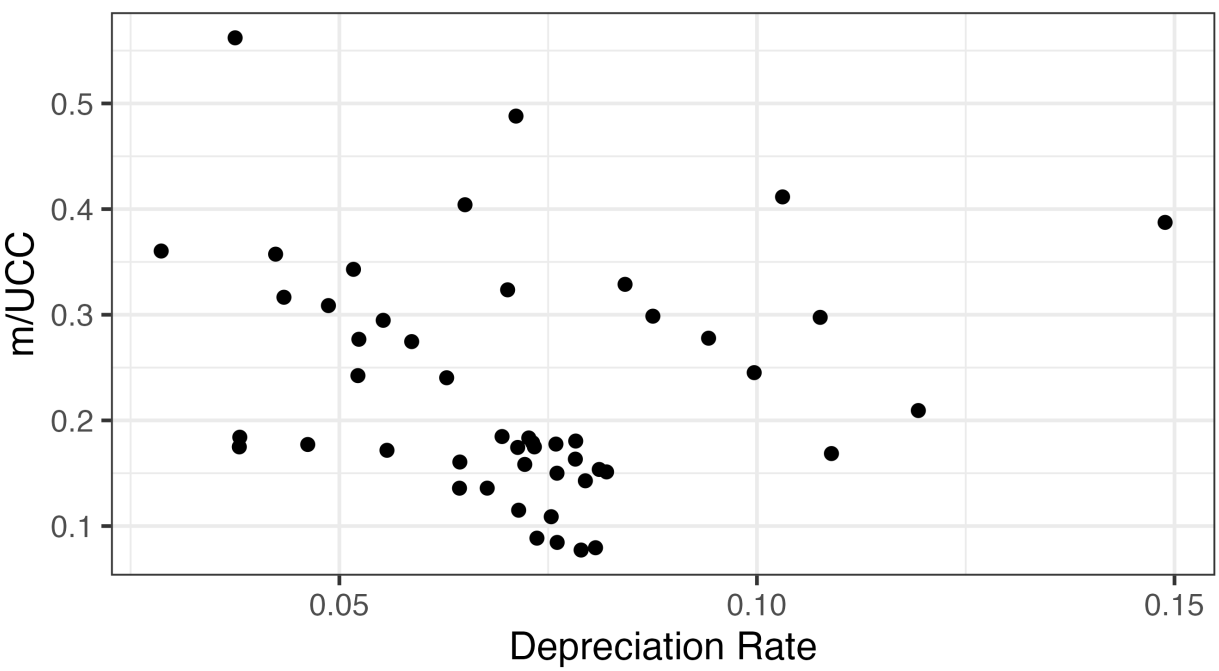
$$\frac{g_L(t)}{g_M(t)} = \frac{w_t}{p_t^X}. \tag{A.54}$$

On the other hand, external maintenance choice is given by

$$-\delta_2(t) = \frac{(1-\tau_t^c)\Big[1 + C_1(t)\Big] + \frac{1}{1+r^k}(1-\tau_{t+1}^c)\, C_2(t+1)}{(1-\tau_t^c z_t)p_t^I}. \tag{A.55}$$

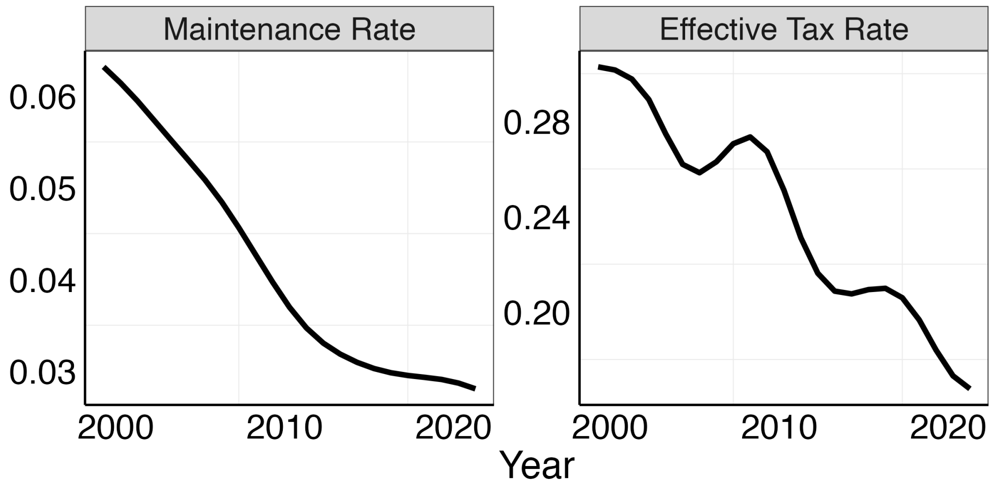
G Additional Figures and Tables

Figure G.1: Industry maintenance share of user cost against the depreciation rate



Notes: Each point is the time series average of maintenance shares and depreciation rates within an industry.

Figure G.2: Trends in Maintenance and Corporate Tax Rates



Notes: The maintenance rate is constructed as gross output in NAICS 811 excluding home repair as a share of current cost private equipment capital. The effective tax rate is the ratio of domestic corporate taxes paid to pre-tax profits from BEA Tables 6.16D and 6.17D. The cyclical component of each series has been removed with a Hodrick-Prescott filter.